

Microeconomic Analysis

Matthew Tan

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1 Pre-requisites

1.1 Basic Analysis: Sets and Sequences, Functions

Sets

Definition 1.1 (Sets). A set S is a collection of elements, where $x \in S$ iff x is in S . Two sets S_1 and S_2 are equal, denoted $S_1 = S_2$ if they contain the same elements.

Definition 1.2 (Operations of Sets). We have the following operations for sets A and B :

- Union:

$$A \cup B := \{x : x \in A \text{ or } x \in B\}$$

- Intersection:

$$A \cap B := \{x : x \in A \text{ and } x \in B\}$$

- Complement:

$$A \setminus B := \{x : x \in A \text{ and } x \notin B\}$$

Definition 1.3 (Countability). A set S is said to be denumerable (or countably infinite) if there exists a bijection of $\mathbb{N}$ onto S . A set S is said to be countable if it is either finite or denumerable. A set S is said to be uncountable if it is not countable

Proposition 1.4 (Countability of certain sets). The following sets are countable (or denumerable)

- The set $\mathbb{N} \times \mathbb{N}$
- The set of all integers $\mathbb{Z}$
- The set of all even natural numbers $E = \{2n : n \in \mathbb{N}\}$
- The set of all rational numbers $\mathbb{Q}$

Proposition 1.5. The following statements are equivalent:

1. S is a countable set
2. There exists a surjection of $\mathbb{N}$ onto S
3. There exists an injection of S into $\mathbb{N}$

Theorem 1.6 (Cantor's Theorem). If A is any set, then there is no surjection of A onto the set $\mathcal{P}(A)$ or all subsets of A

Lemma 1.7. The cardinality of the power set of the natural numbers, $\mathcal{P}(\mathbb{N})$, is equal to the cardinality of the reals, $\mathbb{R}$

Theorem 1.8 (Uncountability of $\mathbb{R}$). The set of real numbers $\mathbb{R}$ is uncountable

Theorem 1.9 (The Density Theorem). If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$

Functions

Definition 1.10 (Function). Let A and B be sets. Then a function from A to B is a set f of ordered pairs in $A \times B$ such that for each $a \in A$ there exists a unique $b \in B$ with $(a, b) \in f$. In other words, if $(a, b) \in f$ and $(a, b') \in f$ then $b = b'$. We also denote this

$$f : A \mapsto B$$

Definition 1.11. Let $f : A \mapsto B$ be a function from A to B . Then

- f is said to be injective (or one-one) if whenever $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$. If f is an injective function, we also say that f is an injection
- f is said to be surjective (onto) if $f(A) = B$ i.e. the range of f , $R(f) = B$. If f is a surjective function, we also say that f is a surjection.
- If f is both injective and surjective, then f is said to be bijective (one-to-one correspondence). If f is bijective, we also say that f is a bijection

To prove that a function f is injective, we must establish that

$$\forall x_1, x_2 \in A, f(x_1) = f(x_2) \implies x_1 = x_2$$

i.e. that there is not more than one element in the domain mapping onto one element in the range. This satisfies the first horizontal line test: every horizontal line $y = b$ with $b \in B$ intersects the graph f in at most one point.

To prove that a function is surjective, we must show that every $b \in B$ has at least one $x \in A$ such that $f(x) = b$. This satisfies the second horizontal line test: every horizontal line $y = b$ with $b \in B$ intersects the graph f in at least one point.

Notice that a bijection implies that $|A| = |B|$ i.e. there are equal number of elements in both the domain and range. To see this, suppose that $|A| > |B|$. By definition of a function, every element in A must map onto some element in B . But since there are more elements in A , at least one $b \in B$ would be mapped onto by more than one element in A , contradicting the fact that the function is injective. Suppose $|B| > |A|$. By the fact that f is a surjection, every $b \in B$ must be mapped onto by some $a \in A$. But since there are more elements in B , there exists a distinct pair $b_1, b_2 \in B$ such that $f(a_1) = b_1$ and $f(a_1) = b_2$ for some $a_1 \in A$, contradicting the definition of a function.

Definition 1.12 (Inverse function). If $f : A \mapsto B$ is injective and surjective, then it has an inverse $f^{-1} : B \mapsto A$ for which

$$f^{-1}(f(x)) = x$$

Definition 1.13 (Monotonic Functions). A function $f : A \mapsto B$ is increasing if for any $x, y \in A$ with $y > x$, $f(y) \geq f(x)$. A function f is strictly increasing if for any $x, y \in A$ with $y > x$, $f(y) > f(x)$.

Definition 1.14 (Limits of functions). Let $f : S \mapsto \mathbb{R}$, where $S \subseteq \mathbb{R}$ and $x_0 \in S$. Then f has a limit, y_0 , at x_0 if and only if for any $\varepsilon > 0$, $\exists \delta > 0$ such that $|x - x_0| < \delta \implies |f(x) - y_0| < \varepsilon$. We denote this

$$\lim_{x \rightarrow x_0} f(x) = y_0$$

Example 1.15. Suppose $k(x) = 3 + 1/x^2$ then

$$\lim_{x \rightarrow 0} k(x) = +\infty, \quad \lim_{x \rightarrow \infty} k(x) = 3$$

So some functions have infinite limits at finite values of x and/or finite limits as x goes to infinity. Note that strictly speaking, $k(x)$ is not defined at $x = 0$.

Sometimes the limit of a function at a point depends on whether you approach the point from below or from above. For example, if $f(x) = 1/x$, then

$$\lim_{x \rightarrow 0^+} f(x) = +\infty, \quad \lim_{x \rightarrow 0^-} f(x) = -\infty$$

Proposition 1.16 (Algebra of Limits). If $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} g(x) = B$ then:

- $\lim_{x \rightarrow a} (f(x) + g(x)) = A + B$
- $\lim_{x \rightarrow a} (f(x)g(x)) = AB$
- $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{A}{B}$ if $B \neq 0$

Definition 1.17 (Continuity). The function $f(x)$ is continuous at x_0 if and only if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. Suppose $f : A \rightarrow \mathbb{R}$ where $A \subseteq \mathbb{R}$. Then, if f is continuous at every point in A , we say that f is continuous on the set A (or a continuous function).

Example 1.18. The function

$$f(x) = \begin{cases} x & \text{if } x < 0 \\ x + 3 & \text{if } x \geq 0 \end{cases}$$

is not continuous at 0. But

$$\lim_{x \rightarrow 0^+} f(x) = 3 = f(0), \quad \lim_{x \rightarrow 0^-} f(x) = 0 \neq f(0)$$

The function is right-continuous at zero but not left-continuous.

Corollary 1.19. If f and g are continuous functions, so are $f + g$ and fg

Proposition 1.20 (Sequential Criterion for Continuity). A function $f : A \rightarrow \mathbb{R}$ is continuous at the point $x_0 \in A$ iff for every sequence $\{x_n\}$ in A that converges to x_0 , the sequence $\{f(x_n)\}$ converges to $f(x_0)$.

Definition 1.21 (Convex and concave functions). A continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex iff, for all $a, b \in \mathbb{R}$:

$$f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b), \forall \lambda \in (0, 1)$$

A function is strictly convex with the inequality being strict. A function f is concave iff $-f$ is convex.

For a function to be convex or concave, its domain must be a convex set.

The following are some important theorems about functions

Theorem 1.22 (Weierstrass Theorem). A continuous function $f : S \rightarrow \mathbb{R}$ where S is a compact subset of $\mathbb{R}$ attains a maximum and minimum value on S

Theorem 1.23 (Intermediate Value Theorem). Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous, and $f(a) < y_0 < f(b)$. Then $\exists x_0 \in (a, b)$ such that $f(x_0) = y_0$

Theorem 1.24 (Brouwer's Fixed Point Theorem). Suppose $S \subset \mathbb{R}$ is a non-empty, compact and convex set, and $f : S \rightarrow S$ is a continuous function. Then f has a fixed point: that is $\exists x_0$ such that $f(x_0) = x_0$

Remark. Brouwer's Fixed Point Theorem also applies when $S \subset \mathbb{R}^n$

1.2 Differentiation

Definition 1.25 (Differentiability). We say that f is differentiable at x_0 if the limit

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. Equivalently, f is differentiable at x_0 if the following limit

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists. We write $f'(x_0)$ or $\frac{df}{dx}$ for the derivative.

Theorem 1.26. If f is differentiable at x , then f is continuous at x

Proof. Suppose

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$$

Then,

$$\begin{aligned} \lim_{h \rightarrow 0} f(x+h) - f(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \cdot h \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \cdot \lim_{h \rightarrow 0} h \\ &= f'(x) \cdot 0 = 0 \end{aligned}$$

■

Theorem 1.27 (Mean Value Theorem). Suppose $f : [a, b] \rightarrow \mathbb{R}$ and f is a differentiable function. Then there is a point $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a)$$

Basic rules for differentiating functions – look it up.

Theorem 1.28 (Taylor's Theorem). Suppose $f : U \rightarrow \mathbb{R}$ where U is a convex subset of $\mathbb{R}$, is $k + 1$ times differentiable. Then for any $a, x \in U$, there exists $z \in (a, x)$ such that:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x - a)^k + \frac{f^{(k+1)}(z)}{(k+1)!}(x - a)^{k+1}$$

where the term

$$R_k(x, a) = \frac{f^{(k+1)}(z)}{(k+1)!}(x - a)^{k+1}$$

is the remainder and is of order $(x - a)^{k+1}$ and converges to 0

If f is infinitely differentiable, we have an infinite series. The Maclaurin's Theorem is a special case with $a = 0$

Theorem 1.29 (L'Hopital Rule). Consider $f : U \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$ where $U, S \subseteq \mathbb{R}$. Then, if either both functions are zero or $\pm\infty$ at a ,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

1.3 Integration

The indefinite integral is

$$\int f(x)dx = F(x) + c$$

where $F'(x) = f(x)$. The definite integral is when you evaluate over an interval i.e. limits of integration

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$

When the limit is $\pm\infty$, you take the limit $b \rightarrow \infty$ or $a \rightarrow -\infty$

Theorem 1.30 (Mean Value Theorem for Integrals). If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, $\exists c \in [a, b]$ such that

$$\int_a^b f(x)dx = f(c)(b - a)$$

Rules for Integration

1. Basic Rules:

$$\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx \text{ and } \int kf(x)dx = k \int f(x)dx$$

2. Common functions

$$\begin{aligned} \int f'(x)f(x)^n dx &= \frac{1}{n+1}f(x)^{n+1} + c, \forall n \in \mathbb{R}, n \neq -1 \\ \int \frac{f'(x)}{f(x)} dx &= \ln f(x) + c \\ \int e^{f(x)} dx &= \frac{1}{f'(x)} e^{f(x)} + c \end{aligned}$$

3. Integration by parts

$$\int u(x)v'(x) = u(x)v(x) - \int v(x)u'(x)dx$$

4. Substitution: Consider $f(y)$ where $y = g(x)$. Then, $dy = g'(x)dx$ and so

$$f(y)dy = f(y)g'(x)dx$$

So,

$$\int f(y)g'(x)dx = \int f(y)dy \text{ or } \int f(y)\frac{dy}{dx}dx = \int f(y)dy$$

Repeated Integrals

If f is a function of two variables x, y , we can integrate with respect to each in turn. If f is continuous, the order of integration does not matter i.e.

$$\int_c^d \left(\int_a^b f(x, y)dx \right) dy = \int_a^b \left(\int_c^d f(x, y)dy \right) dx$$

Differentiating under the integral sign.

Let f be a continuous function of two variables x, y and let p and q be differentiable functions of y . Then consider $I(y) = \int_{p(y)}^{q(y)} f(x, y)dx$. Then,

$$\frac{dI}{dy} = \int_{p(y)}^{q(y)} \frac{\partial}{\partial y} f(x, y)dx + f(q(y), y) \frac{dq}{dy} - f(p(y), y) \frac{dp}{dy}$$

2 Linear Algebra

Definition 2.1 (Vector Space). Let F be a field (e.g., $\mathbb{R}$ or $\mathbb{C}$). A **vector space over F** is a nonempty set V together with two operations:

- **Vector addition:** $+: V \times V \rightarrow V$,
- **Scalar multiplication:** $\cdot: F \times V \rightarrow V$

such that for all $u, v, w \in V$ and all $a, b \in F$, the following axioms hold:

1. **Closure under addition:** $u + v \in V$.
2. **Commutativity of addition:** $u + v = v + u$.
3. **Associativity of addition:** $(u + v) + w = u + (v + w)$.
4. **Additive identity:** There exists $0 \in V$ such that $v + 0 = v$ for all $v \in V$.

5. **Additive inverses:** For each $v \in V$, there exists $-v \in V$ such that $v + (-v) = 0$.

6. **Closure under scalar multiplication:** $av \in V$.

7. **Distributivity over vector addition:** $a(u + v) = au + av$.

8. **Distributivity over scalar addition:** $(a + b)v = av + bv$.

9. **Associativity of scalar multiplication:** $a(bv) = (ab)v$.

10. **Scalar identity:** $1v = v$, where 1 is the multiplicative identity in F .

We typically look at vector spaces that are $\mathbb{R}^n$, or subsets of $\mathbb{R}^n$.

Definition 2.2 (Inner Product). An **inner product** on a vector space V over $\mathbb{R}$ (or $\mathbb{C}$) is a map

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$$

satisfying the following axioms.

Over $\mathbb{R}$ For all $u, v, w \in V$ and all $a \in \mathbb{R}$:

1. **Linearity in the first argument:**

$$\langle au + v, w \rangle = a\langle u, w \rangle + \langle v, w \rangle.$$

2. **Symmetry:**

$$\langle u, v \rangle = \langle v, u \rangle.$$

3. **Positive definiteness:**

$$\langle v, v \rangle \geq 0 \quad \text{and} \quad \langle v, v \rangle = 0 \iff v = 0.$$

The most common inner product is the **dot product** of two vectors i.e. $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i$$

Definition 2.3 (Norm). The norm of a vector $\mathbf{v} \in \mathbb{R}^n$, denoted $\|\mathbf{v}\|$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$$

Of course, the norm a vector is always non-negative (and positive if $\mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$).

2.1 Independence, Span, Basis

Definition 2.4 (Linear combination). A vector $\mathbf{v} \in \mathbb{R}^n$ is a linear combination of a collection of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ if there exists real numbers $\alpha_1, \dots, \alpha_m$ such that

$$\mathbf{v} = \alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \dots + \alpha_m \mathbf{x}_m$$

Definition 2.5 (Linear Independence). A collection of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in \mathbb{R}^n$ is linear independent if none of them is a linear combination of the others (except the trivial linear combination where all $\alpha_1, \alpha_2, \dots, \alpha_m = 0$). Equivalently, a collection of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in \mathbb{R}^n$ is linearly independent if the only solution satisfying

$$\alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \dots + \alpha_m \mathbf{x}_m = \mathbf{0}$$

is $\alpha_1 = \alpha_2 = \dots = \alpha_m = 0$

Remark. The zero vector $\mathbf{0}$ is always linearly dependent since it is a linear combination of any vectors

Definition 2.6 (Orthogonal). Two vectors $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^n$ are orthogonal just when

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$$

Definition 2.7 (Basis). A collection of linearly independent vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in \mathbb{R}^n$ is called a basis of $\mathbb{R}^n$ if any other vector $\mathbf{v} \in \mathbb{R}^n$ is a linear combination of these vectors i.e.

$$\mathbf{v} = \alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \dots + \alpha_m \mathbf{x}_m$$

The standard basis for $\mathbb{R}^n$ is

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$$

Notice that this implies that a basis of $\mathbb{R}^n$ must have exactly n linearly independent vectors

Definition 2.8 (Span). The span of a collection of n -dimensional vectors, $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ is the set of all their linear combinations

$$\text{span}[\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m] = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \dots + \alpha_m \mathbf{x}_m\}$$

Notice that we need at least n vectors to span $\mathbb{R}^n$ and that a basis is the minimal collection of vectors which spans $\mathbb{R}^n$. A collection of n -dimensional vectors, $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$, spans $\mathbb{R}^n$ iff $\text{rank}(\mathbf{X}) = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m) = n$. We may formally state these results (and more).

Proposition 2.9 (Properties of Span). We have the following properties of span

- (All) bases of $\mathbb{R}^n$ have n vectors
- A set of n vectors is a basis if they are linearly independent
- A linear independent collection of vectors is a basis if adding any other vector makes the collection independent
- A set that spans $\mathbb{R}^n$ is a basis if removing any vector makes the remaining collection not to span $\mathbb{R}^n$
- Every spanning set contains a basis
- Every set of independent vectors can be extended to a basis

Definition 2.10 (Subspace). A linear subspace of $\mathbb{R}^n$ is a subset of $\mathbb{R}^n$ that is the span of a finite set of n -vectors. If V is a subspace of $\mathbb{R}^n$, a spanning set for V is a collection of vectors such that their span equals V

Remark. No difference between subspace and span

Definition 2.11 (Dimension). Let V be a nonzero subspace of $\mathbb{R}^n$. The dimension of V , denoted $\dim(V)$ is the smallest number of vectors needed to span V . We define $\dim(\{\mathbf{0}\}) = 0$

2.2 Matrix Algebra

Definition 2.12 (Matrix). An $n \times K$ matrix $\mathbf{A}$ is an array with n rows and K columns

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1K} \\ a_{21} & a_{22} & \cdots & a_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nK} \end{pmatrix}$$

Definition 2.13 (Special Matrices). We have the following matrices with special properties

- **Null Matrix:** The $n \times m$ null matrix is the matrix composed entirely of zeros:

$$\mathbf{0}_{n \times m} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

- **Square Matrix:** A matrix $\mathbf{A}$ is a square matrix when $n = K$ i.e.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

- **Symmetric Matrix:** a square matrix $\mathbf{A}$ that has $a_{ik} = a_{ki}$ for all i, k is called a symmetric matrix. For example

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 9 \\ 2 & 1 & 6 \\ 9 & 6 & 5 \end{pmatrix}$$

- **Diagonal Matrix:** a square matrix that has only non-zero entries on the main diagonal (and hence is necessarily symmetric)

$$\mathbf{A} = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

- **Upper Triangular Matrix:** an upper triangular matrix is an n -square matrix in the form

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

i.e. $a_{ij} = 0$ if $i > j$. Any diagonal matrix is upper (and lower) triangular

- **Scalar Matrix:** a diagonal matrix with the same value in all diagonal elements

$$\mathbf{A} = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

where $a_{11} = a_{22} = \cdots = a_{nn}$

- **Identity Matrix:** a scalar matrix with ones on the diagonal

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

- **Row Echelon Matrix:** An $n \times m$ row echelon matrix is of the form:

$$\mathbf{A} = \begin{pmatrix} \dots & 0 & a_{1k_1} & \dots & \dots & \dots & \dots & a_{1m} \\ \dots & \dots & \dots & 0 & a_{2k_2} & \dots & \dots & a_{2m} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & a_{sk_s} & \dots & a_{sm} \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 \end{pmatrix}$$

where $k_s > \dots > k_2 > k_1 \geq 1$. Upper triangular matrices are row echelon.

Remark. A useful way to think about matrices is that they are linear transformations of vectors. For instance, the identity matrix transforms a vector $\mathbf{x}$ back to itself while the scalar matrix transforms the vector $\mathbf{x}$ to $\lambda\mathbf{x}$ if the scalar matrix contains λ on its main diagonal

Proposition 2.14. Consider an $m \times m$ diagonal matrix $\mathbf{D}$. Then

$$\mathbf{D}^n = \begin{pmatrix} \lambda_1^n & 0 & \dots & 0 \\ 0 & \lambda_2^n & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_m^n \end{pmatrix}$$

Definition 2.15 (Transpose). The transpose of an $n \times K$ matrix $\mathbf{A}$, denoted $\mathbf{A}'$ is obtained by creating the matrix whose k th column is the k th row of $\mathbf{A}$ (and so is a $K \times n$ matrix. An equivalent formulation of this is for a matrix $\mathbf{A}$, we have

$$\mathbf{B} = \mathbf{A}' \Leftrightarrow b_{ik} = a_{ki}, \quad \forall i, k$$

Proposition 2.16. If $\mathbf{A}$ is symmetric, then $\mathbf{A}' = \mathbf{A}$

Example 2.17. Suppose we have

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

then,

$$\mathbf{A}' = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

Definition 2.18 (Algebra of Matrices). We have the following for addition and multiplication

- **Addition:** Let $\mathbf{A}$ and $\mathbf{B}$ be two matrices of the same dimension, then

$$\mathbf{A} + \mathbf{B} = (a_{ij} + b_{ij})$$

For subtraction we similarly have

$$\mathbf{A} - \mathbf{B} = (a_{ij} - b_{ij})$$

though technically we have to define scalar multiplication first and multiply $\mathbf{B}$ by a scalar -1

- **Scalar multiplication:** Let $c \in \mathbb{R}$, then $c\mathbf{A} = (ca_{ij})$ i.e. you multiply every entry in $\mathbf{A}$ by c
- **Matrix Multiplication:** Let $\mathbf{A}$ be $n \times K$ and $\mathbf{B}$ be $K \times m$. Then, $\mathbf{C} = \mathbf{AB} = (c_{ij}) = \sum_{k=1}^K (a_{ik}b_{kj})$ (i.e. the c_{ij} entry is the inner product of the i th row of $\mathbf{A}$ and the k th row of $\mathbf{B}$) where $\mathbf{AB}$ is an $n \times m$ matrix. For example, if

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 9 \\ 2 & 1 & 6 \\ 9 & 6 & 5 \end{pmatrix} \mathbf{B} = \begin{pmatrix} 2 & 1 & 5 \\ 3 & 2 & 7 \\ 1 & 2 & 9 \end{pmatrix}$$

then

$$\mathbf{AB} = \begin{pmatrix} 21 & 25 & 110 \\ 13 & 16 & 71 \\ 41 & 31 & 132 \end{pmatrix}$$

Proposition 2.19 (Properties of algebra of Matrices). We have the following properties

- **Addition:** for addition we have
 1. Adding zero or the null matrix: $\mathbf{A} + \mathbf{0} = \mathbf{A}$
 2. Commutative: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
 3. Associative: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$
- **Scalar Multiplication:** for scalar multiplication we have
 1. $0\mathbf{A} = \mathbf{0}$
 2. $-1\mathbf{A} = -\mathbf{A}$
- **Matrix Multiplication:** for matrix multiplication we have
 1. Not commutative: $\mathbf{AB} \neq \mathbf{BA}$
 2. Associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
 3. Distributive: $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$
 4. Transpose of a product: $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$
 5. Transpose of an extended product: $(\mathbf{ABC})' = \mathbf{C}'\mathbf{B}'\mathbf{A}'$
 6. Identity: $\mathbf{AI}_K = \mathbf{I}_n\mathbf{A} = \mathbf{A}$

Intuition behind Product operation: Say that a linear transformation between $\mathbb{R}^n$ and $\mathbb{R}^m$ is a mapping $\mathbf{x} \mapsto \mathbf{Ax}$, where $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{m \times n}$. We can think of a matrix $\mathbf{A}$ as a linear transformation of some vector $\mathbf{x}$.

Definition 2.20 (Determinant). Let $\mathbf{A}$ be a $K \times K$ square matrix. The determinant of $\mathbf{A}$, denoted $|\mathbf{A}|$, is such that for any row i ,

$$|\mathbf{A}| = \sum_{k=1}^K a_{ik}(-1)^{i+k}|A_{ik}|$$

where $|A_{ik}|$ is a minor of $\mathbf{A}$ i.e. the determinant of A_{ik} where A_{ik} is a matrix obtained after deleting row i and column k .

We call $C_{ik} = (-1)^{i+k}|A_{ik}|$ the **cofactor**.

Note that for a matrix $\mathbf{A}$, if the determinant is 0, then it is not invertible (since it does not have fully rank).

Remark. For a 2×2 matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, a useful rule is $|\mathbf{A}| = ad - bc$

Example 2.21. If $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, then

$$|\mathbf{A}| = a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{12}a_{23}a_{31} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{21}a_{12}a_{33}$$

This is known as the *Rule of Sarrus*

Definition 2.22 (Inverse of a Matrix). Let $\mathbf{A}$ be a square $n \times n$ matrix. The inverse matrix, denoted by $\mathbf{A}^{-1}$, is defined such that

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n$$

In general, we have $\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|}\mathbf{C}'$, where

$$\mathbf{C} = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1K} \\ c_{21} & c_{22} & \cdots & c_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nK} \end{pmatrix}$$

is the matrix of cofactors: $C_{ik} = (-1)^{i+k}|A_{ik}|$

Example 2.23. Let $\mathbf{A}$ be a 2×2 matrix

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

then,

$$\mathbf{A}^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

where $\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$. Notice that the denominator of the fraction is precisely the determinant of $\mathbf{A}$

Definition 2.24 (Trace). The trace of a square matrix $\mathbf{A}$ is the sum of the elements of its diagonal

$$tr(\mathbf{A}) = \sum_i^n a_{ii}$$

Matrices are useful to represent a system of linear equations. We say that a system of n linear equation is of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2m}x_m &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nm}x_m &= b_n \end{aligned}$$

which can be represented as a $n \times m$ matrix $\mathbf{A}$, an $m \times 1$ column matrix $\mathbf{x}$ (i.e. a vector) and an $n \times 1$ column matrix $\mathbf{b}$ as

$$\mathbf{Ax} = \mathbf{b}$$

A linear system must be one of these types:

1. Having exactly one solution (i.e. uniquely solved)
2. Having no solution (i.e. overdetermined)
3. Having infinitely many solutions (i.e. underdetermined/having a freely occurring variable)

Definition 2.25. Consider a system of n linear equation

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2m}x_m &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nm}x_m &= b_n \end{aligned}$$

We say that the system is homogenous if $b_1 = b_2 = \cdots = b_n = 0$. Homogenous systems must have either one or infinite solutions since $x_1 = x_2 = \cdots = x_m = 0$ is always a solution.

Proposition 2.26. Suppose $\mathbf{A}$ is a square matrix. If $\mathbf{A}$ is invertible, we can write

$$\mathbf{A}^{-1}\mathbf{Ax} = \mathbf{Ix} = \mathbf{A}^{-1}\mathbf{b}$$

which provides immediately the unique values of $\mathbf{x}$ that solve the system.

Definition 2.27 (Rank). The rank of an $n \times m$ matrix $\mathbf{A}$ is the maximal number of linearly independent rows or columns. We say that $\mathbf{A}$ is full rank if $\text{rank}(\mathbf{A}) = \min\{n, m\}$

Proposition 2.28 (Properties of rank). Consider an $n \times m$ matrix $\mathbf{A}$. Then

- $\text{rank}(A) = \text{rank}(A')$
- $\text{rank}(A) \leq n$ i.e. number of rows
- $\text{rank}(A) \leq m$ i.e. number of columns

The following are not actually included but I think its super useful to know

Definition 2.29 (Column Space/Image). The Column Space or Image of an $m \times n$ matrix $\mathbf{A}$ is the span of all its column vectors i.e.

$$C(\mathbf{A}) = \text{span}[\mathbf{a}_1, \dots, \mathbf{a}_n]$$

It is a subspace of $\mathbb{R}^n$

Definition 2.30 (Null Space/Kernel). The Null Space or kernel of an $m \times n$ matrix $\mathbf{A}$ is the set of vectors $\mathbf{x}$ that satisfy the homogenous equation $\mathbf{Ax} = \mathbf{0}$. The Null Space is a vector subspace of $\mathbb{R}^n$

Theorem 2.31 (Rank-Nullity Theorem). Consider an $m \times n$ matrix $\mathbf{A}$. Then,

$$\text{Dim}(C(\mathbf{A})) + \text{Dim}(N(\mathbf{A})) = n$$

Connection between Rank and Column Space

Notice that full rank implies $\text{Dim}(C(\mathbf{A})) = n$ since there are no “redundant” columns and so $C(\mathbf{A})$ actually forms a basis.

One purpose of linear algebra is to solve systems of equations. This entails doing two things:

- Finding conditions for when a system has no solution, one unique solution or infinite solutions
- Devising procedures to find explicit solutions to the system

Finding conditions for when a system has no solution, one unique solution or infinite solutions

TLDR: Consider a system $\mathbf{Ax} = \mathbf{b}$

- No solution: $\text{rank}(\mathbf{A}) \neq \text{rank}((\mathbf{A}, \mathbf{b}))$ OR $\mathbf{b} \notin C(\mathbf{A})$ OR $\mathbf{b} \notin \text{span}[\mathbf{a}_1, \dots, \mathbf{a}_n]$
- Unique solution: $\mathbf{A}$ has full rank OR $\mathbf{b} \in C(\mathbf{A})$ and the only vector in the null space $N(\mathbf{A})$ is $\mathbf{0}$
- Infinite solutions: $\text{rank}(\mathbf{A}) = \text{rank}((\mathbf{A}, \mathbf{b}))$ and $\mathbf{A}$ is not full rank OR $\mathbf{b} \in C(\mathbf{A})$ and there exists a non-zero vector in the null space $N(\mathbf{A})$

Way 1: Looking at rank

Proposition 2.32. A system of linear equations $\mathbf{Ax} = \mathbf{b}$ has solutions iff the rank of the matrix (of coefficients) $\mathbf{A}$ is equal to the rank of the augmented matrix $(\mathbf{A}, \mathbf{b})$ where

$$(\mathbf{A}, \mathbf{b}) = \begin{pmatrix} a_{11} & \dots & a_{1m} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \dots & a_{nm} & b_m \end{pmatrix}$$

The system has a unique solution if $\mathbf{A}$ has full rank.

Remark. The rank of the augmented matrix $(\mathbf{A}, \mathbf{b})$ is weakly greater than the rank of $\mathbf{A}$ i.e. $\text{rank}((\mathbf{A}, \mathbf{b})) \geq \text{rank}(\mathbf{A})$. The corollary of this is that if $\mathbf{A}$ is full rank, then $(\mathbf{A}, \mathbf{b})$ is full rank.

Way 2: Look at determinant

Proposition 2.33. The linear system $\mathbf{Ax} = \mathbf{b}$ with n equations and n unknowns has a unique solution iff $\det(\mathbf{A}) \neq 0$ i.e. iff $\mathbf{A}$ has full rank

Thus, if system $\mathbf{Ax} = \mathbf{b}_1$ has a unique solution, then any system $\mathbf{Ax} = \mathbf{b}_2$ has a unique solution.

Way 3: Looking at Column Space and Null Space

Proposition 2.34. Consider the linear system $\mathbf{Ax} = \mathbf{b}$. Then, the linear system has

- No solution iff $\mathbf{b} \notin C(\mathbf{A})$
- Unique solution iff $\mathbf{b} \in C(\mathbf{A})$ and the only vector in the null space $N(\mathbf{A})$ is $\mathbf{0}$
- Infinite solutions iff $\mathbf{b} \in C(\mathbf{A})$ and there exists a non-zero vector in the null space $N(\mathbf{A})$

Way 4: Looking at linear combination

Proposition 2.35. A system of linear equations $\mathbf{Ax} = \mathbf{b}$ has solutions iff $\mathbf{b} \in \text{span}[\mathbf{a}_1, \dots, \mathbf{a}_n]$ i.e. $\mathbf{b}$ is a linear combination of the vectors that compose matrix $\mathbf{A}$.

Additionally, if the vectors that compose $\mathbf{A}$ are linearly independent, then the system has at most one solution

Proof. (Sketch for second part). Notice that linear independence implies that the linear combination is uniquely solved by $x_1 = \dots = x_m = 0$ and hence the homogenous system

$$\mathbf{A} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} = \mathbf{0}$$

has at most one solution ■

This is very obviously equivalent to Way 3.

Finding solutions

Way 1: Gauss Jordan Elimination

Way 2: Cramer's Rule

2.3 Eigenvector and Eigenvalues, Quadratic forms, Definiteness and Spectral Theorem

Definition 2.36 (Quadratic Form). A quadratic form is $\mathbf{x}'\mathbf{Ax}$ where $\mathbf{A}$ is an $n \times n$ symmetric matrix and $\mathbf{x}$ is $n \times 1$

Example 2.37. It is worth memorising for $n = 2, 3$ cases:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & u \\ u & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = ax^2 + by^2 + 2uxy$$

and

$$\begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} a & u & v \\ u & b & w \\ v & w & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = ax^2 + by^2 + cz^2 + 2uxy + 2vzx + 2wyz$$

Definition 2.38 (Definiteness). For a $n \times n$ symmetric matrix $\mathbf{A}$, we say that it is

- Positive Definite if $\mathbf{x}'\mathbf{A}\mathbf{x} > 0$ for all nonzero $\mathbf{x}$
- Negative Definite if $\mathbf{x}'\mathbf{A}\mathbf{x} < 0$ for all nonzero $\mathbf{x}$
- Positive Semidefinite if $\mathbf{x}'\mathbf{A}\mathbf{x} \geq 0$ for all nonzero $\mathbf{x}$
- Negative Semidefinite if $\mathbf{x}'\mathbf{A}\mathbf{x} \leq 0$ for all nonzero $\mathbf{x}$

Proposition 2.39 (Sylvester criterion). Consider a symmetric $n \times n$ matrix $\mathbf{A}$. Then, consider the sequence of corner minors

$$C_1 = |a_{11}|, C_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, C_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \dots$$

Then

- $\mathbf{A}$ is positive semi-definite if the corner minors do not change sign
- $\mathbf{A}$ is negative semi-definite if the corner minors change sign (alternate)

Note that the corner minors are the determinants of the corner minors.

Definition 2.40 (Eigenvalues and Eigenvectors). An eigenvalue and an eigenvector of an n -square matrix $\mathbf{A}$ are a scalar λ and a vector $\mathbf{v} \in \mathbb{R}^n$ such that $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$.

The eigenvalues of $\mathbf{A}$ are the roots of the characteristic polynomial i.e. $\det(\mathbf{A} - \lambda\mathbf{I})$.

There are always n roots which might be complex or repeated (in this course we only consider real roots in symmetric matrices).

Remark. The intuition of this is that we are trying to find λ such that $\mathbf{A} - \lambda\mathbf{I}$ is not invertible hence why the determinant is set to 0 (to find the roots).

Proposition 2.41. For a symmetric n -square matrix $\mathbf{A}$,

- $\det(\mathbf{A}) = \lambda_1 \lambda_2 \dots \lambda_n$
- $\text{tr}(\mathbf{A}) = \lambda_1 + \lambda_2 + \dots + \lambda_n$

Furthermore,

- If $\det(\mathbf{A}) < 0$, eigenvalues have opposite sign
- If $\det(\mathbf{A}) > 0$, eigenvalues have the same sign
 - If $\text{tr}(\mathbf{A}) > 0$, eigenvalues are positive
 - If $\text{tr}(\mathbf{A}) < 0$, eigenvalues are negative.

Note that if $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ then $\mathbf{A}(\alpha\mathbf{x}) = \lambda(\alpha\mathbf{x})$ so eigenvectors are defined up to a multiple.

Theorem 2.42 (Spectral Theorem). Let $\mathbf{A}$ be a symmetric $n \times n$ matrix with linearly independent eigenvectors $\mathbf{w}_1, \dots, \mathbf{w}_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. Let $\mathbf{W}$ be the $n \times n$ matrix whose columns are the unique vectors $\mathbf{w}_j$.

Then, $\mathbf{W}$ is invertible and we have $\mathbf{W}^{-1}\mathbf{A}\mathbf{W} = \mathbf{D}$ with

$$\mathbf{A} = \mathbf{W}\mathbf{D}\mathbf{W}^{-1}$$

where $\mathbf{D}$ is the diagonal matrix formed by eigenvalues i.e.

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$$

Theorem 2.43 (Spectral Theorem II). Let $\mathbf{A}$ be a symmetric $n \times n$ matrix with orthonormal eigenvectors $\mathbf{w}_1, \dots, \mathbf{w}_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. Let $\mathbf{W}$ be the $n \times n$ matrix whose columns are the unit vectors $\mathbf{w}_j / \|\mathbf{w}_j\|$. Then, $\mathbf{W}$ is invertible and $\mathbf{W}^{-1} = \mathbf{W}'$. We also have $\mathbf{W}^{-1}\mathbf{A}\mathbf{W} = \mathbf{D}$ with

$$\mathbf{A} = \mathbf{W}\mathbf{D}\mathbf{W}^{-1} = \mathbf{W}\mathbf{D}\mathbf{W}'$$

where $\mathbf{D}$ is the diagonal matrix formed by eigenvalues i.e.

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$$

Remark. The key difference between the two theorems is that the eigenvectors are orthonormal (a stronger condition than linear independence).

Corollary 2.44. Let $\mathbf{A}$ be a symmetric $n \times n$ matrix with linearly independent eigenvectors $\mathbf{w}_1, \dots, \mathbf{w}_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. Let $\mathbf{W}$ be the $n \times n$ matrix whose columns are the unique vectors $\mathbf{w}_j$. Then

$$\mathbf{A}^n = \mathbf{W}\mathbf{D}^n\mathbf{W}^{-1}$$

Proof. (Sketch) Notice that

$$\mathbf{A}^2 = (\mathbf{W}\mathbf{D}\mathbf{W}^{-1})(\mathbf{W}\mathbf{D}\mathbf{W}^{-1}) = \mathbf{W}\mathbf{D}^2\mathbf{W}^{-1}$$

Proceeding inductively, we yield the claim. ■

Example 2.45.

Definition 2.46 (Definiteness with Eigenvalues). Consider a symmetric $n \times n$ matrix $\mathbf{A}$ with n eigenvalues $\lambda_1, \dots, \lambda_n$. Then

- $\mathbf{A}$ is positive definite if $\lambda_j > 0$ for all $i \in \{1, \dots, n\}$
- $\mathbf{A}$ is negative definite if $\lambda_j < 0$ for all $i \in \{1, \dots, n\}$
- $\mathbf{A}$ is indefinite if there exists $i, j \in \{1, \dots, n\}$ where $\lambda_i > 0$ and $\lambda_j < 0$
- $\mathbf{A}$ is positive semi-definite if $\lambda_j \geq 0$ for all $i \in \{1, \dots, n\}$ with at least one $\lambda_i = 0$
- $\mathbf{A}$ is negative semi-definite if $\lambda_j \leq 0$ for all $i \in \{1, \dots, n\}$ with at least one $\lambda_i = 0$

Determinant Test for Definiteness

Define $\mathbf{A}_k$ as the square submatrix of $\mathbf{A}$ with only the first k rows and columns retained. The k -th leading principal minor is the determinant of $\mathbf{A}_k$. Then, we have the following determinant test

- $\mathbf{A}$ is positive definite iff $|\mathbf{A}_k| > 0$ for all $k = 1, \dots, n$
- $\mathbf{A}$ is positive semi-definite iff $|\mathbf{A}_k| \geq 0$ for all $k = 1, \dots, n$ AND all row+col permutations
- $\mathbf{A}$ is negative definite iff $(-1)^k |\mathbf{A}_k| > 0$ for all $k = 1, \dots, n$
- $\mathbf{A}$ is negative semidefinite iff $(-1)^k |\mathbf{A}_k| \geq 0$ for all $k = 1, \dots, n$ AND all row+col permutations
- $\mathbf{A}$ is indefinite otherwise

Row and Column permutations for the leading minors means you check for every row/column combination. For example,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

The usual leading 2×2 principal minor is

$$\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

But after permuting indices, one must also check minors such as

$$\det \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix}, \quad \det \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix},$$

since these become leading principal minors after a suitable simultaneous permutation of rows and columns.

3 Multivariate Calculus

3.1 Basic Analysis in $\mathbb{R}^n$

3.1.1 Properties of Real Numbers

Definition 3.1 (Supremum, Infimum, Maximum and Minimum). Let S be a subset of $\mathbb{R}$.

- If there is a real number b such that $x \leq b, \forall x \in S$, then b is an upper bound for S and the set is bounded above
- If S is bounded above, then there is an upper bound b^* such that if b is an upper bound for S , $b^* \leq b$. We call b^* the supremum, or least upper bound i.e. $b^* = \sup(S)$
- If S is not bounded above, $\sup(S) = \infty$

Similarly, S may be bounded below; $\inf(S)$ is the infimum or greatest lower bound.

Furthermore, if S is bounded above, and there is a real number m^* such that $m^* \in S$ and $\forall m \in S, m \leq m^*$, then m^* is the maximum, or greatest element, of S . Similarly, if S is bounded below, and there is a real number u^* such that $u^* \in S$ and $\forall u \in S, u^* \leq u$, then u^* is the minimum, or greatest element, of S .

Remark. Notice that in $\mathbb{R}$, $a^* = \min(S) \iff a^* = \inf(S) \wedge a^* \in S$

Example 3.2. We have the following examples

- If $S = [a, b]$, S is both bounded above and below and $\inf(S) = a$ and $\sup(S) = b$. Since $a \in S$, then $\min(S) = \inf(S)$ but since $b \notin S$, $\max(S)$ does not exist.
- If $S = \{1/n : n \in \mathbb{N}\}$, $\inf(S) = 0$ but $\min(S)$ does not exist.

Definition 3.3 (Properties of $\mathbb{R}$). We have the following properties of $\mathbb{R}$

1. $\mathbb{R}$ has the operations $(+, \cdot)$ (i.e. is called a Field) and is closed under $(+, \cdot)$: if $x, y \in \mathbb{R}$, $x + y \in \mathbb{R}$ and $x \cdot y \in \mathbb{R}$ and both addition and multiplication operations are commutative, associative, neutral, with an inverse and multiplication is also distributive.
2. There is a complete order (reflexive, transitive, complete) $\leq$ on $\mathbb{R}$ compatible with $(+, \cdot)$
3. Completeness: Let L and H be two non-empty subsets of $\mathbb{R}$ such that $\forall l \in L, \forall h \in H, l \leq h$. Then, there exists $x \in \mathbb{R}$ such that

$$l \leq x \leq h, \forall l \in L, \forall h \in H$$

Remark. The last property of completeness sets $\mathbb{R}$ apart from others such as $\mathbb{Q}$ and is what allows the field $\mathbb{R}$ to be closed to the limits operation.

Theorem 3.4. Every non-empty subset of $\mathbb{R}$ that is bounded above (below) has a supremum (infimum) in $\mathbb{R}$

Proof. Let $X \subset \mathbb{R}$ be bounded above. Denote by U the set of upper bounds of X . U is non-empty. For all $x \in X$ and $u \in U$, $x \leq u$, therefore by completeness, there exists $s \in \mathbb{R}$ such that

$$x \leq s \leq u, \forall x \in X, \forall u \in U$$

But then, s is the lowest upper bound ■

Lemma 3.5 ($\sqrt{2}$ is irrational). $\sqrt{2} \notin \mathbb{Q}$ where $\mathbb{Q} = \{p/q | p, q \in \mathbb{Z}, q \neq 0\}$

Proof. SFC $\exists a \in \mathbb{Q}$ s.t. $a^2 = 2$. WLOG, $a \equiv \frac{p}{q}$ where $\frac{p}{q}$ have no common factor. Then $a^2 = \frac{p^2}{q^2} = 2$ and so $p^2 = 2q^2$. Since p^2 is even then p is even. Define $p = 2m$ where $m \in \mathbb{Z}$. Then, $2q^2 = 4m^2 \implies q^2 = 2m^2$ so q^2 is even and so q is even. Since p and q are even they have a common factor 2 by definition, a contradiction ■

Proposition 3.6 ($\mathbb{Q}$ is incomplete). $\mathbb{Q}$ does not satisfy the property of completeness

Proof. Let $L, H \subset \mathbb{Q}$ where

$$\begin{aligned} L &= \{l \in \mathbb{Q} : l \geq 0 \text{ and } l^2 < 2\} \\ H &= \{h \in \mathbb{Q} : h \geq 0 \text{ and } h^2 > 2\} \end{aligned}$$

Then, L and H are non-empty with $h^2 > l^2 \implies h \geq l, \forall h \in H, \forall l \in L$ since $h, l \geq 0$. AFC $\exists x \in \mathbb{Q}$ s.t. $l \leq x \leq h, \forall l \in L, \forall h \in H$. Consider

- $x^2 < 2$. SFC $\exists y \in \mathbb{Q}$ s.t. $y = x + \frac{2-x^2}{x+2} > x$ since $x^2 < 2$. Furthermore,

$$y = \frac{x^2 + 2x - 2 - x^2}{x+2} = \frac{2(x+1)}{x+2} \implies y^2 = \frac{4(x+1)^2}{(x+2)^2}$$

Hence,

$$y^2 - 2 = \frac{2(x^2 - 2)}{(x+2)^2}$$

Since $x^2 < 2$ then $y^2 - 2 < 0$ which implies $y^2 < 2$ and so $y^2 \in L$. So, $\exists l \in L$ s.t. $l < x$

- $x^2 > 2$. SFC $\exists y \in \mathbb{Q}$ s.t. $y = x + \frac{2-x^2}{x+2} \leq x$. Then,

$$y^2 - 2 = \frac{2(x^2 - 2)}{(x+2)^2} > 0$$

since $x^2 > 2$. So $y^2 > 2$ which implies $y \in H$. So, there exists $h \in H$ s.t. $h < x$ which is a contradiction

By trichotomy, $x^2 = 2$ but $x \notin \mathbb{Q}$ so $\mathbb{Q}$ is incomplete ■

3.1.2 Convergence and Sequences

Definition 3.7 (Sequences). A sequence $\{\mathbf{x}_k\}_{k=1}^{\infty}$ in $\mathbb{R}^n$ is an ordered list of elements in $\mathbb{R}^n$.

Definition 3.8 (Convergence). A sequence $\{x_k\}$ in $\mathbb{R}$ is said to converge to a limit x if $\forall \varepsilon > 0$, there exists an integer K such that

$$|x_k - x| \leq \varepsilon, \forall k > K$$

A sequence $\{\mathbf{x}_k\}$ in $\mathbb{R}^n$ converges to $\mathbf{x}$ if $\forall \varepsilon > 0, \exists K$ such that

$$\|\mathbf{x}_k - \mathbf{x}\| \leq \varepsilon, \forall k > K$$

Example 3.9. Notice that for the following sequences

- $x_k = k$ does not converge
- $x_k = 1/k$ converges to 0

Remark. The sum of the terms of a sequence in $\mathbb{R}$ is called a series i.e. geometric series

Proposition 3.10. A sequence $\{\mathbf{x}_k\}$ in $\mathbb{R}^n$ converges to $\mathbf{x}_0$ iff $x_i \rightarrow x_i^0$ for all i i.e. each component converges

Theorem 3.11 (Further Results). We have the following results for sequences in $\mathbb{R}^n$

- Every convergent sequence in $\mathbb{R}^n$ is bounded
- Every increasing sequence in $\mathbb{R}^n$ that is bounded above converges to its supremum
- Every sequence in $\mathbb{R}$ contains either an increasing subsequence or decreasing subsequence or both/

Definition 3.12 (Subsequence). A subsequence of a sequence $\{\mathbf{x}_k\}$ is an (countably) infinite subset of its elements in the same order

Example 3.13. $x_k = (-1)^k$ does not converge but has convergent subsequences i.e. $x_k = (-1)^{2k}$ or $x_k = (-1)^{2k+1}$. $x_k = k$ does not converge nor does it have convergent subsequences

Theorem 3.14 (Bolzano-Weierstrass Theorem). If S is a compact subset of $\mathbb{R}^n$, then every sequence of points in S has a subsequence that converges to a point in S .

3.1.3 Properties of Sets

Definition 3.15 (Norms). A function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ is called a norm if for every $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $a \in \mathbb{R}$:

- $\|\mathbf{x}\| \geq 0$ (non-negativity)
- $\|\mathbf{x}\| = 0$ iff $\mathbf{x} = \mathbf{0}$
- $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ (triangle inequality)
- $\|a\mathbf{x}\| = |a|\|\mathbf{x}\|$

The main norm we work with is the Euclidean norm:

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

Definition 3.16 (Open Ball). An open ball $B_r(\mathbf{x})$ in $\mathbb{R}^n$ is a set defined by

$$B_r(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{y}\| < r\}$$

Remark. This can be easily generalised for any metric space (X, d)

Definition 3.17 (Open, Closed, Bounded and Convex Sets). We have the following

- A set S in $\mathbb{R}^n$ is open if for all $\mathbf{x} \in S$, there is some $r > 0$ such that $B_r(\mathbf{x}) \subset S$. (every point in S is safely inside S and not on the boundary so there is an open ball around it contained in S)
- A set is closed if and only if its complement is open. (A closed set contains its boundary)
- $\mathbf{x}$ is an interior point of S if there is some $r > 0$ such that $B_r(\mathbf{x}) \subseteq S$
- A set S in $\mathbb{R}^n$ is bounded if there is some $r > 0$ such that $S \subseteq B_r(\mathbf{0})$ (basically, $\{\mathbf{y} \in S : \|\mathbf{y}\| < r\}$ i.e. its bounded by r)
- A set S in $\mathbb{R}^n$ is convex if, $\forall \mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$, $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in S$

Theorem 3.18. A set S in $\mathbb{R}^n$ is closed iff, for any sequence $\mathbf{x}_k$ such that $\mathbf{x}_k \in S$ for all k and $\mathbf{x}_k \rightarrow \mathbf{x}$, then $\mathbf{x} \in S$

NOTE: to prove that a set is open, it is insufficient to show that there is a sequence where the limit falls out of the set – this only shows that the set is NOT CLOSED. You need to positively show that we can find a ball included in S for any $\mathbf{x} \in S$

Theorem 3.19 (Heine-Borel). A set $S \subseteq \mathbb{R}^n$ is compact iff it is closed and bounded

Example 3.20. Consider the following:

- (a) For each of the following subsets of $\mathbb{R}$, say whether the set is (i) open, closed, or neither; (ii) bounded; (iii) compact; (iv) convex; and write down the supremum, and the maximum if it exists.

$$A = (0, 1), \quad B = (-\infty, -1], \quad C = \{x : |x| > 2\}, \quad \mathbb{R} \setminus C.$$

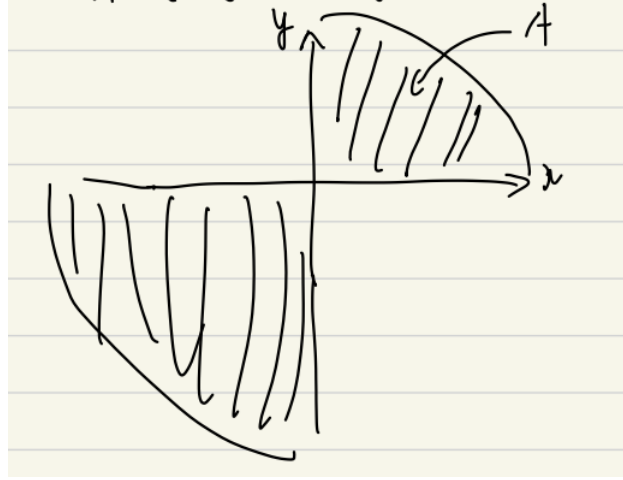
- $A = (0, 1)$.
Open: Let $\varepsilon = \min\{x, 1 - x\}$. Then, for any $x \in A$, $(x - \varepsilon, x + \varepsilon) \subset A$. Then, if $x < 0.5$, $B_\varepsilon(x) = (0, 2x) \subset (0, 1)$ since $2x < 1$. If $x \geq 0.5$, $B_\varepsilon(x) = (2x - 1, 1)$ where $1 > 2x - 1 > 0$ since $2 > 2x > 1$.
Not Closed: Consider a sequence $\{x_k\}$ where $x_k = \frac{1}{2k}$ where $k = 1, \dots, N$. so $x_k \in A$ but as $k \rightarrow \infty$, $x_k \rightarrow x = 0$ but $x \notin A$.
Bounded: since $\inf(A) = 0$ and $\sup(A) = 1$ (alternatively, we can define $\varepsilon = 1.1$ such that $A \subset B_\varepsilon(0)$.
Not Compact: since not closed
Convex: since for any $x, y \in (0, 1)$ and $\lambda \in [0, 1]$, $\lambda x + (1 - \lambda)y \in A$.
 $\inf(A) = 0$, $\sup(A) = 1$ and maximum does not exist
- $B = (-\infty, 1]$
Closed: Consider $B^C = (-1, \infty)$. Then, we can easily choose $\varepsilon = x + 1$ so that $B_\varepsilon(x) \subset B$ as then for any $x \in B$, $B_\varepsilon(x) = (-1, 2x + 1)$ where $2x + 1 < \infty$.
Not Open: Any ball $B_\varepsilon(x) = (x - \varepsilon, x + \varepsilon)$ contains a point greater than -1 when $x = -1$ so no $\varepsilon > 0$ such that $B_\varepsilon(x) \subset B$.
Not Bounded: Cannot choose any r such that $B \subset B_r(0)$ since $r < \infty$ by definition.
Not Compact: since not bounded
Convex: any interval on the real line is convex
 $\inf(B) = -1$, $\sup(B)$ does not exist (or $+\infty$), maximum does not exist
- $C = (-\infty, -2) \cup (2, \infty)$
Not Closed: Consider $x_k = \sqrt{2 + \frac{1}{2}}$, then $x_k \in C$ for all $k \in \mathbb{N}$ but $x = \sqrt{2} \notin C$
Open: Take $\varepsilon = |x| - 2$. Then, for any $x \in C$, $B_\varepsilon(x) = (-2|x| + 2, -2) \cup (2, 2|x| - 2)$ and since $x < \infty$, $-2|x| + 2 > -\infty$ and $2|x| - 2 < \infty$. Hence $B_\varepsilon(x) \subset C$.
Not Bounded: no $r > 0$ such that $C \subset B_r(0)$ since $r < \infty$.
Not Compact: since not closed and not bounded Not Convex: Let $\lambda = \frac{1}{2}$, then

$$-3\left(\frac{1}{2}\right) + \left(1 - \frac{1}{2}\right)(3) = 0 \notin C$$

$\sup$, $\inf$ and maximum does not exist.

- $\mathbb{R} \setminus C = [-2, 2]$
Closed: Consider the complement C which is Open. So $\mathbb{R} \setminus C$ is Closed.
Not Open: Any ball $B_\varepsilon(x) = (x - \varepsilon, x + \varepsilon)$ contains a point greater than 2 when $x = 2$. So no $\varepsilon > 0$ such that $B_\varepsilon(x) \subset \mathbb{R} \setminus C$.
Bounded: Let $r = 3$. Then $B_r(0) = (-3, 3)$ for which $\mathbb{R} \setminus C$ clearly is a subset of
Compact: By Heine-Borel
Convex: interval on real line
 $\sup(D) = 2$, $\inf(D) = -2$, $\max(D) = 2$.

- (b) Let $A = \{(x, y) \in \mathbb{R}^2 : xy \geq 0\}$. Sketch the set, and say whether it is (i) open, closed or neither; (ii) bounded; (iii) compact; (iv) convex.



- Closed: Define $f(x, y) = xy$ which is continuous on $\mathbb{R}^2$ and notice that $[0, \infty)$ is a closed set (since its complement $(-\infty, 0)$ is clearly open since we can take $\varepsilon = x + 1$ such that $B_\varepsilon(x) \subset (-\infty, 0)$ for any $x \in (-\infty, 0)$). Then since $A = f^{-1}([0, \infty))$, i.e. A is the preimage of a closed set, it is also closed.
Not Open: For any $r > 0$, $B_r(x, y)$ will always contain a point outside A when $x = y = 0$.
- Not Bounded: $\sim \exists r > 0$ s.t. $A \subset B_r(\mathbf{0})$ since geometrically we observe that $B_r(\mathbf{0})$ will always contain points from A^C .
- Not Compact: since not bounded
- Not Convex: Consider $\lambda = \frac{1}{2}$

$$\lambda \begin{pmatrix} 5 \\ 1 \end{pmatrix} + (1 - \lambda) \begin{pmatrix} -1 \\ -5 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} \notin A$$

Proposition 3.21. If $\{A_i \subset \mathbb{R}^n : i \in I\}$ is a collection of open sets, $\cup_{i \in I} A_i$ is an open set. Analogously, if $\{A_i \in \mathbb{R}^n : i = 1, \dots, m\}$ is a finite collection of open sets, then $\cap_{i=1}^m A_i$ is an open set

Proof. First part: Consider any $\mathbf{x} \in \cup_{i \in I} A_i$. Then, there is at least one t such that $\mathbf{x} \in A_t$. Since A_t is an open set, there exists an $r > 0$ s.t. $B_r(\mathbf{x}) \subset A_t$. Since $A_t \subset \cup_{i \in I} A_i$, $B_r(\mathbf{x}) \subset \cup_{i \in I} A_i$.

Second part: Consider any $\mathbf{x} \in \cap_{i=1}^m A_i$. Then, since each set is open, $B_{r_i}(\mathbf{x}) \subset A_i$. Choose $r = \min\{r_1, \dots, r_m\} > 0$. Then, $B_r(\mathbf{x}) \subset \cap_{i=1}^m A_i$ ■

3.1.4 Continuity and Limits

Definition 3.22 (Functions in $\mathbb{R}^n$). Given $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}$, a real-valued function $f : S \rightarrow T$ is a set of ordered pairs $S \times T$ such that for each $s \in S$, there exists a unique $t \in T$ s.t. $(s, t) \in f$. In other words, if $(s, t) \in f$ and $(s, t') \in f$, then $t = t'$.

Given $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$, a vector-valued function $f : S \rightarrow T$ is a set of ordered pairs $S \times T$ such that for each $s \in S$, there exists a unique $t \in T$ s.t. $(s, t) \in f$. In other words, if $(s, t) \in f$ and $(s, t') \in f$, then $t = t'$. We may write

$$f(\mathbf{x}) = \begin{pmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{pmatrix} = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))^T$$

Definition 3.23 (Limits of Functions in $\mathbb{R}^n$). Let $f : S \rightarrow T$ where $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$ and $\mathbf{x}_0 \in S$. Then f has a limit $\mathbf{y}_0$ at $\mathbf{x}_0$ iff for any $\varepsilon > 0$, $\exists \delta > 0$ such that

$$\|\mathbf{x} - \mathbf{x}_0\| < \delta \implies \|f(\mathbf{x}) - \mathbf{y}_0\| < \varepsilon$$

This is expressed as

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = \mathbf{y}_0$$

We can express this in terms of open balls

Proposition 3.24. Let $f : S \rightarrow T$ where $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$ and $\mathbf{x}_0 \in S$. Then, f has a limit $\mathbf{y}_0$ iff for all $\varepsilon > 0$, $\exists \delta > 0$

$$f(B_\delta(\mathbf{x}_0)) \subset B_\varepsilon(\mathbf{y}_0)$$

Proposition 3.25. Let $f : S \rightarrow T$ where $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$ and $\mathbf{x}_0 \in S$. Then, f has a limit $\mathbf{y}_0$ at $\mathbf{x}_0$ iff for any sequence $\{\mathbf{x}_k\}_{k=1}^\infty \subseteq S$ converging to $\mathbf{x}_0$, the sequence $\{f(\mathbf{x}_k)\}_{k=1}^\infty$ converges to $\mathbf{y}_0$

The usefulness of this proposition is that it shows us that limits inherit the properties of sequences. It also helps us show counterexamples to the existence of a limit by showing that there is at least one such sequence that fails the criteria

Example 3.26. Show that the limit $\lim_{\mathbf{x} \rightarrow (0,0)} f(\mathbf{x})$ does not exist for

$$f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0) \end{cases}$$

Here, we want to show that there are two sequences that give different values. First, consider $\{(x_1, x_2)\}_k^\infty = (0, \frac{1}{k})$. This sequence converges to $(0, 0)$ and $f(0, \frac{1}{k})$ also converges to 0 as $k \rightarrow \infty$ (or rather, is constant 0). But then, consider $\{(x_1, x_2)\}_k^\infty = (\frac{1}{k}, \frac{1}{k})$. Clearly, it also converges to $(0, 0)$. But then, $f(1/k, 1/k) = \frac{1/k^2}{2/k^2} = \frac{1}{2}$ for all k . So it does not converge to 0 as the sequence converges to 0. So clearly, two sequences converge to different values, and hence it does not have a limit 0 at $(0, 0)$

Proposition 3.27 (Properties of Limits). Let $f : S \rightarrow T$ where $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$ and $\mathbf{x}_0 \in S$. Then, consider $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})$. We have the following properties

- Suppose the limit exists i.e. $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = \mathbf{b}$. Then, the limit is unique
- Given $f = (f_1, f_2, \dots, f_m)^T$, then

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = \mathbf{A} \iff \lim_{\mathbf{x} \rightarrow \mathbf{a}} f_i(\mathbf{x}) = A_i, \forall i \in 1, \dots, m$$

We also have the following algebra of limits (but only for real-valued functions). Let $f, g : S \rightarrow T$ where $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}$. Suppose $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = A$ and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x}) = B$. Then

- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} [f(\mathbf{x}) + g(\mathbf{x})] = A + B$
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} [f(\mathbf{x})g(\mathbf{x})] = AB$
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} [f(\mathbf{x})/g(\mathbf{x})] = A/B$ if $B \neq 0$

Proposition 3.28 (Limits of Compositions of functions). Let $f : S \rightarrow T$ and $g : T \rightarrow U$ with $S \subseteq \mathbb{R}^n, T \subseteq \mathbb{R}^m$ and $U \subseteq \mathbb{R}^k$. Suppose $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = \mathbf{b}$ and $\lim_{\mathbf{y} \rightarrow \mathbf{b}} g(\mathbf{y}) = \mathbf{c}$, then

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} g(f(\mathbf{x})) = \mathbf{c}$$

Definition 3.29 (Continuity of functions in $\mathbb{R}^n$). Let $f : S \rightarrow T$ where $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$. f is continuous at $\mathbf{x}_0$ iff

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0)$$

If f is continuous at every point in its domain S , then it is said to be continuous on S .

Proposition 3.30 (Properties of Continuity). We have the following properties

- Let $f, g : S \rightarrow T$ with $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$. Suppose that f and g are continuous at $\mathbf{x}$. Then we have the fo $f + g, f - g$ are continuous at $\mathbf{x}$.
- Let $f, g : S \rightarrow T$ with $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}$. Suppose that f and g are continuous at $\mathbf{x}$. Then $f \cdot g$ is continuous at $\mathbf{x}$. Moreover, if $g(\mathbf{x}) \neq 0$, then f/g is continuous at $\mathbf{x}$
- Let $f : S \rightarrow T$ with $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$. Then, f is continuous at $\mathbf{x}$ iff each of its component functions $f_i : S \rightarrow U$ with $U \subseteq \mathbb{R}$ is continuous at $\mathbf{x}$
- Let $f : S \rightarrow T$ and $g : T \rightarrow U$ with $S \subseteq \mathbb{R}^n, T \subseteq \mathbb{R}^m$ and $U \subseteq \mathbb{R}^k$. Suppose that f is continuous at $\mathbf{x}$ and g is continuous at $f(\mathbf{x})$. Then the composition $g \circ f : T \rightarrow U$ is continuous at $\mathbf{x}$

Example 3.31. Show, using the (ε, δ) definition of continuity, that $f(x, y) = xy$ is continuous at $(0, 0)$. Consider $\varepsilon > 0, \delta = \sqrt{\varepsilon}$. Then

$$\left\| \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\| = \sqrt{x^2 + y^2} < \delta$$

Then

$$||f(x, y) - f(0, 0)|| = ||xy|| = \sqrt{x^2 y^2} = |xy| = |x||y| < \delta^2 = \varepsilon$$

since $|x| \leq \sqrt{x^2 + y^2} < \delta$ with $|y| < \delta$. Hence, $f(x, y)$ is continuous at $(0, 0)$.

Proposition 3.32. If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous, then if A is an open set in $\mathbb{R}^m$, $f^{-1}(A)$ is open in $\mathbb{R}^n$

Proof. Choose any $x_0 \in f^{-1}(A)$. Then $f(x_0) \in A$. Since A is open by assumption, $\exists \varepsilon > 0$ s.t. $B_\varepsilon(f(x_0)) \subset A$. By continuity, $\exists \delta > 0$ s.t.

$$f(B_\delta(x_0)) \subset B_\varepsilon(f(x_0)) \subset A$$

Hence $f(B_\delta(x_0)) \subset A$. Hence, $\exists \delta > 0$ s.t.

$$B_\delta(x_0) \subset f^{-1}(A)$$

Hence $f^{-1}(A)$ is open in $\mathbb{R}^n$ since x_0 is arbitrarily chosen from $x_0 \in f^{-1}(A)$ ■

Proposition 3.33. If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous, then if A is a closed set in $\mathbb{R}^m$, then $f^{-1}(A)$ is closed in $\mathbb{R}^n$

Proof. The idea of the proof is to show that for any arbitrary sequence $\mathbf{x}_k \rightarrow \mathbf{x}_0$ where $\mathbf{x}_k \in f^{-1}(A)$ for all k , we also have $\mathbf{x}_0 \in f^{-1}(A)$. Since $\mathbf{x}_k \in f^{-1}(A)$, we also have $f(\mathbf{x}_k) \in A$ for all k . By continuity of f , since $\mathbf{x}_k \rightarrow \mathbf{x}_0$, $f(\mathbf{x}_k) \rightarrow f(\mathbf{x}_0)$. Since A is closed by assumption, we have $f(\mathbf{x}_0) \in A$. Hence, we establish $\mathbf{x}_0 \in f^{-1}(A)$ ■

Theorem 3.34 (Weierstrass Theorem). A continuous function $f : S \rightarrow \mathbb{R}$, where $S \subset \mathbb{R}$ is compact, attains a (global) maximum and minimum value on S

Proof. Since $S \subset \mathbb{R}$ is compact, it is closed and bounded. Let $M = \sup_{x \in S} f(x)$. By definition of supremum, there exists a sequence $(x_n) \subset S$ such that $f(x_n) \rightarrow M$. Since S is bounded, by Bolzano-Weierstrass there exists a convergent subsequence $x_{n_k} \rightarrow x^* \in S$ (and $x^* \in S$ since S is closed). By continuity of f , $f(x_{n_k}) \rightarrow f(x^*)$, so $f(x^*) = M$. The same argument applies to the infimum. ■

Remark. The Weierstrass Theorem is an extreme value theorem as it says that the extremums are achieved. Useful to show that optimal points exists under the two main assumptions of compactness and continuity

Theorem 3.35 (Intermediate Value Theorem). Suppose $f[a, b] \rightarrow \mathbb{R}$ is continuous and $f(a) < y < f(b)$. Then, $\exists c \in (a, b)$ such that $f(c) = y$

Proof. Let $f(a) < y < f(b)$ and define $S = \{x \in [a, b] : f(x) \leq y\}$. Since $a \in S$, S is non-empty, and S is bounded above by b , so $c = \sup S$ exists. Since f is continuous and $f(x_n) \leq y$ for any sequence $x_n \rightarrow c$ with $x_n \in S$, we have $f(c) \leq y$. If $f(c) < y$, then by continuity there exists $\delta > 0$ such that $f(x) < y$ for all $x \in (c, c + \delta)$, contradicting $c = \sup S$. Hence $f(c) = y$, and $c \in (a, b)$ since $f(a) < y$ and $f(b) > y$. ■

Remark. Useful to prove the fixed-point theorems

Theorem 3.36 (Brouwer's Fixed Point Theorem). Suppose $f : [a, b] \rightarrow [a, b]$ is a continuous function on a closed interval in $\mathbb{R}$. Then f has a fixed point: that is $\exists x_0 \in [a, b]$ such that $f(x_0) = x_0$.

Suppose $S \subset \mathbb{R}^n$ is a non-empty, compact and convex set and $f : S \rightarrow S$ is a continuous function. Then f has a fixed point: that is $\exists \mathbf{x}_0$ such that $f(\mathbf{x}_0) = \mathbf{x}_0$

Proof. Define $g : [a, b] \rightarrow \mathbb{R}$ by $g(x) = f(x) - x$. Since $f : [a, b] \rightarrow [a, b]$, we have $f(a) \geq a$ and $f(b) \leq b$, so $g(a) = f(a) - a \geq 0$ and $g(b) = f(b) - b \leq 0$. If $g(a) = 0$ or $g(b) = 0$ then $x_0 = a$ or $x_0 = b$ is a fixed point. Otherwise $g(a) > 0 > g(b)$, and since g is continuous, by the Intermediate Value Theorem there exists $x_0 \in (a, b)$ such that $g(x_0) = 0$, i.e. $f(x_0) = x_0$.

For the general case, suppose $S \subset \mathbb{R}^n$ is non-empty, compact and convex and $f : S \rightarrow S$ is continuous. Define $g : S \rightarrow \mathbb{R}^n$ by $g(\mathbf{x}) = f(\mathbf{x}) - \mathbf{x}$. Suppose for contradiction that $f(\mathbf{x}) \neq \mathbf{x}$ for all $\mathbf{x} \in S$, so $g(\mathbf{x}) \neq \mathbf{0}$ for all $\mathbf{x} \in S$. Then one can construct a continuous retraction from S onto its boundary ∂S by mapping each $\mathbf{x}$ to the intersection of the ray from $f(\mathbf{x})$ through $\mathbf{x}$ with ∂S . However, no such retraction exists for a compact convex set, a contradiction. Hence there exists $\mathbf{x}_0 \in S$ such that $f(\mathbf{x}_0) = \mathbf{x}_0$. ■

3.2 Differentiation in $\mathbb{R}^n$

Definition 3.37 (Partial Derivative). Let $f : S \rightarrow T$ where $S \subset \mathbb{R}^n, T \subset \mathbb{R}$ be a real-valued function of n variables. The k -th partial derivative of f at $\mathbf{x} \in S$ is

$$\frac{\partial f}{\partial x_k}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_k + h, \dots, x_n) - f(x_1, \dots, x_k, \dots, x_n)}{h}$$

If the limit does not exist, then there is no k -th partial derivative.

Definition 3.38 (Gradient and Jacobian). Let $f : S \rightarrow T$ where $S \subset \mathbb{R}^n, T \subset \mathbb{R}$ be a real-valued function of n variables. The gradient of f at $\mathbf{x} \in S$ is the column vector of first derivatives evaluated at $\mathbf{x}$, denoted

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \frac{\partial f}{\partial x_2}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{pmatrix}$$

Let $f : S \rightarrow T$ where $S \subset \mathbb{R}^n, T \subset \mathbb{R}^m$ i.e. $f = (f_1, \dots, f_m)$ The Jacobian matrix is the $m \times n$ matrix of first derivatives for each function

$$Df(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \dots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

Remark. $Df(\mathbf{x})$ points in the direction in which the rate of change of f at $\mathbf{x}$ is greatest and is perpendicular to the tangent plane of the level curve through a given point $\mathbf{x}$

Definition 3.39 (Differentiability in $\mathbb{R}^n$). Let $f : S \rightarrow T$ where $S \subset \mathbb{R}^n, T \subset \mathbb{R}^m$ be a vector-valued function. The function f is differentiable at $\mathbf{x}$ if there exists an $m \times n$ matrix $\mathbf{A}$ such that

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = 0$$

Note that the matrix $\mathbf{A}$ is the total derivative of f at $\mathbf{x}$.

Theorem 3.40. Let $f : S \rightarrow T$ where $S \subset \mathbb{R}^n, T \subset \mathbb{R}^m$. If f is differentiable at $\mathbf{x}$, then it is continuous at $\mathbf{x}$, all its partial derivatives exist and $\mathbf{A} = Df(\mathbf{x})$, the Jacobian matrix of f

Theorem 3.41 (Sufficient Condition for Differentiability). Let $f : S \rightarrow T$ where $S \subset \mathbb{R}^n, T \subset \mathbb{R}^m$. If all the partial derivatives of f exist in a neighbourhood of $\mathbf{x}$ and are continuous at $\mathbf{x}$, then f is differentiable at $\mathbf{x}$. If this sufficient condition holds on S , f is said to be continuously differentiable, or C^1 , on S

Example 3.42. Consider two functions

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{otherwise} \end{cases}$$

$$g(x, y) = \begin{cases} x^{\frac{1}{2}}y^{\frac{1}{2}} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Use the formal definition to compute the partial derivatives of the functions at $(0, 0)$.
We compute the following

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h - 0}{h} = 1$$

and

$$\frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

and

$$\frac{\partial g}{\partial x} = \lim_{h \rightarrow 0} \frac{g(h, 0) - g(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h}$$

and

$$\frac{\partial g}{\partial y} = \lim_{h \rightarrow 0} \frac{g(0, h) - g(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

- (b) Are the functions of class C^1 ?

Claim: f is not C^1 on $\mathbb{R}^2$ but C^1 on $\mathbb{R}^2 \setminus \{0\}$.

First, consider $(x, y) \in \mathbb{R} \setminus \{0\}$. Then,

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{x^2(x^2 + 3y^2)}{(x^2 + y^2)^2} \\ \frac{\partial f}{\partial y} &= -\frac{2x^3y}{(x^2 + y^2)^2} \end{aligned}$$

Furthermore, each partial derivative is continuous on $\mathbb{R}^2 \setminus \{0\}$ because their limits exists for any $(x, y) \rightarrow (x_0, y_0)$.

However, f is not C^1 on $\mathbb{R}^2$. Consider $\lim_{(x,y) \rightarrow (0,0)} \frac{\partial f}{\partial x}$. If $x = 0$, then

$$\lim_{y \rightarrow 0} \frac{\partial f}{\partial x} = \lim_{y \rightarrow 0} \frac{0 \cdot 3y^2}{y^4} = 0$$

However, if $y = 0$, then

$$\lim_{x \rightarrow 0} \frac{\partial f}{\partial x} = \lim_{x \rightarrow 0} \frac{x^2(x^2 + 0)}{(x^2 + 0)^2} = 1$$

Since limits are unique and the limits along y, x are different, so the limit

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\partial f}{\partial x}$$

does not exist. Hence the partial derivative wrt to x is not continuous at $(0, 0)$ and hence not continuous on $\mathbb{R}^2$. So f is not C^1 on $\mathbb{R}^2$.

Claim: g is not C^1 on $\mathbb{R}^2$. Consider $(x, y) = (0, 1)$. Then,

$$\frac{\partial g}{\partial x}(0, 1) = \lim_{h \rightarrow 0} \frac{g(h, 1) - g(0, 1)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{h}}{h} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{h}} \rightarrow +\infty$$

So $\frac{\partial g}{\partial x}$ does not exist at $(0, 1)$

(c) Are the functions differentiable at $(0, 0)$?

f is differentiable at $\mathbf{x}$ if $\exists \mathbf{A}$ s.t.

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = 0$$

At $(0, 0)$,

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(\mathbf{h}) - f(0, 0) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = \lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(\mathbf{h}) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|}$$

Consider $\mathbf{h} = \begin{pmatrix} t \\ 0 \end{pmatrix}$ and so

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(t, 0) - f(0, 0) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = \lim_{\|\mathbf{h}\| \rightarrow 0} \frac{|t - at|}{|t|} = |1 - a| = 0 \implies a = 1$$

Consider $\mathbf{h} = \begin{pmatrix} 0 \\ t \end{pmatrix}$ and so

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(0, t) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = \lim_{\|\mathbf{h}\| \rightarrow 0} \frac{|0 - bt|}{|t|} = |b| = 0 \implies b = 0$$

Hence $\mathbf{A} = \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Consider $\mathbf{h} = \begin{pmatrix} t \\ t \end{pmatrix}$. Then

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(t, t) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = \lim_{\|\mathbf{h}\| \rightarrow 0} \frac{|\frac{t}{2} - t|}{\sqrt{2}|t|} = \frac{1}{2\sqrt{2}} \neq 0$$

so there does not exist $\mathbf{A}$ so f is not differentiable.

At $(0, 0)$,

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|g(\mathbf{h}) - g(0, 0) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = \lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|g(\mathbf{h}) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|}$$

Consider $\mathbf{h} = \begin{pmatrix} t \\ 0 \end{pmatrix}$,

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|g(t, 0) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = \lim_{\|\mathbf{h}\| \rightarrow 0} \frac{|at|}{|t|} = |a| = 0 \implies a = 0$$

Consider $\mathbf{h} = \begin{pmatrix} 0 \\ t \end{pmatrix}$,

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|g(0, t) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = \lim_{\|\mathbf{h}\| \rightarrow 0} \frac{|bt|}{|t|} = |b| = 0 \implies b = 0$$

Hence $\mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Consider $\mathbf{h} = \begin{pmatrix} t \\ t \end{pmatrix}$, then

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|g(t, t) - \mathbf{A}\mathbf{h}\|}{\|\mathbf{h}\|} = \lim_{\|\mathbf{h}\| \rightarrow 0} \frac{|t|}{\sqrt{2}|t|} = \frac{1}{\sqrt{2}} \neq 0$$

Hence there does not exist $\mathbf{A}$ satisfying the equation and g is not differentiable.

Proposition 3.43. Let $f : S \rightarrow T$ where $S \subset \mathbb{R}^n, T \subset \mathbb{R}^m$. If f is differentiable, it might not be C^1 or continuous differentiable

Theorem 3.44 (Chain Rule for vector-valued functions). If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $\mathbf{a}$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^p$ is differentiable at $f(\mathbf{a})$ then $g \circ f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is differentiable at $\mathbf{a}$ and

$$D(g \circ f)(\mathbf{a}) = Dg(f(\mathbf{a}))Df(\mathbf{a})$$

Corollary 3.45 (Total Differential). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^1 function. If there is an infinitesimal change $d\mathbf{x}$ in $\mathbf{x}$, the corresponding change in f is given by the total differential

$$df = \frac{\partial f}{\partial x_1}dx_1 + \frac{\partial f}{\partial x_2}dx_2 + \cdots + \frac{\partial f}{\partial x_n}dx_n$$

Definition 3.46 (Directional Derivative). Given a C^1 function f , the directional derivative of f in a direction $\mathbf{v}$ is

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{v}) - f(\mathbf{x})}{t} = \frac{d}{dt}f(\mathbf{x} + t\mathbf{v})_{t=0} = Df(\mathbf{x})\mathbf{v}$$

Example 3.47. Let $f(x, y) = x^2 + 2xy, \mathbf{v} = (1, 2)$. Then,

$$\frac{d}{dt}f(\mathbf{x} + t\mathbf{v})_{t=0} = (2x + 2y, 2x) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 2x + 2y + 4x = 6x + 2y$$

Definition 3.48 (Second-Order Partial Derivatives). The 2nd order partial derivatives are

$$\frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

for $i, j = 1, \dots, n$.

We can easily generalise this for k -th order partial derivatives. If all the partial derivatives of order less than or equal to k exist and are continuous on S , then f is said to be k times continuously differentiable or C^k on S .

Theorem 3.49 (Young's Theorem). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^2 function. Then

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}, \text{ for } i, j = 1, \dots, n$$

If the second partial derivatives are not continuous, this need not be true

Example 3.50. Consider

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Then one can check that f is C^1 but $\partial y \partial x f(0, 0) = -1$ but $\partial x \partial y f(0, 0) = 1$ since $\partial x f(0, y) = -y$ while $\partial y f(x, 0) = x$

Definition 3.51 (Hessian). The matrix of second order partial derivatives is called the Hessian matrix i.e.

$$D^2 f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

If f is a C^2 function, the Hessian matrix is symmetric by Young's Theorem.

Theorem 3.52 (Taylor's Theorem in $\mathbb{R}^n$). Suppose $f : U \rightarrow \mathbb{R}$ where U is an open subset of $\mathbb{R}^n$ is a C^2 function. Then, for any $\mathbf{a}, \mathbf{x} \in U$:

$$f(\mathbf{x}) = f(\mathbf{a}) + Df(\mathbf{a})(\mathbf{x} - \mathbf{a}) + R_1(\mathbf{x}, \mathbf{a})$$

$$\text{and } f(\mathbf{x}) = f(\mathbf{a}) + Df(\mathbf{a})(\mathbf{x} - \mathbf{a}) + \frac{1}{2}(\mathbf{x} - \mathbf{a})^T D^2 f(\mathbf{a})(\mathbf{x} - \mathbf{a}) + R_2(\mathbf{x}, \mathbf{a})$$

where

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} \frac{R_1(\mathbf{x}, \mathbf{a})}{\|\mathbf{x} - \mathbf{a}\|} = 0, \quad \lim_{\mathbf{x} \rightarrow \mathbf{a}} \frac{R_2(\mathbf{x}, \mathbf{a})}{\|\mathbf{x} - \mathbf{a}\|^2} = 0$$

Not including the remainder terms give us the first and second order Taylor approximations in $\mathbb{R}^n$

Motivation for Implicit Function Theorem Basically, we want to know how the solution or equilibrium changes when one parameter changes. Sometimes we do not have an explicit functional form of the solution but that it satisfies some equations.

Consider the equation $F(x, y) = 0$. For each value x you could try to solve the equation to obtain a corresponding value of y . If this is possible i.e. there is a unique y for each x , we say that the equation defines y as an implicit function of x .

Example 3.53. These equations are such examples

- $F(x, y) = 3x^2 - 2y$
- $F(x, y) = y + \ln(y + 1) - x$

The derivative of an implicit function $y(x)$ defined by $F(x, y) = 0$ can be obtained by implicit differentiation: differentiate the whole equation wrt x , allowing for the fact that y is a function of x .

Example 3.54. Consider $F(x, y) = y + \ln(y + 1) - x = 0$. Then

$$\frac{dy}{dx} + \frac{1}{y + 1} \frac{dy}{dx} - 1 = 0 \implies \frac{dy}{dx} = \frac{y + 1}{y + 2}$$

Theorem 3.55 (Implicit Function Theorem). Suppose $\mathbf{x} \in \mathbb{R}^m$ and $(\mathbf{x}^*, y^*)$ is a solution to $F(\mathbf{x}, y) = 0$. Suppose that F is a C^1 in an open ball around $(\mathbf{x}^*, y^*)$ with $\frac{\partial F}{\partial y}(\mathbf{x}^*, y^*) \neq 0$. Then, there is a C^1 function $y = y(\mathbf{x})$ defined on an open ball around $\mathbf{x}^*$ such that:

- (i) $y(\mathbf{x}^*) = y^*$
- (ii) $F(\mathbf{x}, y(\mathbf{x})) = 0$
- (iii) $\frac{\partial y}{\partial x_i}(\mathbf{x}, y) = -\frac{\frac{\partial F}{\partial x_i}(\mathbf{x}, y)}{\frac{\partial F}{\partial y}(\mathbf{x}, y)}$

If $F(\mathbf{x}, y) = 0$ has a solution y^* for a particular $\mathbf{x}^*$, then it is legitimate to think of y as a function of $\mathbf{x}$ around that point and you can determine how y changes from y^* when $\mathbf{x}$ changes a little from $\mathbf{x}^*$ by implicitly differentiating.

We want to generalise the IFT to consider a system of n equations.

Theorem 3.56. Consider $\mathbf{F} : \mathbb{R}^{m+n} \rightarrow \mathbb{R}^n$. Suppose that $\mathbf{F}(\mathbf{x}^*, \mathbf{y}^*) = \mathbf{0}$ and that each F_i is C^1 around $(\mathbf{x}^*, \mathbf{y}^*)$, and $D_{\mathbf{y}}\mathbf{F}$ is invertible at this point. Then, there exists a C^1 function $\mathbf{y}(\mathbf{x})$ defined on an open ball around $\mathbf{x}^*$ such that:

- (i) $\mathbf{y}(\mathbf{x}^*) = \mathbf{y}^*$
- (ii) $\mathbf{F}(\mathbf{x}, \mathbf{y}(\mathbf{x})) = \mathbf{0}$
- (iii) $D_{\mathbf{x}}\mathbf{y}(\mathbf{x}) = -[D_{\mathbf{y}}\mathbf{F}(\mathbf{x}, \mathbf{y}(\mathbf{x}))]^{-1}D_{\mathbf{x}}\mathbf{F}(\mathbf{x}, \mathbf{y}(\mathbf{x}))$

Remark. This is useful for quite many questions

Example 3.57. Consider the system of equations:

$$\begin{aligned}x^2 - y^2 - u^3 + v^2 + 4 &= 0 \\ 2xy + y^2 - 2u^2 + 3v^4 + 8 &= 0\end{aligned}$$

and the solution $s^* = (x^*, y^*, u^*, v^*) = (2, -1, 2, 1)$.

- (a) Show that you can express the values of (u, v) that solve the equations as a function of the values of (x, y) , in a neighborhood of $(2, -1)$.

Need to show IFT is satisfied

$$F = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \begin{pmatrix} x^2 - y^2 - u^3 + v^2 + 4 \\ 2xy + y^2 - 2u^2 + 3v^4 + 8 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Then

$$DF = \begin{pmatrix} 2x & -2y & -3u^2 & 2v \\ 2y & 2x + 2y & -4u & 12v^3 \end{pmatrix}$$

and each partial derivative is continuous in a neighbourhood around $(2, -1, 2, 1)$. So F_1 and F_2 are C^1 . Consider

$$D_{[u,v]}F = \begin{pmatrix} -3u^2 & 2v \\ -4u & 12v^3 \end{pmatrix}$$

Hence at $(2, -1, 2, 1)$, the matrix is

$$D_{[u,v]}F(2, -1, 2, 1) = \begin{pmatrix} -12 & 2 \\ -8 & 12 \end{pmatrix}$$

and so $\det(D_{[u,v]}F(2, -1, 2, 1)) = -12 \cdot 12 + 16 = -128 \neq 0$. So $D_{[u,v]}F$ is invertible at $(2, -1, 2, 1)$.

By the IFT there is a function $\varphi(x, y) = \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} g_1(x, y) \\ g_2(x, y) \end{pmatrix}$ such that $F(x, y, \varphi(x, y)) = 0$ in the neighbourhood of $(2, -1)$

- (b) At $(2, -1)$, how do u and v change when x changes?

Consider

$$\begin{aligned}D_{[x,y]}\varphi(x, y) &= -[D_{[u,v]}F]^{-1}D_{[x,y]}F \\ &= -\begin{pmatrix} -3u^2 & 2v \\ -4u & 12v^3 \end{pmatrix}^{-1} \begin{pmatrix} 2x & -2y \\ 2y & 2x + 2y \end{pmatrix} \\ &= -\begin{pmatrix} -12 & 2 \\ -8 & 12 \end{pmatrix}^{-1} \begin{pmatrix} 4 & 2 \\ -2 & 2 \end{pmatrix} \\ &= \frac{1}{128} \begin{pmatrix} 12 & -2 \\ 8 & -12 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ -2 & 2 \end{pmatrix} \\ &= \frac{1}{128} \begin{pmatrix} 52 & 20 \\ 56 & -8 \end{pmatrix}\end{aligned}$$

Hence

$$\frac{\partial u}{\partial x}(2, -1) = \frac{52}{128} = \frac{13}{32}, \quad \frac{\partial v}{\partial x}(2, -1) = \frac{56}{128} = \frac{7}{16}$$

4 Optimisation

The Weierstrass Theorem guarantees the existence of a global maximum and minimum on S attained by a continuous f .

Two types of optima:

- Interior Optima: occurs at an interior point of S
- Constrained Optima: occurs on the boundary of S where the constraint binds

If we have an unconstrained optimisation problem or if the set S is open, we need to only look for interior optima.

4.1 Unconstrained Optimisation

Definition 4.1 (Unconstrained Optimisation). We say that the problem of maximising $f(\mathbf{x})$ on S is an unconstrained optimisation problem just when S is an open set (or $S = \mathbb{R}^n$)

Theorem 4.2 (First-order Condition). If f has a local maximum or minimum at an interior point $\mathbf{x}^*$ of S , then

$$Df(\mathbf{x}^*) = 0$$

Note that this is only a necessary (not sufficient) condition.

Theorem 4.3 (Second Order Sufficient Conditions). Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{x}^*$ satisfies the FOCs. Then,

- If $D^2f(\mathbf{x}^*)$ is negative definite, $\mathbf{x}^*$ is a strict local maximum
- If $D^2f(\mathbf{x}^*)$ is positive definite, $\mathbf{x}^*$ is a strict local minimum
- If $D^2f(\mathbf{x}^*)$ is indefinite, $\mathbf{x}^*$ is a saddle point

We can test this via the definition of definiteness or the eigenvalue test. We also have second order necessary conditions

Theorem 4.4 (Second Order Necessary Conditions). Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Then,

- If f has a local maximum at an interior point $\mathbf{x}^*$ of S , then $D^2f(\mathbf{x}^*)$ is negative semi-definite
- If f has a local minimum at an interior point $\mathbf{x}^*$ of S , then $D^2f(\mathbf{x}^*)$ is positive semi-definite

Cookbook Approach to Find Optima in Unconstrained Problems:

1. Solve the FOCs to find all candidates for interior optima
2. Check the SOC: points where the Hessian is negative/positive definite are local maxima/minima
3. Evaluate the function at these points to find a possible global maximum/minimum

However, this does not tell you (i) what happens at a critical point where the Hessian is negative or positive semi-definite (may or may not be a local max/min) or (ii) whether the interior local optimum where f is the largest or smallest is actually a global optimum – this is because $S = \mathbb{R}^n$ or S is open and thus not compact so the global optimum does not exist.

Example 4.5. Find all the critical points of the function

$$f(x, y) = x^4 + x^2 - 6xy + 3y^2$$

and classify each one (if possible) as a local maximum, local minimum, or saddle point. The first-order conditions are:

$$\begin{aligned}\frac{\partial f}{\partial x} &= 4x^3 + 2x - 6y = 0 \\ \frac{\partial f}{\partial y} &= -6x + 6y = 0\end{aligned}$$

with Hessian $D^2f = \begin{pmatrix} 12x^2 + 2 & -6 \\ -6 & 6 \end{pmatrix}$.

From $\frac{\partial f}{\partial y} = 0 \implies x = y$. Substituting:

$$4x^3 - 4x = 4x(x^2 - 1) = 0 \implies x = \pm 1, x = 0$$

Check $(0, 0)$: $f(0, 0) = 0$ and

$$D^2f(0, 0) = \begin{pmatrix} 2 & -6 \\ -6 & 6 \end{pmatrix}, \quad \det(D^2f(0, 0)) = 12 - 36 = -24 < 0$$

so $\lambda_1, \lambda_2 < 0$, different signs $\therefore$ indefinite, i.e. a saddle point.

$f(1, 1) = f(-1, -1) = -1$ and let

$$A = D^2f(1, 1) = D^2f(-1, -1) = \begin{pmatrix} 14 & -6 \\ -6 & 6 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 14 - \lambda & -6 \\ -6 & 6 - \lambda \end{vmatrix} = (14 - \lambda)(6 - \lambda) - 36 = \lambda^2 - 20\lambda + 48$$

$$\lambda = \frac{20 \pm \sqrt{400 - 4 \cdot 48}}{2} = \frac{20 \pm 4\sqrt{13}}{2} = 10 \pm 2\sqrt{13} > 0$$

$\therefore$ positive definite $\implies$ local minimum. Since the domain is not compact, this does not guarantee a global minimum.

4.2 Constrained Optimisation (Equality/Inequality Constraints)

It is important to note that all of the first-order conditions provided by the Lagrange Theorem and Kuhn-Tucker Theorem are (i) only necessary conditions and (ii) only apply to candidate optima which satisfy the constraint qualification

4.2.1 Equality Constraints

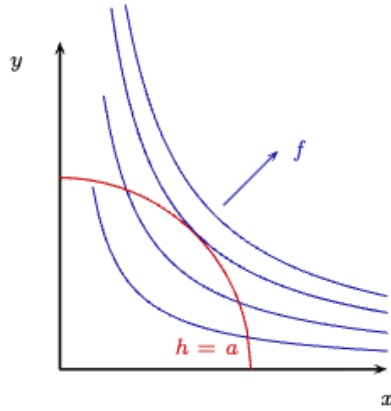
The general form of our problem is to suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a C^2 function and $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbf{R}^m$ is a C^2 (vector-valued) function on $\mathbb{R}^n$. Then, we have the following constrained optimisation problem

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{h}(\mathbf{x}) = \mathbf{a}$$

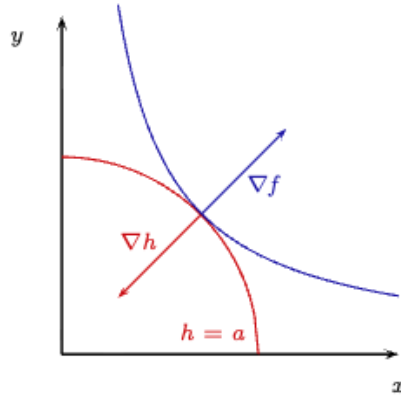
or equivalently

$$\begin{aligned} &\max f(x_1, \dots, x_n) \text{ s.t.} \\ &\begin{cases} h_1(x_1, \dots, x_n) = a_1 \\ \dots \\ h_m(x_1, \dots, x_n) = a_m \end{cases} \end{aligned}$$

Geometric Intuition: Consider the problem of maximising $f(x, y)$ s.t. $h(x, y) = a$.



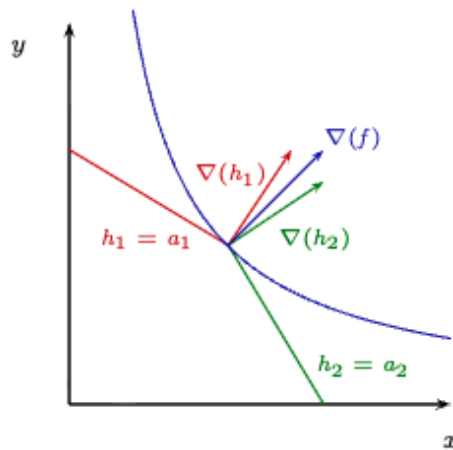
At the optimum, it must not be possible to move in any direction of the constraint set which increases f hence the level curve of f must be tangent to the curve $h = a$ at the optimum. Equivalently, if the level curves are tangent, the normal vectors of the curves must point in the same or opposite directions:



Hence, we have

$$\nabla f = \lambda \nabla h$$

More generally, ∇f must be a linear combination of the gradients of the constraints at the optimal point: $\nabla f = \lambda_1 \nabla h_1 + \cdots + \lambda_m \nabla h_m$. Consider maximising $f(x, y)$ s.t. $h_1(x, y) = a_1$ and $h_2(x, y) = a_2$



Method of the Lagrangian Multipliers

Consider $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a C^2 function and $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbf{R}^m$ is a C^2 (vector-valued) function on $\mathbb{R}^n$ and the problem

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{h}(\mathbf{x}) = \mathbf{a}$$

Then, define the Lagrangian $\mathcal{L} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}$ as

$$\mathcal{L}(\mathbf{x}; \lambda) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (h_j(\mathbf{x} - a_j))$$

Let $D_{\mathbf{x}}\mathcal{L}(\mathbf{x}; \lambda) = Df(\mathbf{x}) - \sum_j \lambda_j Dh_j(\mathbf{x})$.

Theorem 4.6 (Necessary Conditions: Lagrange's Theorem). If $\mathbf{x}^*$ is a local maximum of f subject to the constraint $\mathbf{h}(\mathbf{x}) = \mathbf{a}$ and the matrix $D\mathbf{h}(\mathbf{x}^*)$ is of full rank, m , there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \dots, \lambda_m^*)$ such that

$$D_{\mathbf{x}}\mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*) = \mathbf{0}$$

Remark. The condition that $D\mathbf{h}(\mathbf{x}^*)$ is of full rank is called the constraint qualification

Proposition 4.7. Consider the Lagrangian $\mathcal{L} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}$ as

$$\mathcal{L}(\mathbf{x}; \lambda) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (h_j(\mathbf{x} - a_j))$$

Then, λ_j measures the sensitivity of the value of the objective function at $\mathbf{x}^*$ to a relaxation of the constraint $h_j(\mathbf{x}) = a_j$. That is

$$\lambda_j = \frac{\partial f(\mathbf{x}^*(\mathbf{a}))}{\partial a_j}$$

Let

$$D_{\mathbf{x}}^2\mathcal{L}(\mathbf{x}; \lambda) = D^2f(\mathbf{x}) - \sum_j \lambda_j D^2h_j(\mathbf{x})$$

Theorem 4.8 (Second Order Sufficient Conditions for Lagrange's Method). Suppose $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ satisfies the constraint qualification and the first order conditions. Then:

- (a) if $\mathbf{v}^T D_{\mathbf{x}}^2\mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*)\mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ for which $D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = \mathbf{0}$ then $\mathbf{x}^*$ is a strict local maximum
- (b) if $\mathbf{v}^T D_{\mathbf{x}}^2\mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*)\mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ for which $D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = \mathbf{0}$ then $\mathbf{x}^*$ is a strict local minimum

An equivalent way to test the SOC is the Bordered Hessian test. The bordered matrix $\mathbf{C}$ is the $(m+n) \times (m+n)$ square matrix (at least the ones we consider) is formed by bordering the matrix $\mathbf{A}$ with $\mathbf{B}$ such that

$$\mathbf{C} = \begin{pmatrix} \mathbf{0} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{A} \end{pmatrix}$$

and let $|\mathbf{C}_r|$ denote the leading principal minor that ends with a_{rr} so that each $|\mathbf{C}_r|$ is the determinant of the square matrix of order $m+r$.

Definition 4.9 (Bordered Hessian). Consider $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a C^2 function and $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbf{R}^m$ is a C^2 (vector-valued) function on $\mathbb{R}^n$ and the problem

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{h}(\mathbf{x}) = \mathbf{a}$$

Then, define the Lagrangian $\mathcal{L} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}$ as

$$\mathcal{L}(\mathbf{x}; \lambda) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (h_j(\mathbf{x} - a_j))$$

The bordered Hessian is

$$\begin{pmatrix} \mathbf{0} & D\mathbf{h}(\mathbf{x}^*) \\ D\mathbf{h}(\mathbf{x}^*)^T & D_{\mathbf{x}}^2\mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*) \end{pmatrix}$$

In general there will be n variables and m constraints so there will be $n - m$ degrees of freedoms (directions to move) and hence $n - m$ conditions to check

Theorem 4.10 (Bordered Hessian Test). Consider the Bordered Hessian

$$\mathbf{C} = \begin{pmatrix} \mathbf{0} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{A} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & D\mathbf{h}(\mathbf{x}^*) \\ D\mathbf{h}(\mathbf{x}^*)^T & \mathbf{D}_{\mathbf{x}}^2 \mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*) \end{pmatrix}$$

Then

1. $\mathbf{v}^T \mathbf{A} \mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ where $\mathbf{B} \mathbf{v} = \mathbf{0}$ iff $(-1)^m |\mathbf{C}_r| > 0$ for $r = m + 1, \dots, n$, the last $n - m$ leading principal minors have the same sign, namely $(-1)^m$
2. $\mathbf{v}^T \mathbf{A} \mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ where $\mathbf{B} \mathbf{v} = \mathbf{0}$ iff $(-1)^r |\mathbf{C}_r| > 0$ for $r = m + 1, \dots, n$, the last $n - m$ leading principal minors alternate in sign ending with $(-1)^n$

When the Lagrangian Method fails: We must take seriously that the constraint qualification is satisfied at the critical points – otherwise, the approach fails.

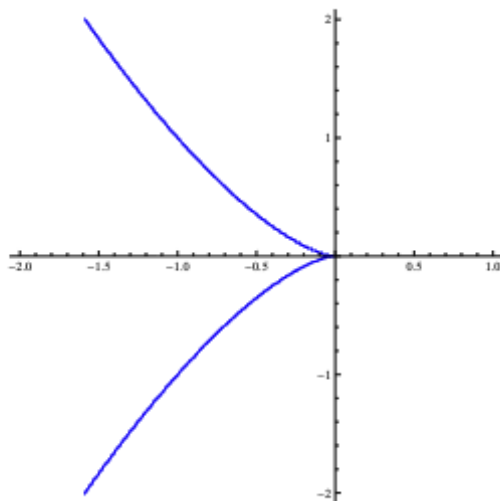
Example 4.11. Consider

$$\max x \text{ s.t. } y^2 + x^3 = 0$$

The optimum is clearly $(x, y) = (0, 0)$. But then, $Df = (1, 0)$ and $Dh = (3x^2, 2y)$. At the critical point $(0, 0)$,

$$Df(0, 0) \neq \lambda Dh(0, 0), \forall \lambda \in \mathbb{R}$$

This is because $Dh(0, 0) = (0, 0)$ so $\text{rank}(Dh(0, 0)) = 0 \neq 1$ and the constraint qualification fails. Geometrically,



Notice that the critical point is precisely at the boundary of the constrained area (feasible region), and so the geometric idea of the tangency area fails at the boundary of the feasible region

Cookbook Approach to Solving equality-constrained maximisation problems

1. Start by considering whether the required global maximum exists (guaranteed by Weierstrauss Theorem if region is compact)
2. Write down the matrix $D\mathbf{h}(\mathbf{x})$ and check whether there are any points where it does not have full rank – if so, these are candidates for maximum points

3. Write down the Lagrangian and obtain the first order conditions. Solve to obtain the other candidate maxima
4. If you already know that the global maximum exists, it is one of the candidates you have found. If you evaluate f at each of these points, you can identify it immediately
5. Otherwise, write down the bordered Hessian and check the SOC for each point satisfying the first order conditions. Then reconsider the global maximum.

Remark. For minimisation problems, just maximise $-f$ instead.

4.2.2 Inequality Constraints

Simple Inequality Constraints

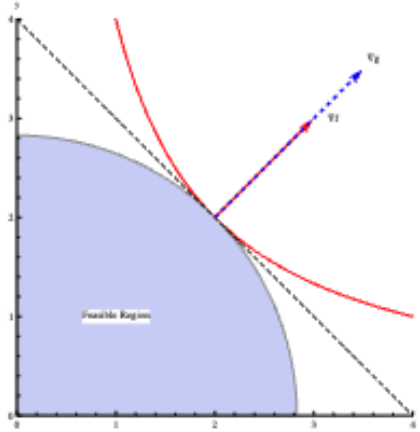
Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are C^2 functions. Consider the problem

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \leq \mathbf{b}$$

or equivalently,

$$\begin{aligned} &\max f(x_1, \dots, x_n) \text{ s.t.} \\ &\begin{cases} g_1(x_1, \dots, x_n) \leq b_1 \\ \dots \\ g_m(x_1, \dots, x_n) \leq b_m \end{cases} \end{aligned}$$

The geometric intuition is similar to equality constraints except that the gradient vector ∇f must point outside the region $\mathbf{g}(\mathbf{x}) \leq \mathbf{b}$. So, consider maximising $f(x, y)$ subject to $g(x, y) \leq b$ which would be



If a constraint is binding at $\mathbf{x}^*$, one must have $\nabla f = \lambda \nabla g$ with $\lambda \geq 0$. However, we can have interior optimum $\mathbf{x}^*$ in which case $\nabla f = 0$.

Kuhn Tucker Method:

We define the Lagrangian before

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (g_j(\mathbf{x}) - b_j)$$

Theorem 4.12 (Kuhn-Tucker Theorem). Suppose $\mathbf{x}^*$ is a local maximum of f subject to the constraints $\mathbf{g}(\mathbf{x}) \leq \mathbf{b}$ and the constraint qualification is satisfied at $\mathbf{x}^*$. Then, there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \dots, \lambda_m^*)$ such that

$$\begin{aligned} D_{\mathbf{x}}(\mathbf{x}^*; \boldsymbol{\lambda}^*) &= 0 \\ \lambda_j^* &\geq 0, g_j(\mathbf{x}^*) - b_j \leq 0 \text{ and } \lambda_j^* (g_j(\mathbf{x}^*) - b_j) = 0 \text{ for } j = 1, \dots, m \end{aligned}$$

The constraint qualification for $\mathbf{x}^*$ is as follows: Let $\mathbf{g}_E$ be the constraints which bind at $\mathbf{x}^*$ and e the number of binding constraints. The constraint qualification is satisfied if the matrix $D\mathbf{g}_E(\mathbf{x}^*)$ is of full rank e

Remark. I included the primal feasibility constraints (i.e. the inequality constraints) because we need to consider them explicitly too – the theorem embeds them in the set up

The conditions

$$\lambda_j^* \geq 0 \text{ and } \lambda_j^*(g_j(\mathbf{x}^*) - b_j) = 0 \text{ for } j = 1, \dots, m$$

are called the complementary slackness conditions.

1. λ_j is nonnegative for all j as the agent only needs to be penalised if he wishes $g_j(\mathbf{x})$ to exceed b_j
2. Note that if $g_j(\mathbf{x}) < b_j$, then $\lambda_j = 0$. So λ_j is only positive if the j -th constraint is satisfied with equality

These FOCs capture interior optima i.e. $\lambda_j^* = 0$ for all $j = 1, \dots, m$ so $D_{\mathbf{x}}\mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*) = Df(\mathbf{x}^*) = 0$ and constrained optima.

Essentially, this method boils down to checking cases where (when there are 2 constraints): (i) both constraints bind, (ii) one constraint binds and (iii) no constraint binds.

The Kuhn-Tucker Theorem means that (provided the constraint qualification is satisfied) you can look for a solution to the optimisation problem by looking for a solution to the following $n + m$ FOCs:

$$\begin{aligned} D_{\mathbf{x}}(\mathbf{x}^*; \boldsymbol{\lambda}^*) &= 0 \\ \lambda_j^* &\geq 0 \text{ and } \lambda_j^*(g_j(\mathbf{x}^*) - b_j) = 0 \text{ for } j = 1, \dots, m \end{aligned}$$

Proposition 4.13 (Kuhn-Tucker multipliers). λ_j^* measures the sensitivity of the value of the objective function at $\mathbf{x}^*$ to a relaxation of the constraint $g_j(\mathbf{x}^*) \leq b_j$. If the j -th constraint does not bind, then $\lambda_j^* = 0$

Theorem 4.14 (Kuhn-Tucker Second Order Conditions). Suppose $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ satisfy the constraint qualification and the first order conditions. Then

1. If some of the constraints, $\mathbf{g}_E$, bind at $\mathbf{x}^*$ and $\mathbf{v}^T D_{\mathbf{x}}^2 \mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*) \mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ for which $D\mathbf{g}_E(\mathbf{x}^*)\mathbf{v} = 0$ then $\mathbf{x}^*$ is a strict local maximum
2. If no constraints bind at $\mathbf{x}^*$ and $D^2 f(\mathbf{x}^*)$ is negative definite, then $\mathbf{x}^*$ is a strict local maximum

The first case is equivalent to checking the bordered Hessian

$$H = \begin{pmatrix} \mathbf{0} & D\mathbf{g}_E(\mathbf{x}^*) \\ D\mathbf{g}_E(\mathbf{x}^*)^T & D_{\mathbf{x}}^2 \mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*) \end{pmatrix}$$

and checking if the sign of the last $n - e$ leading principal minors alternate, ending with $(-1)^n$.

Remark. For minimisation problems

$$\text{minimise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}^*) \geq \mathbf{b}$$

Define the Lagrangian as before

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (g_j(\mathbf{x}) - b_j)$$

Then the Kuhn-Tucker Theorem holds exactly as before and the FOCs and the constraint qualifications are identical. The SOC for a constrained strict local minimum is $\mathbf{v}^T D_{\mathbf{x}}^2 \mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*) \mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ for which $D\mathbf{g}_E(\mathbf{x}^*)\mathbf{v} = 0$

Inequality Constraints with Positivity Constraints

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are C^2 functions. Consider the problem

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}$$

We call the second set of constraints positivity constraints and we can construct the Lagrangian analogously

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}; \boldsymbol{\mu}) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (g_j(\mathbf{x}) - b_j) + \sum_{i=1}^n \mu_i x_i$$

Note that the constraint qualification on the positivity constraints never fail since linear constraints never fail the constraint qualification.

We have two ways of solving the problem:

EITHER

We use the standard Kuhn-Tucker Theorem and just include the positivity constraints in the Lagrangian

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}; \boldsymbol{\mu}) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (g_j(\mathbf{x}) - b_j) + \sum_{i=1}^n \mu_i x_i$$

with FOCS:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_i}(\mathbf{x}^*; \boldsymbol{\lambda}^*; \boldsymbol{\mu}^*) &= \frac{\partial f}{\partial x_i}(\mathbf{x}) - \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i}(\mathbf{x}) + \mu_i = 0, \text{ for } i = 1, \dots, n \\ \lambda_j &\geq 0, g_j(\mathbf{x}^*) \leq b_j \text{ and } \lambda_j (g_j(\mathbf{x}^*) - b_j) = 0 \text{ for } j = 1, \dots, m \\ \mu_i &\geq 0, x_i \geq 0 \text{ and } \mu_i x_i = 0 \text{ for } i = 1, \dots, n \end{aligned}$$

We can rearrange the first and third conditions to get rid of μ

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_i}(\mathbf{x}^*; \boldsymbol{\lambda}^*; \boldsymbol{\mu}^*) &= \frac{\partial f}{\partial x_i}(\mathbf{x}) - \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i}(\mathbf{x}) \leq 0, \text{ for } i = 1, \dots, n \\ \lambda_j &\geq 0, g_j(\mathbf{x}^*) \leq b_j \text{ and } \lambda_j (g_j(\mathbf{x}^*) - b_j) = 0 \text{ for } j = 1, \dots, m \\ x_i &\geq 0 \text{ and } \left(\frac{\partial f}{\partial x_i}(\mathbf{x}) - \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i}(\mathbf{x}) \right) x_i = 0 \text{ for } i = 1, \dots, n \end{aligned}$$

OR

We use the Kuhn-Tucker reformulation. Consider the modified Lagrangian

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}; \boldsymbol{\mu}) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (g_j(\mathbf{x}) - b_j)$$

where we exclude the positivity constraints.

Theorem 4.15 (Kuhn-Tucker Reformulation for Positivity Constraints). Suppose $\mathbf{x}^*$ is a local maximum of f subject to the constraints $\mathbf{g}(\mathbf{x}) \leq \mathbf{b}$ and $\mathbf{x} \geq \mathbf{0}$. Suppose that the constraint qualification is satisfied at $\mathbf{x}^*$. Then, there exists $\boldsymbol{\lambda}^*$ such that

$$\begin{aligned} D_{x_i} \mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*) &\leq 0 \text{ and } x_i^* D_{x_i} \mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*) = 0 \text{ for } i = 1, \dots, n \\ \lambda_j^* &\geq 0 \text{ and } \lambda_j^* (g_j(\mathbf{x}^*) - b_j) = 0 \text{ for } j = 1, \dots, m \end{aligned}$$

The constraint qualification for $\mathbf{x}^*$ is as follows: Let $\mathbf{g}_E$ be the constraints which bind at $\mathbf{x}^*$ and e the number of binding constraints. The constraint qualification is satisfied if the matrix $D\mathbf{g}_E(\mathbf{x}^*)$ is of full rank e

Two ways the Kuhn Tucker Theorem can fail:

The first way is that the constraint qualification fails for the optimum.

Example 4.16. Consider the following problem

$$\max f(x, y) = -(x^2 + y^2) \text{ s.t. } h(x, y) = (x - 1)^3 - y^2 \geq 0$$

Then, notice that the constraint qualification fails at $(1, 0)$ (and binds there) since

$$Dh(x, y) = (3(x - 1)^2 \quad -2y)$$

which implies $Dh(1, 0) = (0, 0)$ which is rank $0 < 1$. But $(1, 0)$ is the global maximum solving the problem. Notice that there are indeed no solutions to the equations that define the critical points of $\mathcal{L}$

$$\begin{aligned} -2x + 3\lambda(x - 1)^2 &= 0 \\ -2y - 2\lambda y &= 0 \\ (x - 1)^3 - y^2 \geq 0, \lambda \geq 0, \lambda((x - 1)^3 - y^2) &= 0 \end{aligned}$$

The moral of the story is that you have to check all points for which the CQ fails and compare the output. The second way is that the constraint qualification is satisfied everywhere but the optimum does not exist (notice that the Kuhn-Tucker theorem says nothing about the maximum existing).

Example 4.17. Consider the problem

$$\max f(x) = x^2 - x \text{ s.t. } g(x) = x \geq 0$$

Notice that $g'(x) = 1$ everywhere and so the CQ cannot fail. Define $\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$. The critical points of $\mathcal{L}$ are the solutions (x, λ) to

$$\begin{aligned} 2x - 1 + \lambda &= 0 \\ x \geq 0, \lambda \geq 0, \lambda x &= 0 \end{aligned}$$

This system admits two solutions $(x, \lambda) = (0, 1)$ and $(x, \lambda) = (\frac{1}{2}, 0)$. However, neither point is a solution to the given maximisation problem since at $x = 2$ we have $f(2) = 3$ but $f(0) = 0$ and $f(\frac{1}{2}) = -\frac{1}{4}$. In fact, the maximum is not guaranteed to obtain (and indeed does not) because the set is not compact and $f(x) \uparrow \infty$ as $x \uparrow \infty$.

Remark. Only if the constraint set is compact, then would a maximum/minimum exist (i.e. by the Weierstrass Theorem)

Mixed Equality and Inequality Constraints

Our general problem is: Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^k$. Define the Lagrangian as

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}; \boldsymbol{\mu}) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (h_j(\mathbf{x}) - a_j) - \sum_{j=1}^k \mu_j (g_j(\mathbf{x}) - b_j)$$

Theorem 4.18 (First-Order Conditions for Mixed Constraints). If $\mathbf{x}^*$ is a local maximum of f on S and the constraint qualification holds, then $\mathbf{x}^*$ is a solution to

1. the first order conditions

$$\frac{\partial \mathcal{L}}{\partial x_i}(\mathbf{x}^*; \boldsymbol{\lambda}^*; \boldsymbol{\mu}^*) = 0, \text{ for } i = 1, \dots, n$$

2. the equality constraints

$$h_j(\mathbf{x}^*) = a_j \text{ for } j = 1, \dots, m$$

3. the complementary slackness conditions for the inequality constraints

$$\mu_j^* \geq 0, g_j(\mathbf{x}^*) \leq b_j \text{ and } \mu_j^* (g_j(\mathbf{x}^*) - b_j) = 0 \text{ for } j = 1, \dots, k$$

The constraint qualification is to require that the matrix of binding constraints

$$\begin{pmatrix} D\mathbf{h}(\mathbf{x}^*) \\ D\mathbf{g}_E(\mathbf{x}^*) \end{pmatrix}$$

has full rank $m + e$ where the matrix is $(m + e) \times n$ i.e. $m + e$ rows and n columns.

Theorem 4.19 (Second Order Sufficient Conditions). Suppose $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ satisfy the constraint qualification and the first order condition. If $\mathbf{v}^T D_{\mathbf{x}}^2 \mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*; \boldsymbol{\mu}^*) \mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbf{R}^n$ for which $D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = 0$ and $D\mathbf{g}_E(\mathbf{x}^*)\mathbf{v} = 0$ then $\mathbf{x}^*$ is a strict local maximum.

The bordered Hessian is

$$H = \begin{pmatrix} \mathbf{0} & \mathbf{0} & D\mathbf{h}(\mathbf{x}^*) \\ \mathbf{0} & \mathbf{0} & D\mathbf{g}_E(\mathbf{x}^*) \\ D\mathbf{h}(\mathbf{x}^*)^T & D\mathbf{g}_E(\mathbf{x}^*)^T & D_{\mathbf{x}}^2 \mathcal{L}(\mathbf{x}^*; \boldsymbol{\lambda}^*; \boldsymbol{\mu}^*) \end{pmatrix}$$

and we can check that the signs of the determinants of the last $n - (m + e)$ principal minors alternate, ending with $(-1)^n$. We can construct analogous FOC and SOC (i.e. the quadratic form being positive) for strict local minima.

Worked Examples

Example 4.20 (Equality Constraints). Use the Lagrangian Method to find the maximum and minimum values of

$$x^2 + y^2 \text{ subject to } x^2 + xy + y^2 = 3$$

checking the constraint qualification and the second-order conditions.

Let $f(x, y) = x^2 + y^2$ and $h(x, y) = x^2 + xy + y^2$. First, we check where the CQ fails

$$Dh(x, y) = (2x + y \quad 2y + x)$$

so the CQ fails only at $(0, 0)$ where $f(0, 0) = 0$. However, $(0, 0)$ does not lie on the constraint region so we can ignore it. Now, define the Lagrangian

$$\mathcal{L}(x, y, \lambda) = x^2 + y^2 - \lambda(x^2 + xy + y^2 - 3)$$

Our FOCs are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= 2x - 2\lambda x - \lambda y = (2 - 2\lambda)x - \lambda y = 0 \\ \frac{\partial \mathcal{L}}{\partial y} &= 2y - 2\lambda y - \lambda x = (2 - 2\lambda)y - \lambda x = 0 \end{aligned}$$

Notice we can write this as

$$\begin{pmatrix} 2 - 2\lambda & -\lambda \\ -\lambda & 2 - 2\lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Since $h(x, y) = 3$, then $(x, y) \neq (0, 0)$ so the null space of

$$\begin{pmatrix} 2 - 2\lambda & -\lambda \\ -\lambda & 2 - 2\lambda \end{pmatrix}$$

must contain non-zero vectors and hence is not invertible. In other words, the determinant of the matrix is 0 and we can solve

$$\begin{vmatrix} 2 - 2\lambda & -\lambda \\ -\lambda & 2 - 2\lambda \end{vmatrix} = (2 - 2\lambda)(2 - \lambda) - \lambda^2 = (2 - 3\lambda)(2 - \lambda) = 0$$

which is solved by $\lambda = 2, \frac{2}{3}$. Then,

- Suppose $\lambda = \frac{2}{3}$. Solving the FOCs give us $(x, y) = (1, 1)$ or $(x, y) = (-1, -1)$ which yield $f(1, 1) = f(-1, -1) = 2$.

- Suppose $\lambda = 2$. Solving the FOCs give us $(x, y) = (\sqrt{3}, \sqrt{3})$ or $(-\sqrt{3}, -\sqrt{3})$. This yields $f(\sqrt{3}, \sqrt{3}) = f(-\sqrt{3}, -\sqrt{3}) = 6$

To check the SOC, we have two ways.

Way #1: Bordered Hessian

$$B\mathcal{H} = \begin{pmatrix} 0 & 2x + y & x + 2y \\ 2x + y & 2 - 2\lambda & -\lambda \\ x + 2y & -\lambda & 2 - 2\lambda \end{pmatrix}$$

Since there are 2 variables and 1 constraint, we only need to check the whole determinant, i.e. 1 minor to check. With $n = 2, m = 1$, we need $(-1)^m |\mathbf{C}_r| > 0$ for a minimum and $(-1)^r |\mathbf{C}_r| > 0$ for a maximum, where $r = m + 1 = 2$.

For $\lambda = \frac{2}{3}$, $(x, y) = (\pm 1, \pm 1)$: here $2x + y = x + 2y = \pm 3$, so $D_x^2 \mathcal{L} = \begin{pmatrix} 2/3 & -2/3 \\ -2/3 & 2/3 \end{pmatrix}$ and:

$$B\mathcal{H} = \begin{pmatrix} 0 & 3 & 3 \\ 3 & 2/3 & -2/3 \\ 3 & -2/3 & 2/3 \end{pmatrix}$$

$$\det(B\mathcal{H}) = -3(3 \cdot \frac{2}{3} - (-\frac{2}{3}) \cdot 3) + 3(3 \cdot (-\frac{2}{3}) - \frac{2}{3} \cdot 3) = -3(4) + 3(-4) = -24 < 0$$

Since $(-1)^m |B\mathcal{H}| = (-1)^1 (-24) = 24 > 0$, by Theorem 4.10 $\mathbf{v}^T D_x^2 \mathcal{L} \mathbf{v} > 0$ for all feasible $\mathbf{v} \neq \mathbf{0}$, confirming a **local minimum** with $f = 2$.

For $\lambda = 2$, $(x, y) = (\sqrt{3}, -\sqrt{3})$ or $(-\sqrt{3}, \sqrt{3})$: here $2x + y = \sqrt{3}$, $x + 2y = -\sqrt{3}$, so $D_x^2 \mathcal{L} = \begin{pmatrix} -2 & -2 \\ -2 & -2 \end{pmatrix}$ and:

$$B\mathcal{H} = \begin{pmatrix} 0 & \sqrt{3} & -\sqrt{3} \\ \sqrt{3} & -2 & -2 \\ -\sqrt{3} & -2 & -2 \end{pmatrix}$$

$$\begin{aligned} \det(B\mathcal{H}) &= -\sqrt{3}(-2\sqrt{3} - 2\sqrt{3}) - \sqrt{3}(-2\sqrt{3} - 2\sqrt{3}) \cdot (-1)^2 \\ &= -\sqrt{3}(-4\sqrt{3}) - \sqrt{3}(-4\sqrt{3}) \cdot (-1) = 12 + 12 = 24 > 0 \end{aligned}$$

Since $(-1)^r |B\mathcal{H}| = (-1)^2 (24) = 24 > 0$, by Theorem 4.10 $\mathbf{v}^T D_x^2 \mathcal{L} \mathbf{v} < 0$ for all feasible $\mathbf{v} \neq \mathbf{0}$, confirming a **local maximum** with $f = 6$.

Way #2: Directly testing

Let

$$D^2 \mathbf{L} = \begin{pmatrix} 2 - 2\lambda & -\lambda \\ -\lambda & 2 - 2\lambda \end{pmatrix}$$

Then, when $\lambda = 2$,

$$D^2 \mathcal{L} = \begin{pmatrix} -2 & -2 \\ -2 & -2 \end{pmatrix} = -2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Hence

$$\begin{aligned} \mathbf{v}^T D^2 \mathcal{L} \mathbf{v} &= -2 \begin{pmatrix} v_1 & v_2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\ &= -2 \begin{pmatrix} v_1 + v_2 & v_1 + v_2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\ &= -2(v_1 + v_2) < 0 \end{aligned}$$

for all $\mathbf{v} \in \mathbb{R}^2 \setminus \{0\}$. So this is a strict local maximum.

For $\lambda = \frac{2}{3}$,

$$D^2 \mathcal{L} = \begin{pmatrix} \frac{2}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix} = \frac{2}{3} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

Notice that $Dh(1,1)\mathbf{v} = Dh(-1,-1)\mathbf{v} = 0$ implies that $\mathbf{v} = \begin{pmatrix} t \\ -t \end{pmatrix}$. So,

$$\begin{aligned}\mathbf{v}^T D^2 \mathcal{L} \mathbf{v} &= \frac{2}{3} \begin{pmatrix} t & -t \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} t \\ -t \end{pmatrix} \\ &= \frac{8}{3} t^2 > 0\end{aligned}$$

for all $t \neq 0$ so strict local minimum.

Example 4.21 (Inequality and Positivity Constraints). A household has preferences represented by the utility function $u(x_1, x_2)$ where

$$u(x_1, x_2) = x_1^\beta + x_2^\beta \quad \text{with } 0 < \beta < 1;$$

it is constrained by $x_1 \geq 0, x_2 \geq 0$ and also by

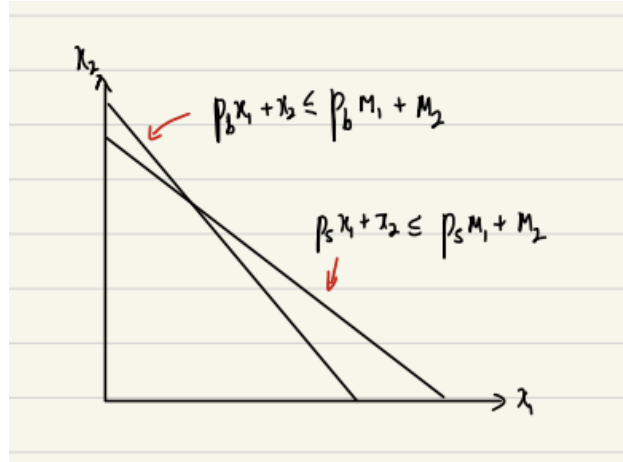
$$p_s x_1 + x_2 \leq p_s m_1 + m_2, \quad p_b x_1 + x_2 \leq p_b m_1 + m_2$$

where m_1, m_2 are fixed endowments, both strictly positive. The price of good 2 is normalised to 1, and we also have $0 < p_s < p_b$.

Draw the budget set. What is the interpretation of p_s & p_b ?

What choices should the household make if it wants to maximise utility? (Give bounds(s) on p_s and/or p_b in terms of β, m_1 and m_2 under which the solutions you have found are valid.)

The budget set is



The problem is

$$\max u(x_1, x_2) = x_1^\beta + x_2^\beta, \quad 0 < \beta < 1 \quad \text{s.t.} \quad \begin{cases} x_1 \geq 0, x_2 \geq 0 \\ p_s x_1 + x_2 \leq p_s m_1 + m_2 \\ p_b x_1 + x_2 \leq p_b m_1 + m_2 \end{cases}$$

The constraint matrix is $Dg_e = \begin{pmatrix} p_s & 1 \\ p_b & 1 \end{pmatrix}$, which has $\text{rank}(Dg_e) = 2$ since $p_b > p_s$, so the constraint qualification holds everywhere on the constraint set.

The Lagrangian is:

$$\mathcal{L}(x, \lambda, \mu) = x_1^\beta + x_2^\beta - \lambda_1(p_s x_1 + x_2 - p_s m_1 - m_2) - \lambda_2(p_b x_1 + x_2 - p_b m_1 - m_2) + \mu_1 x_1 + \mu_2 x_2$$

FOC:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x_1} &= \beta x_1^{\beta-1} - \lambda_1 p_s - \lambda_2 p_b + \mu_1 = 0 \\ \frac{\partial \mathcal{L}}{\partial x_2} &= \beta x_2^{\beta-1} - \lambda_1 - \lambda_2 + \mu_2 = 0\end{aligned}$$

CS: $\lambda_1, \lambda_2, \mu_1, \mu_2 \geq 0$ and

$$\begin{aligned}\lambda_1(p_s x_1 + x_2 - p_s m_1 - m_2) &= \lambda_1(p_s + 1)x_1 - p_s m_1 - m_2 = 0 \\ \lambda_2(p_b x_1 + x_2 - p_b m_1 - m_2) &= \lambda_2((p_b + 1)x_1 - p_b m_1 - m_2) = 0 \\ \mu_1 x_1 &= 0, \quad \mu_2 x_2 = 0\end{aligned}$$

Since $\mu_1 = \mu_2 = 0$ (utility is strictly increasing) $\therefore x_1, x_2 > 0$.

Case 1: Neither constraint binds. $\therefore \lambda_1 = \lambda_2 = 0$, which implies $\frac{\partial \mathcal{L}}{\partial x_2} = \beta x_2^{\beta-1} = 0$. Since $0 < \beta < 1$, $x_2 \neq \infty$, so this is a contradiction since $x_2 > 0$.

Case 2: Both constraints bind.

$$\begin{aligned}p_s x_1 + x_2 &= p_s m_1 + m_2 \\ p_b x_1 + x_2 &= p_b m_1 + m_2 \\ (p_s - p_b)x_1 &= (p_s - p_b)m_1 \implies x_1 = m_1, \quad \therefore x_2 = m_2\end{aligned}$$

Since $\mu_2 = 0$, $\beta x_2^{\beta-1} = \beta m_2^{\beta-1} = \lambda_1 + \lambda_2 > 0$ since $0 < \beta < 1$, $m_2 > 0$. Write $\lambda_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2} \beta m_2^{\beta-1}$, $\lambda_2 = \frac{\lambda_2}{\lambda_1 + \lambda_2} \beta m_2^{\beta-1}$, then:

$$\begin{aligned}\beta m_1^{\beta-1} &= p_s \frac{\lambda_1}{\lambda_1 + \lambda_2} \beta m_2^{\beta-1} + p_b \frac{\lambda_2}{\lambda_1 + \lambda_2} \beta m_2^{\beta-1} \\ \implies \left(\frac{m_1}{m_2}\right)^{\beta-1} &= \frac{\lambda_1}{\lambda_1 + \lambda_2} p_s + \frac{\lambda_2}{\lambda_1 + \lambda_2} p_b \\ \implies \left(\frac{m_2}{m_1}\right)^{1-\beta} &= p_s + \frac{\lambda_2}{\lambda_1 + \lambda_2} (p_b - p_s)\end{aligned}$$

Since $\frac{\lambda_2}{\lambda_1 + \lambda_2} \in [0, 1]$, the equation has a solution iff $p_s \leq \left(\frac{m_2}{m_1}\right)^{1-\beta} \leq p_b$.

Case 3: Only one constraint binds.

Suppose $p_s x_1 + x_2 = p_s m_1 + m_2$ s.t. $\lambda_1 > 0$, $\lambda_2 = 0$:

$$\begin{aligned}\beta x_1^{\beta-1} &= \lambda_1 p_s, \quad \beta x_2^{\beta-1} = \lambda_1 \\ \implies \left(\frac{x_1}{x_2}\right)^{\beta-1} &= p_s \implies \frac{x_2}{x_1} = p_s^{\frac{1}{1-\beta}}\end{aligned}$$

Substituting into the binding constraint: $p_s x_1 + p_s^{\frac{1}{1-\beta}} x_1 = p_s m_1 + m_2$, giving:

$$x_1^* = \frac{p_s m_1 + m_2}{p_s + p_s^{\frac{1}{1-\beta}}}, \quad x_2^* = p_s^{\frac{1}{1-\beta}} \left[\frac{p_s m_1 + m_2}{p_s + p_s^{\frac{1}{1-\beta}}} \right]$$

The constraint binds when $x_1^* \leq m_1$. To see this, we define the slack functions

$$\begin{aligned}z_s(x) &= p_s x_1 + x_2 - (p_s m_1 + m_2) \\ z_b(x) &= p_b x_1 + x_2 - (p_s m_1 + m_2)\end{aligned}$$

Notice that for $j \in \{s, b\}$, $z_j(x) = 0$ if (x_1, x_2) satisfies the budget constraint, and $z_j(x) < 0$ if it is below the constraint. Then, if the first constraint binds first, check $z_s(x) - z_b(x) = (p_s - p_b)(x_1 - m_1) > 0$ which holds iff if $x_1 \leq m_1$ since $p_s - p_b < 0$.

$$\frac{p_s m_1 + m_2}{p_s + p_s^{\frac{1}{1-\beta}}} \leq m_1 \implies \frac{m_2}{m_1} \leq p_s^{\frac{1}{1-\beta}} \implies \left(\frac{m_2}{m_1}\right)^{1-\beta} \leq p_s$$

Suppose $\lambda_2 > 0$, $\lambda_1 = 0$ with $p_b x_1 + x_2 = p_b m_1 + m_2$. Then $\beta x_1^{\beta-1} = \lambda_2 p_b$, $\beta x_2^{\beta-1} = \lambda_2$:

$$\left(\frac{x_2}{x_1}\right)^{1-\beta} = p_b \implies x_2 = p_b^{\frac{1}{1-\beta}} x_1$$

$$x_1 = \frac{p_b m_1 + m_2}{p_b + p_b^{\frac{1}{1-\beta}}}, \quad x_2 = p_b^{\frac{1}{1-\beta}} \left[\frac{p_b m_1 + m_2}{p_b + p_b^{\frac{1}{1-\beta}}} \right]$$

Constraint binds when $x_1 > m_1$:

$$\frac{p_b m_1 + m_2}{p_b + p_b^{\frac{1}{1-\beta}}} > m_1 \implies \frac{m_2}{m_1} > p_b^{\frac{1}{1-\beta}} \implies \left(\frac{m_2}{m_1}\right)^{1-\beta} > p_b$$

Summary of solutions:

$$(x_1^*, x_2^*) = (m_1, m_2) \quad \text{when } p_s \leq \left(\frac{m_2}{m_1}\right)^{1-\beta} \leq p_b$$

$$(x_1^*, x_2^*) = \left(\frac{p_s m_1 + m_2}{p_s + p_s^{\frac{1}{1-\beta}}}, p_s^{\frac{1}{1-\beta}} \left[\frac{p_s m_1 + m_2}{p_s + p_s^{\frac{1}{1-\beta}}} \right] \right), \quad p_s \geq \left(\frac{m_2}{m_1}\right)^{1-\beta}$$

$$(x_1^*, x_2^*) = \left(\frac{p_b m_1 + m_2}{p_b + p_b^{\frac{1}{1-\beta}}}, p_b^{\frac{1}{1-\beta}} \left[\frac{p_b m_1 + m_2}{p_b + p_b^{\frac{1}{1-\beta}}} \right] \right), \quad p_b \leq \left(\frac{m_2}{m_1}\right)^{1-\beta}$$

Check SOC:

$$D_x^2 \mathcal{L}(x^*, \hat{\lambda}) = \begin{pmatrix} \beta(\beta-1)x_1^{\beta-2} & 0 \\ 0 & \beta(\beta-1)x_2^{\beta-2} \end{pmatrix}$$

Since $0 < \beta < 1$, $\mathbf{v}^T D_x^2 \mathcal{L}(x^*, \hat{\lambda}) \mathbf{v} < 0$ for all $\mathbf{v} \in \mathbb{R}^2 \setminus \{0\}$, i.e. negative definite, so all solutions are local maxima since negative definite $\forall \mathbf{v} \in \mathbb{R}^2 \setminus \{0\}$.

4.3 Convex/Concave Problems

Main goal: ensure that the local optimum is also a global one.

Definition 4.22 (Convexity/Concavity). Consider a convex set $S \subseteq \mathbb{R}^n$. A real valued function is concave if for any $\mathbf{x}, \mathbf{y} \in S$ and any $t \in (0, 1)$

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \geq tf(\mathbf{x}) + (1-t)f(\mathbf{y})$$

A real valued function is convex if for any $\mathbf{x}, \mathbf{y} \in S$ and any $t \in (0, 1)$

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq tf(\mathbf{x}) + (1-t)f(\mathbf{y})$$

The functions are strictly concave/convex if the inequalities are strict

If the function $f : S \rightarrow \mathbb{R}$ is C^2 , another way of checking whether the function is concave/convex is to look at its hessian

Proposition 4.23. If $f : S \rightarrow \mathbb{R}$ is C^2 , then

- $D^2 f$ is negative semi-definite for all $\mathbf{x} \in S$ iff f is concave on S
- $D^2 f$ is negative definite for all $\mathbf{x} \in S$ iff f is strictly concave on S
- $D^2 f$ is positive semi-definite for all $\mathbf{x} \in S$ iff f is convex on S

- D^2f is positive definite for all $\mathbf{x} \in S$ iff f is convex on S

Lemma 4.24. If $f : S \rightarrow \mathbb{R}$ is C^1 , then, f is concave iff

$$f(\mathbf{y}) - f(\mathbf{x}) \leq Df(\mathbf{x})(\mathbf{y} - \mathbf{x}), \forall \mathbf{x}, \mathbf{y} \in S$$

If $f : S \rightarrow \mathbb{R}$ is C^1 , then, f is convex iff

$$f(\mathbf{y}) - f(\mathbf{x}) \geq Df(\mathbf{x})(\mathbf{y} - \mathbf{x}), \forall \mathbf{x}, \mathbf{y} \in S$$

Corollary 4.25. Suppose $\mathbf{x}^*$ satisfies $Df(\mathbf{x}^*) = \mathbf{0}$ then

- if f is concave, $\mathbf{x}^*$ is a global maximum
- if f is convex, $\mathbf{x}^*$ is a global minimum

Theorem 4.26. Let f be (strictly) concave/convex in S

- If a stationary point exists, it is a (unique) global maximum/minimum
- If a local maximum/minimum exists, it is a (unique) global maximum/minimum

Note that concavity and convexity only give sufficient conditions for unconstrained optimisation problems i.e. interior critical points. For constrained candidates, we still need to check SOC to see whether they are local constrained optima.

Theorem 4.27 (Lagrangian Sufficiency). Consider the following constrained optimisation problem

$$P : \text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \leq \mathbf{b}$$

If there exists $\mathbf{x}^*$ and $\boldsymbol{\lambda}^* \geq \mathbf{0}$ such that

$$\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) \geq \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*), \forall \mathbf{x} \in \mathbb{R}^n \quad (1)$$

$$\boldsymbol{\lambda}^*(\mathbf{g}(\mathbf{x}^*) - \mathbf{b}) = 0 \quad (2)$$

and $\mathbf{x}^*$ is feasible ($\mathbf{g}(\mathbf{x}^*) \leq \mathbf{b}$), then $\mathbf{x}^*$ is optimal for P

Proof. Consider any $\mathbf{x}$ which is feasible for P (i.e., $\mathbf{g}(\mathbf{x}) \leq \mathbf{b}$),

$$f(\mathbf{x}^*) = \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) \geq \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*) \geq f(\mathbf{x})$$

The equality follows from (2) and the second inequality from the fact that $\mathbf{x}$ is feasible and $\boldsymbol{\lambda}^* \geq \mathbf{0}$. Hence $\mathbf{x}^*$ is optimal. ■

Remark. Note that

- The Lagrangian sufficiency says that if we are able to find a feasible $\mathbf{x}^*$ and a $\boldsymbol{\lambda}^* \geq \mathbf{0}$ with $\boldsymbol{\lambda}^*(\mathbf{g}(\mathbf{x}^*) - \mathbf{b}) = 0$, such that $\mathbf{x}^*$ maximises the Lagrangian $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*)$, then $\mathbf{x}^*$ is a global optimum
- We might not be able to find such a $\boldsymbol{\lambda}^*$ and $\mathbf{x}^*$...or in other words, it might be that the global maximum of the problem does not maximise the Lagrangian
- Note that the Lagrangian sufficiency result does not make any concavity or convexity assumption so can be used to prove global optimality even when these assumptions do not hold
- Note that we do not require any Constraint Qualification either

Definition 4.28 (Concave and Convex Problems). Consider the problem

$$P : \text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \leq \mathbf{b}$$

P is a Concave Problem if f is concave and each g_i is convex.

Analogously, the problem

$$P : \text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \geq \mathbf{b}$$

is a Convex Problem if f is convex and each g_j is concave

For such problems, the FOCs are sufficient

Proposition 4.29 (Sufficient FOC). Let P be a concave (convex) problem. Then the Kuhn-Tucker first order conditions are sufficient for a global maximum (minimum)

Proof. If $\lambda \geq 0$ then since f is concave and g_j is convex, the Lagrangian

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(\mathbf{g}(\mathbf{x}) - \mathbf{b})$$

is a concave function of $\mathbf{x}$. Let $(\mathbf{x}^*, \lambda^*)$ satisfy the Kuhn-Tucker first-order conditions. These imply that $\mathbf{x}^*$ is a stationary point of $\mathcal{L}(\mathbf{x}, \lambda^*)$ with respect to $\mathbf{x}$. Since $\mathcal{L}$ is concave this implies that $\mathbf{x}^*$ maximises $\mathcal{L}(\mathbf{x}, \lambda^*)$ globally with respect to $\mathbf{x}$. By Lagrangian Sufficiency, $\mathbf{x}^*$ is optimal for P ■

Remark. Note that the Constraint Qualification is not required for this result.

Furthermore,

- Without the concavity assumptions, a solution to the Kuhn-Tucker conditions need not maximise $\mathcal{L}(\mathbf{x}, \lambda^*)$ and the Lagrangian sufficiency does not apply
- These are sufficient conditions. It could be that the Kuhn-Tucker conditions do not have a solution. However, if f is concave and g_j is convex, then if in addition $\mathbf{g}(\mathbf{x}) << \mathbf{b}$ for some feasible $\mathbf{x}$, the global maximum will also maximise the Lagrangian

Proposition 4.30 (Lagrangian Necessity). Let f be concave and let each g_j be convex. Suppose, in addition, $\mathbf{g}(\mathbf{x}) << \mathbf{b}$ for some feasible $\mathbf{x}$. If $\mathbf{x}^*$ is a global maximum there exists a vector $\lambda^* \geq 0$, with

$$\lambda^*(\mathbf{g}(\mathbf{x}^*) - \mathbf{b}) = 0$$

such that $\mathbf{x}^*$ maximises the Lagrangian

$$\mathcal{L} = f(\mathbf{x}) - \lambda^*(\mathbf{g}(\mathbf{x}) - \mathbf{b})$$

Here, we consider weakening the concavity and convexity assumptions.

Definition 4.31 (Upper/Lower Contour Sets). Suppose that the domain S is convex

- f is quasi-concave if the upper contour set $\{\mathbf{x} : f(\mathbf{x}) \geq c\}$ is a convex set for any real c
- f is quasi-convex if the lower contour set $\{\mathbf{x} : f(\mathbf{x}) \leq c\}$ is a convex set for any real c

The properties are strict if the upper (lower) contour sets are strictly convex.

Lemma 4.32. A function is quasi-concave iff

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\}, \forall t \in (0, 1)$$

A function f is quasi-convex iff

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq \max\{f(\mathbf{x}), f(\mathbf{y})\}, \forall t \in (0, 1)$$

The properties are strict if the inequalities are strict for $\mathbf{x} \neq \mathbf{y}$

Lemma 4.33. If f is a C^1 function,

- f is quasi-concave iff

$$f(\mathbf{y}) \geq f(\mathbf{x}) \implies Df(\mathbf{x})(\mathbf{y} - \mathbf{x}) \geq 0$$

- f is quasi-convex iff

$$f(\mathbf{y}) \leq f(\mathbf{x}) \implies Df(\mathbf{x})(\mathbf{y} - \mathbf{x}) \leq 0$$

The conditions for strict quasi-concavity are as follows:

If for all $\mathbf{x} \neq \mathbf{y}$, $Df(\mathbf{x})(\mathbf{y} - \mathbf{x}) > 0$ whenever $f(\mathbf{y}) \geq f(\mathbf{x})$, then f is strictly quasi-concave

If f is strictly quasi-concave and $Df(\mathbf{x}) \neq 0$, then $Df(\mathbf{x})(\mathbf{y} - \mathbf{x}) > 0$ whenever $f(\mathbf{y}) \geq f(\mathbf{x})$

Proposition 4.34. We have the following properties for quasi-concavity and quasi-convexity

- f is quasi-concave $\iff -f$ is quasi-convex
- f is concave $\implies f$ is quasi-concave
- f is convex $\implies f$ is quasi-convex
- Any (strictly) monotonic function of one variable is both (strictly) quasi-concave and (strictly) quasi-convex.

Example 4.35. The following examples are instructive

- $\ln(x)$ is concave, so is quasi-concave; increasing, so also quasi-convex (but not convex)
- $\ln(x) + \ln(y)$ is concave, so also quasi-concave; but not quasi-convex, so also not convex

Quasi-concavity and quasi-convexity provide sufficient conditions for optimisation.

Theorem 4.36 (Global Optima). Consider the problem

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{x} \in S$$

Let f be strictly quasi-concave. If a local maximum exists, it is a unique global maximum.

Theorem 4.37 (Sufficient FOC). Consider the problem

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \leq \mathbf{b}$$

Let f be quasi-concave and let each g_j be quasi-convex.

The Kuhn-Tucker first order conditions are sufficient for a global maximum (at $\mathbf{x}^*$) provided at least one of the following holds:

$$(1) f \text{ is concave, } (2) Df(\mathbf{x}^*) \neq 0$$

4.4 Envelope Theorem

The envelope theorem provides us with a way to examine how the maximised value of an objective function change as parameters change.

Definition 4.38 (Maximum Value Function). Suppose the objective function and the constraints depend on a set of parameters $\mathbf{a}$:

$$\max_{\mathbf{x}} f(\mathbf{x}; \mathbf{a}) \text{ subject to } \mathbf{g}(\mathbf{x}; \mathbf{a}) \leq \mathbf{b}$$

The maximum value function of the parameters is

$$M(\mathbf{a}) = \{\max_{\mathbf{x}} f(\mathbf{x}; \mathbf{a}) | \mathbf{g}(\mathbf{x}; \mathbf{a}) \leq \mathbf{b}\}$$

If $\mathbf{x}^*(\mathbf{a})$ is the optimal value of $\mathbf{x}$ as a function of $\mathbf{a}$ then

$$M(\mathbf{a}) = f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})$$

Unconstrained Problems

The motivation for the envelope theorem is that if there are no constraints, $\mathbf{x}^*(\mathbf{a})$ satisfies the FOCs such that

$$\frac{\partial f}{\partial x_i}(\mathbf{x}^*(\mathbf{a}); \mathbf{a}) = 0, \text{ for } i = 1, \dots, n$$

If the conditions for the Implicit Function Theorem hold,

- f_{x_i} is C^1 (guaranteed if f is C^2)
- $\frac{\partial^2 f}{\partial x_i \partial a}(\mathbf{x}^*(\mathbf{a}); \mathbf{a}) \neq 0$

then $\mathbf{x}^*$ is a C^1 function of $\mathbf{a}$

Theorem 4.39 (Envelope Theorem (Unconstrained Optimisation)). Let

$$M(\mathbf{a}) = f(\mathbf{x}^*(\mathbf{a}), \mathbf{a})$$

Then,

$$\frac{\partial M(\mathbf{a})}{\partial a_j} = \frac{\partial f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})}{\partial a_j}$$

Proof. By the Chain Rule

$$\frac{\partial M(\mathbf{a})}{\partial a_j} = \sum_i \frac{\partial f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})}{\partial x_i} \frac{\partial x_i^*(\mathbf{a})}{\partial a_j} + \frac{\partial f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})}{\partial a_j}$$

But $\frac{\partial f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})}{\partial x_i} = 0$ by construction ■

Only the direct change of a_j on f matters.

Example 4.40. Consider $f(x : a, b) = 4 + 2b \ln(x) - ax^2$ with $a \neq 0$. The FOC is

$$f_x(x^*; a, b) = 0 \implies \frac{2b}{x^*} - 2ax^* = 0$$

which yields

$$(x^*(a, b))^2 = \frac{b}{a}$$

By the Envelope Theorem

$$\frac{\partial M(a, b)}{\partial a} = \frac{\partial f(x^*(a); a)}{\partial a} = -(x^*(a, b))^2 = -\frac{b}{a}$$

since the partial derivative of f wrt a is $-x^2$.

Theorem 4.41 (Envelope Theorem (Constrained Problems)). Assume that $\mathbf{x}^*(\mathbf{a})$ satisfies the Kuhn-Tucker first order conditions and that the conditions of the Implicit Function Theorem hold so that $\mathbf{x}^*(\mathbf{a})$ and $\boldsymbol{\lambda}^*(\mathbf{a})$ are C^1 functions. Construct the Lagrangian

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}; \mathbf{a}) = f(\mathbf{x}; \mathbf{a}) - \boldsymbol{\lambda}(\mathbf{g}(\mathbf{x}; \mathbf{a}) - \mathbf{b})$$

Then,

$$\frac{\partial M(\mathbf{a})}{\partial a_i} = \frac{\partial \mathcal{L}(\mathbf{x}^*(\mathbf{a}); \boldsymbol{\lambda}^*(\mathbf{a}); \mathbf{a})}{\partial a_i}$$

where the maximum value function is defined as

$$M(\mathbf{a}) = \max_{\mathbf{x}} f(\mathbf{x}; \mathbf{a}) | \mathbf{g}(\mathbf{x}; \mathbf{a}) \leq \mathbf{b}$$

Proof. ■

We know that λ_j represents the effect on the objective function of relaxing the constraint $g_j(\mathbf{x}) \leq b_j$. This can be derived from the Envelope Theorem

Corollary 4.42. Consider

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}; \mathbf{b}) = f(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{g}(\mathbf{x}) - \mathbf{b})$$

Then, define

$$M(\mathbf{b}) = f(\mathbf{x}^*(\mathbf{b})) \equiv \mathcal{L}(\mathbf{x}^*(\mathbf{b}), \boldsymbol{\lambda}^*(\mathbf{b}); \mathbf{b})$$

Then,

$$\frac{\partial M}{\partial b_j} = \frac{\partial \mathcal{L}}{\partial b_j} = \lambda_j^*$$

5 Information Economics

5.1 Hidden Action

Hidden Action Principal-Agent Problems are problems when the principal cannot observe how much effort the agent exerts i.e. moral hazard.

The basic problem is that a risk neutral principal hires a risk averse agent for a one-time project, who then chooses an action that affects the project's profit. We focus on the simple case of two possible levels of effort: $e = e_L$ or e_H with $e_H > e_L$. The effort is unobservable to the principal and is not perfectly deducible from the observation of the profits of the project π . The profits π are distributed on $[\pi_L, \pi_H]$ and the distribution depends on the effort e .

Assumption 5.1. The distribution of π conditional on e_H first order stochastically dominates the distribution of π conditional on e_L

Definition 5.2 (First Order Stochastic Dominance). The distribution $F(\cdot)$ first order stochastically dominates $G(\cdot)$ if, for every non-decreasing function $u : \mathbb{R} \rightarrow \mathbb{R}$, we have

$$\mathbb{E}[u(x)|F] \geq \mathbb{E}[u(x)|G]$$

In particular, taking $u(x) = x$, this implies that $\mathbb{E}[x|F] > \mathbb{E}[x|G]$ which also means in our setting

$$\mathbb{E}[\pi|e_H] > \mathbb{E}[\pi|e_L]$$

Proposition 5.3. $F(\cdot)$ first order stochastically dominates $G(\cdot)$ iff

$$F(x) \leq G(x)$$

with strict inequality for some x

This implies

$$F(\pi|e_H) \leq F(\pi|e_L), \forall \pi \in [\pi_L, \pi_H]$$

in our setting.

Further Primitives:

- Agent's Preferences: Agent's utility from wage and effort:

$$u(w, e) = v(w) - g(e)$$

with $v' > 0, v'' \leq 0, g' > 0$

- If $v'' \leq 0$: agent is risk averse
- If $g' > 0$: higher level of effort is more costly to the agent

The agent has ex ante reservation utility $\bar{u}$

- Principal's Preferences: Principal's utility from wage w and effort e is the expected net profit:

$$\mathbb{E}[\pi - w|e]$$

where the principal is taken to be risk neutral

Definition 5.4 (Principal's Problem in Hidden Action). Choose $e \in \{e_L, e_H\}$ and the wage scheme conditional on the profits $\pi(w)$ to maximise

$$\mathbb{E}[\pi - w(\pi)|e]$$

such that the agent wishes to participate (Individual Rationality)

$$\mathbb{E}[v(w(\pi))|e] - g(e) \geq \bar{u}$$

and that the agent wishes to choose e (Incentive Compatibility) i.e. e maximises

$$\mathbb{E}[v(w(\pi))|e] - g(e)$$

Remark. In our simple two effort setting, the IC constraint is equivalent to

$$\mathbb{E}[v(w(\pi))|e_H] - g(e_H) \geq \mathbb{E}[v(w(\pi))|e_L] - g(e_L)$$

Our general approach is two stage (for a general discrete effort case)

1. for each e_i , find the optimal $w_i(\pi)$ that implements e_i :

$$\begin{aligned} & \min_{w(\pi)} \mathbb{E}[w(\pi)|e_i] \text{ s.t.} \\ & \mathbb{E}[v(w(\pi))|e_i] - g(e_i) \geq \bar{u} \\ & e_i = \operatorname{argmax}_e \mathbb{E}[v(w(\pi))|e] - g(e) \end{aligned}$$

2. find the optimal e^* :

$$e^* = \operatorname{argmax}_e \mathbb{E}[\pi - w_i(\pi)|e_i]$$

The general trade-off is between risk-sharing and incentives.

First Best: Contractible (Observable) Effort

When effort is observable/contractible, a contract specifies the agent's effort and the wage scheme so incentives are not a concern and we just need to make sure the agent will take the contract. So,

Stage 1: for each e

$$\min_{w(\pi)} \mathbb{E}[w(\pi)|e] \text{ such that } \mathbb{E}[v(w(\pi))|e] - g(e) \geq \bar{u}$$

Note that the participation constraint has to bind because if it holds strictly, then we can decrease the wage while satisfying the constraint and hence achieve a lower expected wage. Hence, the problem reduces to

$$\min_{w(\pi)} \mathbb{E}[w(\pi)|e] \text{ such that } \mathbb{E}[v(w(\pi))|e] - g(e) = \bar{u}$$

Consider a discrete distribution of π

$$\begin{aligned} \mathbb{E}[w(\pi)|e] &= \sum_{\pi} w(\pi) \mathbb{P}[\pi|e] \\ \mathbb{E}[v(w(\pi))|e] &= \sum_{\pi} v(w(\pi)) \mathbb{P}[\pi|e] \end{aligned}$$

We construct the Lagrangian

$$\mathcal{L}(\{w(\pi)\}_{\pi}, \gamma; e) = \sum_{\pi} w(\pi) \mathbb{P}[\pi|e] - \gamma \left(\sum_{\pi} v(w(\pi)) \mathbb{P}[\pi|e] - g(e) - \bar{u} \right)$$

For each π , the FOC wrt $\pi(w)$ is:

$$\mathbb{P}[\pi|e] - \gamma v'(w(\pi)) \mathbb{P}[\pi|e] = 0$$

so,

$$\gamma = \frac{1}{v'(w^e(\pi))} > 0$$

This means that for all π , $v'(w^e(\pi)) = \frac{1}{\gamma}$ or in other words if $v'' < 0$, the optimal wage scheme will be constant across profit level π . This is because if $v'' < 0$ holds, then v' is strictly decreasing so that for any pair of wage schemes

$$w_1 \neq w_2 \implies v'(w_1) \neq v'(w_2)$$

Then

$$v'(w_1) = v'(w_2) = \frac{1}{\gamma} \implies w_1 = w_2$$

for optimal wage schemes to implement a specific e .

If $v'' = 0$, there might be multiple solutions.

Intuition: The agent is risk averse and using a variable wage scheme introduces risk and hence on average it is more costly to induce the reservation utility.

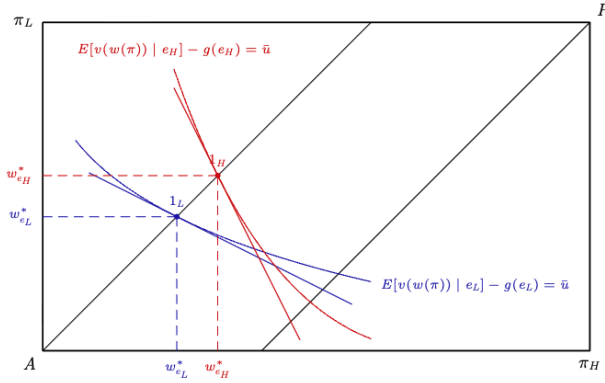
So, $w^e(\pi) \equiv w_e^*$ for each e . And the constraint implies that

$$v(w_{e_H}^*) - g(e_H) = \bar{u}$$

and

$$v(w_{e_L}^*) - g(e_L) = \bar{u}$$

Note that $w_{e_H}^* > w_{e_L}^*$. We may illustrate this for the case of two values of profit π_L and π_H



Stage 2: This is a simple trade-off between expected profits and wages:

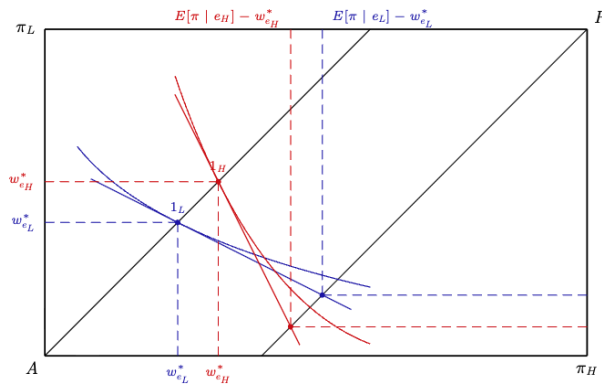
$$\max_{e \in \{e_L, e_H\}} \{ \mathbb{E}[\pi | e_L] - w_{e_L}^*, \mathbb{E}[\pi | e_H] - w_{e_H}^* \}$$

This contract achieves (i) efficient effort choice by the agent, (ii) no risk to the agent. With full information the payoffs are

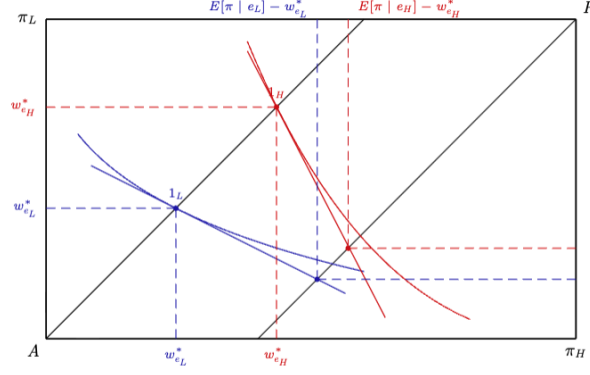
- The agent receives reservation utility $\bar{u}$
- The principal gets:

$$\mathbb{E}[\pi | e^*] - v^{-1}(\bar{u} + g(e^*))$$

This case is where $e^* = e_H$



This case is where $e^* = e_L$



First Best: Unobservable Effort, Risk-Neutral Agent

If effort is unobservable, but the agent is risk neutral, then **risk sharing is not a concern**. In other words, the agent's utility is $v(w) \equiv w$.

First best: if the effort was observable, the optimal effort e^* would maximise

$$\mathbb{E}[\pi - w_e^*|e] = \mathbb{E}[\pi|e] - g(e) - \bar{u}$$

which coincides with the effort that maximises social surplus, since the expected wage, $\mathbb{E}[w_e^*(\pi)|e]$ is merely a transfer between principal and agent.

Proposition 5.5. If the agent is risk-neutral, the principal can achieve her full information payoff when the effort is unobservable by selling the project

Consider the contract where the principal sells the project to the agent at price α . Selling the project at price α is equivalent to setting a wage scheme:

$$\pi(w) = \pi - \alpha$$

In that case, the agent chooses e to maximise

$$\mathbb{E}[\pi - \alpha|e] - g(e)$$

The agent's incentives are aligned to social surplus maximisation (i.e. the agent chooses effort e^*); and the principal can choose α optimally, si that the agent's participation constraint binds i.e.

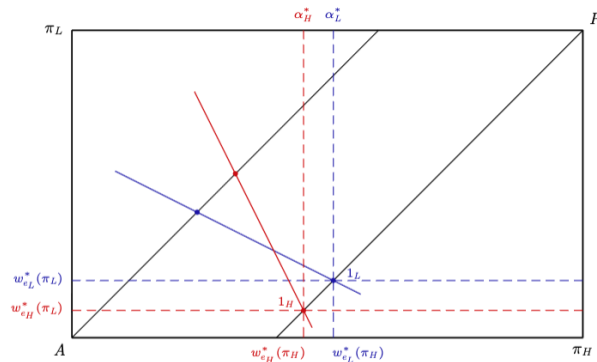
$$\alpha^* = \mathbb{E}[\pi|e^*] - g(e^*) - \bar{u}$$

Hence, the principal and the agent get the same payoff as when the effort was observable:

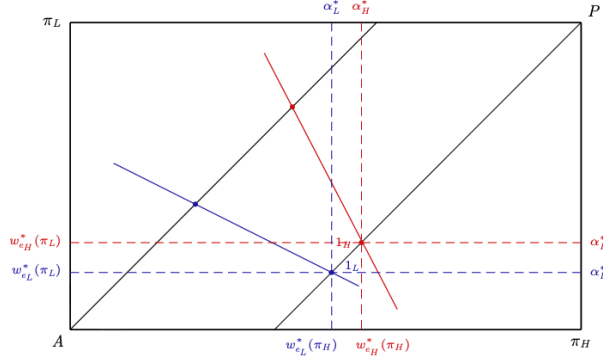
- Agent gets $\mathbb{E}[\pi|e^*] - (\mathbb{E}[\pi|e^*] - g(e^*) - \bar{u}) - g(e^*) = \bar{u}$
- Principal gets: $\alpha^* = \mathbb{E}[\pi|e^*] - g(e^*) - \bar{u}$

We can illustrate the following cases:

This case is where $e^* = e_H$



This case is where $e^* = e_L$



Second Best: Unobservable Effort, Risk-Adverse Agent

This is the main object of focus where there is a trade-off as incentives for high effort can be provided only at the cost of risk to the agent.

Step 1: To implement e_L it is sufficient for the principal to offer the fixed wage $w_{e_L}^*$ so that

$$v(w_{e_L}^*) - g(e_L) = \bar{u}$$

which means (i) the participation constraint binds; when facing a fixed wage, the agent strictly prefers to choose e_L rather than e_H .

To implement e_H we must find the optimal $w(\pi)$ that minimises $\mathbb{E}[w(\pi)|e_H]$ subject to IR and IC

$$\begin{aligned} \mathbb{E}[v(w(\pi))|e_H] - g(e_H) &\geq \bar{u} \\ \mathbb{E}[v(w(\pi))|e_H] - g(e_H) &\geq \mathbb{E}[v(w(\pi))|e_L] - g(e_L) \end{aligned}$$

The Kuhn-Tucker FOCs imply that for every π , the wage $w(\pi)$ satisfies:

$$\mathbb{P}[\pi|e_H] - \gamma v'(w(\pi))\mathbb{P}[\pi|e_H] - \mu v'(w(\pi))(\mathbb{P}[\pi|e_H] - \mathbb{P}[\pi|e_L]) = 0$$

or

$$\frac{1}{v'(w(\pi))} = \gamma + \mu \left[1 - \frac{\mathbb{P}[\pi|e_L]}{\mathbb{P}[\pi|e_H]} \right]$$

and the complementarity slackness conditions. We now investigate the full solution with the K-T FOCs.

Proposition 5.6. In any solution of this problem, both $\gamma > 0$ and $\mu > 0$

Proof. Suppose $\mu = 0$, then we have a constant wage schedule. But then the agent will always choose e_L since it leads to the same constant wage and is less costly. Mathematically, we see that

$$\frac{1}{v'(w(\pi))} = \gamma \implies v'(w(\pi)) = \frac{1}{\gamma}$$

and since $v(\cdot)$ is strictly increasing, we have that w must be constant across all wages.

Suppose $\gamma = 0$. Because the distribution of π conditional on e_H FOSD the distribution of π conditional on e_L , for some low values of π we must have

$$\mathbb{P}[\pi|e_L] > \mathbb{P}[\pi|e_H]$$

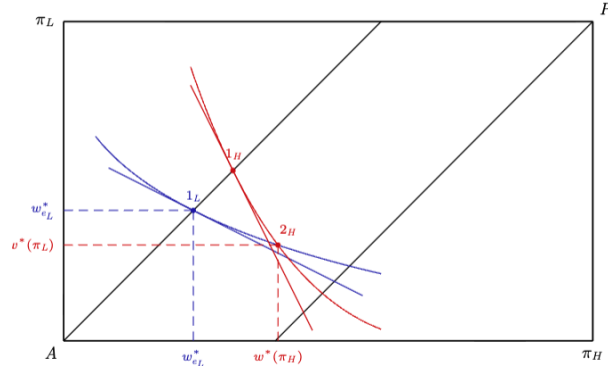
i.e. the low outcomes will have more probability mass.

But then for those π we would have

$$\frac{1}{v'(w(\pi))} = \mu \left[1 - \frac{\mathbb{P}[\pi|e_L]}{\mathbb{P}[\pi|e_H]} \right] < 0$$

which implies $v'(w(\pi)) < 0$ since $\mu > 0$ which is a contradiction. Hence, by the complementarity slackness conditions, both the participation constraint and the incentive constraint are binding. ■

We may illustrate this like so:



In order to implement e_H the expected wage is higher under non-observability: for the risk averse agent to accept the gamble, the expected utility needs to be at least $\bar{u}$ which means that the certainty equivalence must be at least $w_{e_H}^*$. The risk premium associated to the wage scheme, is called the agency cost.

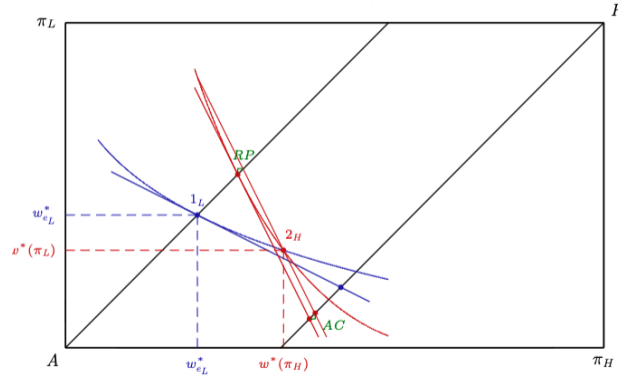
Definition 5.7 (Agency Cost). The agency cost is the risk premium associated to the wage scheme

$$A = \mathbb{E}[w|e_H] - w_{e_H}^*$$

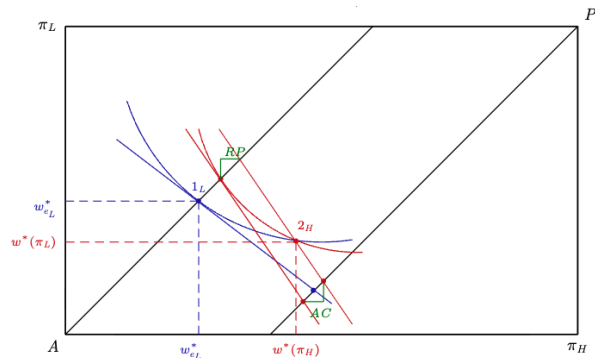
Stage 2: Here, the problem is to

$$\max_{e \in \{e_L, e_H\}} \{ \mathbb{E}[\pi|e_L] - w_{e_L}^*, \mathbb{E}[\pi - w|e_H] \}$$

The illustration for the case of two values for π : π_L, π_H



where in this particular case it is still optimal to induce e_H . It can also be optimal to induce e_L



In sum

- The optimal compensation scheme that implements e_L is the same as when effort is observable: fixed wage and the agent's utility is $\bar{u}$
- The optimal compensation scheme that implements e_H also gives the agent an expected utility of $\bar{u}$ but involves a higher expected wage than is required when effort is observable
- Whenever the optimal effort level with observable effort is e_H , there is a welfare loss (to the principal) under non-observability

Further properties of the optimal wage scheme.

Proposition 5.8. Consider some optimal wage scheme π_1 and π_2 . If π_1 and π_2 are such that

$$\mathbb{P}[\pi_1|e_L]/\mathbb{P}[\pi_1|e_H] = \mathbb{P}[\pi_2|e_L]/\mathbb{P}[\pi_2|e_H]$$

then, $w(\pi_1) = w(\pi_2)$. And if

$$\mathbb{P}[\pi_1|e_L]/\mathbb{P}[\pi_1|e_H] > \mathbb{P}[\pi_2|e_L]/\mathbb{P}[\pi_2|e_H]$$

then, $w(\pi_1) < w(\pi_2)$

Proof. This follows by examining the Kuhn-Tucker FOC:

$$\frac{1}{v'(w(\pi))} = \gamma + \mu \left[1 - \frac{\mathbb{P}[\pi|e_L]}{\mathbb{P}[\pi|e_H]} \right]$$

■

Moreover, if $\hat{w}$ is the fixed wage s.t. $(1/v(\hat{w})) = \gamma$:

$$w(\pi) > \hat{w} \text{ if } \frac{\mathbb{P}[\pi|e_L]}{\mathbb{P}[\pi|e_H]} < 1$$

and

$$w(\pi) < \hat{w} \text{ if } \frac{\mathbb{P}[\pi|e_L]}{\mathbb{P}[\pi|e_H]} > 1$$

So,

- If a realisation of π is relatively more likely to occur under e_H than under e_L , then the optimal reward scheme pays more
- If a realisation of π is relatively less likely to occur under e_H than under e_L , then the optimal reward scheme pays less

Therefore, we have

Proposition 5.9 (Monotone Likelihood Ratio Property). If

$$\frac{\mathbb{P}[\pi|e_H]}{\mathbb{P}[\pi|e_L]}$$

is increasing in π , then the wage schedule $w(\pi)$ will be increasing in π

Remark. FOSD does not imply the Monotone Likelihood Ratio Property

Example 5.10 (Linear compensation schemes). Consider a risk-neutral owner (principal) and a risk-averse manager (agent). Output (profit) is

$$\pi = 1 - q_1 - q_2 \quad (\text{not used here}), \quad \text{and in this example } \pi | e \sim \mathcal{N}(e, \sigma^2).$$

The manager chooses effort $e \geq 0$ and has mean-variance preferences

$$\mathbb{E}[u(w, e)] = \mathbb{E}[w] - \phi \text{Var}(w) - g(e),$$

where $\phi > 0$ and $g', g'' > 0$ with $\lim_{e \rightarrow \infty} g'(e) = \infty$. Restrict attention to linear wage schemes

$$w(\pi) = a + b\pi.$$

1. **Manager's expected utility under a linear contract.** Since $\mathbb{E}[\pi | e] = e$ and $\text{Var}(\pi | e) = \sigma^2$,

$$\mathbb{E}[w | e] = a + b \mathbb{E}[\pi | e] = a + be, \quad \text{Var}(w | e) = b^2 \text{Var}(\pi | e) = b^2 \sigma^2.$$

Hence

$$\mathbb{E}[u(w, e)] = (a + be) - \phi b^2 \sigma^2 - g(e).$$

2. **Effort observable (contractible).** If e is observable and enforceable, incentives are unnecessary, so the optimal contract fully insures the manager i.e. constant wage:

$$b^* = 0, \quad w(\pi) \equiv a.$$

Normalise the reservation utility to 0. Then the participation constraint (IR) is

$$a - g(e) \geq 0 \quad \Rightarrow \quad a = g(e) \quad \text{at optimum.}$$

The owner then chooses e to maximise expected profit:

$$\max_e \mathbb{E}[\pi - w | e] = \max_e (e - g(e)).$$

The FOC (necessary and sufficient) is

$$1 - g'(e^*) = 0 \quad \Rightarrow \quad g'(e^*) = 1,$$

so

$$e^* = (g')^{-1}(1), \quad a^* = g(e^*), \quad w(\pi) \equiv g(e^*).$$

3. **Effort not observable (moral hazard).** Fix an effort level e that the owner wants to implement. The owner chooses (a, b) to minimise expected wage subject to (IR) and (IC):

$$\min_{a, b} \mathbb{E}[w | e] = a + be$$

subject to

$$\text{(IR)} \quad a + be - \phi b^2 \sigma^2 - g(e) \geq 0,$$

$$\text{(IC)} \quad e \in \arg \max_{e'} \left\{ a + be' - \phi b^2 \sigma^2 - g(e') \right\}.$$

The IC reduces to maximising $be' - g(e')$, so the FOC gives

$$g'(e) = b.$$

(IR) binds at optimum, hence

$$a + be - \phi b^2 \sigma^2 - g(e) = 0 \quad \Rightarrow \quad a = g(e) + \phi b^2 \sigma^2 - be = g(e) + \phi \sigma^2 g'(e)^2 - g'(e)e.$$

Therefore the expected wage when implementing e is

$$\mathbb{E}[w | e] = a + be = g(e) + \phi \sigma^2 g'(e)^2.$$

Now the owner chooses e to maximise expected profit:

$$\max_e \mathbb{E}[\pi - w | e] = \max_e \left(e - g(e) - \phi \sigma^2 g'(e)^2 \right).$$

The FOC is

$$1 - g'(e) - 2\phi \sigma^2 g'(e)g''(e) = 0,$$

equivalently

$$g'(e) = \frac{1}{1 + 2\phi \sigma^2 g''(e)}.$$

Since $\phi > 0$, $\sigma^2 > 0$, and $g''(e) > 0$, we get

$$b^* = g'(e^*) \in (0, 1),$$

so incentives are not fully aligned (relative to the first-best $g'(e) = 1$).

4. **How do outputs change with ϕ or σ^2 ? (Comparative statics)** Define

$$H(e, \phi, \sigma^2) := g'(e)(1 + 2\phi\sigma^2g''(e)) - 1.$$

At the optimum, $H(e^*, \phi, \sigma^2) = 0$. Moreover,

$$\frac{\partial H}{\partial e} = g''(e)(1 + 2\phi\sigma^2g''(e)) + 2\phi\sigma^2g'(e)g'''(e),$$

which is positive under standard regularity assumptions (so the IFT applies). Also,

$$\frac{\partial H}{\partial \phi} = 2\sigma^2g'(e)g''(e) > 0, \quad \frac{\partial H}{\partial \sigma^2} = 2\phi g'(e)g''(e) > 0.$$

Hence,

$$\frac{\partial e^*}{\partial \phi} = -\frac{H_\phi}{H_e} < 0, \quad \frac{\partial e^*}{\partial \sigma^2} = -\frac{H_{\sigma^2}}{H_e} < 0.$$

Since $b^* = g'(e^*)$ and $g'' > 0$, b^* also decreases in ϕ and σ^2 .

Intuition. Higher risk aversion (ϕ) or higher output risk (σ^2) makes performance pay more costly (because it imposes risk on the manager). The owner reduces the slope b (risk exposure), which weakens incentives, so equilibrium effort/output falls.

In summary,

- The optimal linear compensation scheme when the agent is risk neutral is to sell the project, $b = 1$ and $a = \bar{u} - e^* + g(e^*)$ where $g'(e^*) = 1$.
- If the agent is risk averse but the effort is observable, then $b = 0$ and $a = g(e^*)$ where $g'(e^*) = 1$.
- If the agent is risk averse and effort is unobservable, then the optimal linear contract
 - Involves positive (but not fully aligned) incentives $b \in (0, 1)$
 - Induces inefficient level of effort: $\hat{e} < e^*$
 - The level of effort decreases with risk aversion or with the noise of the signal
- Extensions include multiple actions, additional signals, repeated interaction, multi-tasking etc.

Example 5.11 (PSet4 Q1). A risk neutral principal hires a risk averse agent to undertake a project. There are two possible effort levels that the agent can exert, $e = 0$ or $e = 1$ and three possible outcomes for the gross revenue to the principal, $\pi_1 < \pi_2 < \pi_3$. If $e = 0$, the probabilities of π_1, π_2, π_3 are $1/3, 1/3, 1/3$ respectively; if $e = 1$, the probabilities of π_1, π_2, π_3 are $1/4, 1/2, 1/4$ respectively.

The agent's utility from receiving a wage w and exerting effort e is $\sqrt{w} - e$ and the agent has a reservation utility of 10.

In the contract between the two parties, w cannot depend on e directly but can depend on the outcome π_k . Thus a contract will be of the form $\{w_1, w_2, w_3\}$ where w_k is the wage paid by the principal to the agent if the gross revenue is π_k .

Assume that the gross revenues are such that the principal wants to induce the agent to choose $e = 1$.

- (a) Establish that $w_1^* = w_3^*$ is the optimal contract.
(You may assume that the Kuhn-Tucker FOCs are necessary and sufficient for optimality. Also, there is no need to reproduce the general proof but you should establish the result in this particular case.)
Hence, or otherwise, provide a complete characterisation of the optimal contract.

Claim: $w_1^* = w_3^*$ in the optimal contract.

The principal minimises $\frac{1}{4}w_1 + \frac{1}{2}w_2 + \frac{1}{4}w_3$ subject to:

$$\frac{1}{4}\sqrt{w_1} + \frac{1}{2}\sqrt{w_2} + \frac{1}{4}\sqrt{w_3} - 1 \geq 10 \quad (\text{IR})$$

$$\frac{1}{4}\sqrt{w_1} + \frac{1}{2}\sqrt{w_2} + \frac{1}{4}\sqrt{w_3} - 1 \geq \frac{1}{3}\sqrt{w_1} + \frac{1}{3}\sqrt{w_2} + \frac{1}{3}\sqrt{w_3} \quad (\text{IC})$$

Let $(\lambda_1, \lambda_2, \lambda_3) = (\sqrt{w_1}, \sqrt{w_2}, \sqrt{w_3})$. The Lagrangian with multipliers (μ, κ) on (IR) and (IC) respectively is:

$$\mathcal{L} = \lambda_1^2 + 2\lambda_2^2 + \lambda_3^2 - \mu\left(\frac{1}{4}\lambda_1 + \frac{1}{2}\lambda_2 + \frac{1}{4}\lambda_3 - 11\right) - \kappa(2\lambda_2 - \lambda_1 - \lambda_3 - 12)$$

Since the objective is convex and constraints are linear, KKT FOCs are necessary and sufficient. The FOCs are:

$$\begin{aligned} 2\lambda_1 - \frac{\mu}{4} + \kappa &= 0 \implies \mu = 8\lambda_1 + 4\kappa \\ 4\lambda_2 - \frac{\mu}{2} - 2\kappa &= 0 \implies \mu = 8\lambda_2 - 4\kappa \\ 2\lambda_3 - \frac{\mu}{4} + \kappa &= 0 \implies \mu = 8\lambda_3 + 4\kappa \end{aligned}$$

Comparing the first and third equations gives $\lambda_1 = \lambda_3$, i.e. $w_1^* = w_3^*$. Since KKT FOCs are necessary and sufficient, this is the optimum. $\square$

Full characterisation. We consider all cases for which constraints bind.

Case 1: No constraint binds. $(\lambda_1, \lambda_2, \lambda_3) = (0, 0, 0)$, which violates (IR). $\times$

Case 2: Only (IR) binds ($\mu > 0, \kappa = 0$). The FOCs give $\lambda_1 = \lambda_2 = \lambda_3 = 11$. Checking (IC): $2(11) - 2(11) = 0 < 12$. $\times$

Case 3: Only (IC) binds ($\mu = 0, \kappa > 0$). One of $\lambda_1, \lambda_2, \lambda_3$ becomes negative. $\times$

Case 4: Both bind. Using $\lambda_1 = \lambda_3$, the binding constraints give:

$$2\lambda_1 + 2\lambda_2 = 44 \quad (\text{IR}), \quad 2\lambda_2 - 2\lambda_1 = 12 \quad (\text{IC})$$

Solving: $\lambda_1 = \lambda_3 = 8, \lambda_2 = 14$, so $(\lambda_1, \lambda_2, \lambda_3) = (8, 14, 8)$. Therefore:

$$(w_1^*, w_2^*, w_3^*) = (64, 196, 64)$$

- (b) (i) Under the contract $(w_1, w_2, w_3) = (64, 196, 64)$, when $\pi_k = \pi_3$ the agent receives wage 64. If the agent secretly destroys revenue to report π_2 instead, they receive 196. Since $196 > 64$, the agent has an incentive to destroy revenue whenever π_3 is realised in order to obtain the higher wage associated with π_2 .

- (ii) The new problem is:

$$\min_{w_1, w_2, w_3} \frac{1}{4}w_1 + \frac{1}{2}w_2 + \frac{1}{4}w_3$$

subject to (IR), (IC), and the no-destruction constraints $w_3 \geq w_2$ and $w_2 \geq w_1$. Writing $\lambda_k = \sqrt{w_k}$ as before, the Lagrangian with multipliers (μ, κ, ν) on (IR), (IC), and the no-destruction constraint $\lambda_2 \leq \lambda_3$ respectively is:

$$\mathcal{L} = \lambda_1^2 + 2\lambda_2^2 + \lambda_3^2 - \mu\left(\frac{1}{4}\lambda_1 + \frac{1}{2}\lambda_2 + \frac{1}{4}\lambda_3 - 11\right) - \kappa(2\lambda_2 - \lambda_1 - \lambda_3 - 12) - \nu(\lambda_3 - \lambda_2)$$

In the optimum, the cheapest way to prevent revenue destruction is to set $w_2 = w_3$, so we expect the no-destruction constraint to bind. The KKT FOCs are:

$$\begin{aligned} 2\lambda_1 - \frac{\mu}{4} + \kappa &= 0 \\ 4\lambda_2 - \frac{\mu}{2} - 2\kappa + \nu &= 0 \\ 2\lambda_3 - \frac{\mu}{4} + \kappa - \nu &= 0 \end{aligned}$$

with complementary slackness $\mu, \kappa, \nu \geq 0$ and:

$$\mu\left(\frac{1}{4}\lambda_1 + \frac{1}{2}\lambda_2 + \frac{1}{4}\lambda_3 - 11\right) = 0, \quad \kappa(2\lambda_2 - \lambda_1 - \lambda_3 - 12) = 0, \quad \nu(\lambda_3 - \lambda_2) = 0$$

Suppose no constraint binds. Then $(\lambda_1, \lambda_2, \lambda_3) = (0, 0, 0)$, violating (IR). $\times$

Suppose all constraints bind. Then $\lambda_2 = \lambda_3$. Intuitively, IR and IC must bind (otherwise the principal could reduce costs). With $\lambda_2 = \lambda_3$, the binding IR and IC give:

$$\frac{1}{4}\lambda_1 + \frac{3}{4}\lambda_2 = 11, \quad \lambda_2 - \lambda_1 = 12$$

From IC: $\lambda_1 = \lambda_2 - 12$. Substituting into IR: $\frac{1}{4}(\lambda_2 - 12) + \frac{3}{4}\lambda_2 = 11 \implies \lambda_2 = 14, \lambda_1 = 2$. Therefore $(\lambda_1, \lambda_2, \lambda_3) = (2, 14, 14)$ and:

$$(w_1^*, w_2^*, w_3^*) = (4, 196, 196)$$

(c) The expected wage costs given $e = e_H$ are:

- **Part (a):** $E[w \mid e_H] = \frac{1}{4}(64) + \frac{1}{2}(196) + \frac{1}{4}(64) = 130$. Agency cost = $130 - 121 = 9$.
- **Part (b):** $E[w \mid e_H] = \frac{1}{4}(4) + \frac{3}{4}(196) = 148$. Agency cost = $148 - 121 = 27$.

The agency cost in (b) is higher because the additional no-destruction constraint further restricts the set of feasible contracts, resulting in a less efficient outcome.

5.2 Hidden Information

Setting: A risk neutral principal hires a risk averse agent for a one time project. The agent exerts observable effort $e \geq 0$ and gross profits are deterministic: $\pi(e)$ with $\pi(0) = 0, \pi' > 0, \pi'' < 0$.

Information: After the contract is signed, the agent discovers her disutility of effort i.e. the agent's utility is

$$u(w, e, \theta) = v(w - g(e, \theta))$$

with $v' > 0, v'' \leq 0$. Here, θ is the state of nature that is realised after the contract is signed, and is privately observed by the agent. In other words, ex ante, when the contract is signed, there is symmetric information.

Assumption 5.12. The function g measures the cost of effort in monetary terms, with the following assumptions

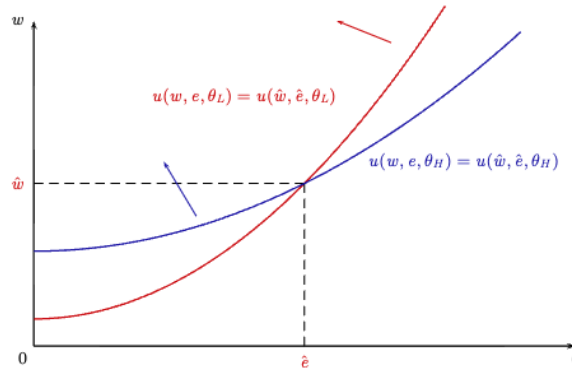
- No disutility if no effort: $g(0, \theta) = 0$, for all θ
- Increasing and convex cost function: $g_e(e, \theta) > 0, g_{ee}(e, \theta) > 0$ for all θ and $e > 0$ (i.e. the first and second partial derivatives wrt e)
- Single crossing property: $g_\theta(e, \theta) < 0, g_{e\theta}(e, \theta) < 0$ for all θ and $e > 0$

The agent is assumed to have ex ante reservation utility $\bar{u}$

For simplicity, focus on the case of two states:

$$\theta \in \{\theta_L, \theta_H\}; \theta_L < \theta_H; \lambda = \mathbb{P}[\theta = \theta_H]$$

We may illustrate the single crossing property in this case like so

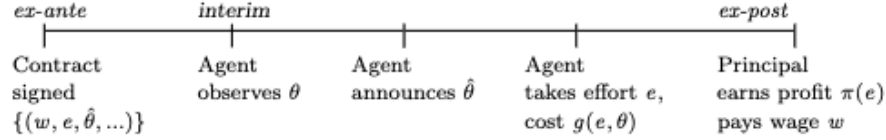


The contract specifies the wage w and the effort e and could be complex i.e. the details of the contract can depend on an announcement, $\hat{\theta}$, that the agent makes after observing θ .

The objectives of the contract are

- The risk neutral owner insures the manager against fluctuations in income
- Managerial effort should respond to true disutility of effort θ

An illustration of the timing/sequence of events:



First Best: state θ is observable

If θ is observable, a contract can specify (w, e) as a function of θ so

$$\{(w_H, e_H), (w_L, e_L)\}$$

The Principal's problem is hence

$$\begin{aligned} \max_{(w_L, e_L), (w_H, e_H)} & \lambda[\pi(e_H) - w_H] + (1 - \lambda)[\pi(e_L) - w_L] \\ \text{s.t.} & \\ & \lambda v(w_H - g(e_H, \theta_H)) + (1 - \lambda)v(w_L - g(e_L, \theta_L)) \geq \bar{u} \end{aligned}$$

To this end, we construct the Lagrangian

$$\mathcal{L}(w_L, e_L, w_H, e_H, \gamma) = \lambda[\pi(e_H) - w_H] + (1 - \lambda)[\pi(e_L) - w_L] + \gamma(\lambda v(w_H - g(e_H, \theta_H)) + (1 - \lambda)v(w_L - g(e_L, \theta_L)) - \bar{u})$$

where $\gamma \geq 0$ is the Kuhn-Tucker multiplier on the agent's ex-ante participation constraint.

The FOCs for the solution to this first best problem are

$$\begin{aligned} \lambda &= \gamma \lambda v'(w_H - g(e_H, \theta_H)) & [\text{FOC}_{w_H}] \\ (1 - \lambda) &= \gamma (1 - \lambda) v'(w_L - g(e_L, \theta_L)) & [\text{FOC}_{w_L}] \\ \lambda \pi'(e_H) &= \gamma \lambda v'(w_H - g(e_H, \theta_H)) g_e(e_H, \theta_H) & [\text{FOC}_{e_H}] \\ (1 - \lambda) \pi'(e_L) &= \gamma (1 - \lambda) v'(w_L - g(e_L, \theta_L)) g_e(e_L, \theta_L) & [\text{FOC}_{e_L}] \end{aligned}$$

and the complementarity slackness conditions:

$$\begin{aligned} \gamma &\geq 0, \lambda v(w_H - g(e_H, \theta_H)) + (1 - \lambda)v(w_L - g(e_L, \theta_L)) \geq \bar{u}, \\ \gamma(\lambda v(w_H - g(e_H, \theta_H)) + (1 - \lambda)v(w_L - g(e_L, \theta_L)) - \bar{u}) &= 0 \end{aligned}$$

The first two FOCs imply that

$$\gamma v'(w_H^* - g(e_H^*, \theta_H)) = \gamma v'(w_L^* - g(e_L^*, \theta_L)) = -1$$

In particular we have

- $\gamma > 0$ so the Participation Constraint binds
- so v' is equalised across states and since $v' > 0$, we have optimal risk sharing:

$$w_H^* - g(e_H^*, \theta_H) = w_L^* - g(e_L^*, \theta_L)$$

- so v is equalised across states and

$$v(w_H^* - g(e_H^*, \theta_H)) = v(w_L^* - g(e_L^*, \theta_L)) = \bar{u}$$

Considering effort levels in the two states:

Substituting $\gamma v' = 1$ into the last two FOCs yields

$$\pi'(e_H^*) = g_e(e_H^*, \theta_H)$$

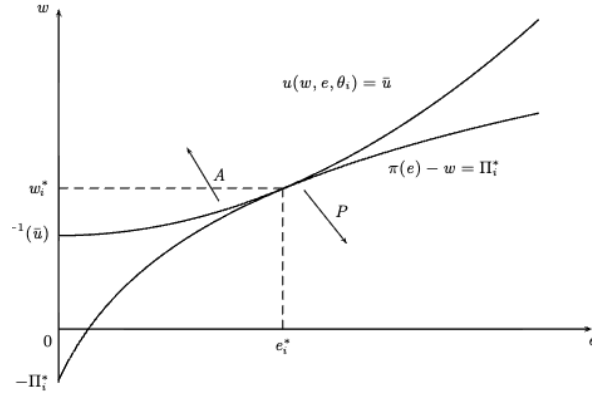
$$\pi'(e_L^*) = g_e(e_L^*, \theta_L)$$

which shows that optimal effort levels equal marginal benefits to marginal costs, implying efficient effort.

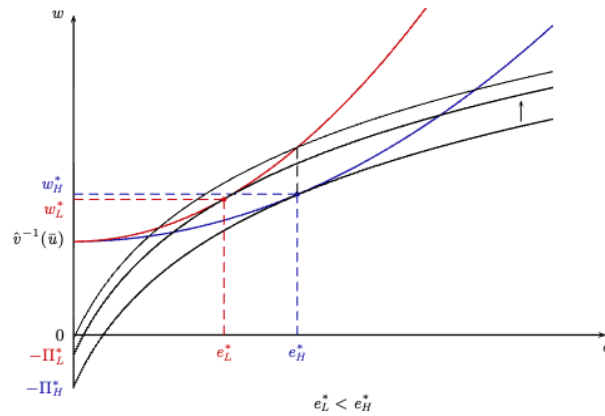
Note that $\pi'(e_i^*) = g_e(e_i^*, \theta_i)$ and $v(w_i^* - g(e_i^*, \theta_i)) = \bar{u}$ uniquely determine the contract (w_i^*, e_i^*) . The profits achieved by the owner are

$$\Pi_i^* = \pi(e_i^*) - w_i^* = \pi(e_i^*) - v^{-1}(\bar{u}) + g(e_i^*, \theta_i)$$

We can illustrate the principal's/owner's profit curves and the agent's utility as



Comparing contracts between e_L^* and e_H^* where $e_L^* < e_H^*$



First Best: state θ is unobservable; the agent is risk neutral

Proposition 5.13. If the agent is risk neutral, the principal can achieve its full information profits by selling the project to the agent even when θ is unobservable

Take $v(w) \equiv w$ (since agent is risk neutral). If the agent keeps all the profits and so the utility function is

$$u(e_i) = \pi(e_i) - g(e_i, \theta_i) - w(e)$$

, she chooses effort to maximise (disregarding wage since it is an upfront transfer to the principal)

$$\pi(e) - g(e, \theta_L) \text{ or } \pi(e) - g(e, \theta_H)$$

Hence, $\pi'(e_i^*) = g_e(e_i^*, \theta_i)$ as before. The agent's expected net (of outside option) surplus ex ante is

$$(1 - \lambda)[\pi(e_L^*) - g(e_L^*, \theta_L)] + \lambda[\pi(e_H^*) - g(e_H^*, \theta_H)] - \bar{u}$$

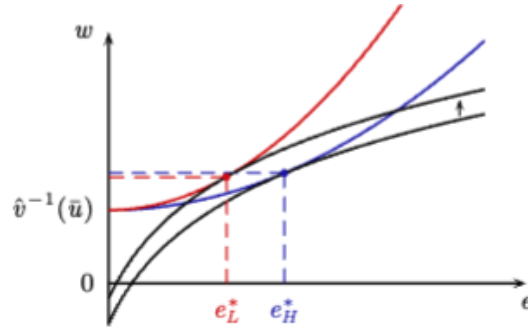
The above is ALSO what the principal charges for the project i.e. the agent's entire ex ante net surplus (so that the agent participates since their PC binds in this case).

So, in effect, the agent's wage is the profits (which they keep) minus what the principal charges:

$$w(e) \equiv \pi(e) - [(1 - \lambda)[\pi(e_L^*) - g(e_L^*, \theta_L)] + \lambda[\pi(e_H^*) - g(e_H^*, \theta_H)] - \bar{u}]$$

Second Best: state θ is unobservable; the agent is risk averse

When θ is private information, the fundamental problem is that there is a conflict between risk sharing and effort elicitation. The principal could ask the manager to report the state, but the first best contract is not incentive compatible: in state θ_H , the agent would pretend that θ_L had occurred



What contracts are possible and how do we restrict our search space for the possible contracts? Here is a fundamental result

Theorem 5.14 (Revelation Principle). Given a mechanism M for which there exists an equilibrium strategy profile $\mathbf{s}$, we can construct an equivalent direct mechanism M^* for which truth-telling is likewise an equilibrium.

Proof. Given a (possibly indirect) mechanism M , with an equilibrium $\mathbf{s}$, we construct a direct mechanism M^* with the corresponding truth-telling equilibrium as follows:

- Elicit all players' types $\theta_1, \theta_2, \dots, \theta_n$
- Simulate M by performing i 's equilibrium action on his behalf, assuming i 's type is θ_i
- Return the outcome produced by M

We can think of the construction as a machine that first asks all players for their types and then runs the equilibrium strategy on their behalf. Each agent reports a (possibly false) type θ_i to the direct mechanism M^* which simulates $s_i(\theta_i)$. The agents are incentivised to report their true types because by doing so the outcome of M^* is the same as the outcome of M which by assumption, is an equilibrium.

If player i lies to the robot (and if they are the only one lying), the robot will run everyone else's equilibrium strategy based on their true types, except for player i 's. The outcome will be the exact same outcome as running M , assuming i deviates from its equilibrium strategy. But by assumption, this deviation is not a best response in M and so it is not in i 's best interest in M^* ■

Definition 5.15 (Direct Mechanism). A direct mechanism is one in which the space of possible actions is equal to the space of possible types

By equivalence of two mechanisms M and M' , given two corresponding strategies $\mathbf{s}$ and $\mathbf{s}'$, we mean that the same type profile θ yield the same equilibrium outcome i.e. $g(\mathbf{s}(\theta)) = g'(\mathbf{s}'(\theta))$ for all θ . If M' is assumed to be a direct mechanism, then $g(\mathbf{s}(\theta)) = g'(\theta)$.

In our setting, this is equivalent as translating a possibly very complex contract:

- in state θ_H , the agent exerts effort e_H and results in wage $w(e_H)$
- in state θ_L , the agent exerts effort e_L , resulting in wage $w(e_L)$

The agent makes these choices because

$$v(w(e_H) - g(e_H, \theta_H)) \geq v(w(e) - g(e, \theta_H))$$

for any other choice of e in state θ_H and similarly in state θ_L . Now, we may translate the complex contract into a simple one where

- The agent reports if he is in state θ_H or θ_L
- If the agent reports θ_H , then he will exert e_H and receive wage $w(e_H)$
- If the agent reports θ_L , the agent will exert e_L and receive wage $w(e_L)$

By the revelation principle, if the agent found the original complex contract acceptable, the agent finds this alternative acceptable. Moreover, it will be optimal for the agent to report the state θ truthfully.

The principal's expected net profit from the alternative contracts is the same as from the original one

$$\pi = \lambda[\pi(e_H) - w(e_H)] + (1 - \lambda)[\pi(e_L) - w(e_L)]$$

The upshot of the revelation principle is that we can *restrict the type of contract offered to a direct mechanism* that induces truth-telling without loss of generality. So, we restrict the attention to the set of contracts of the form

1. After the interim stage, the agent makes an announcement $\hat{\theta}$ about the state
2. The contract specifies an outcome $(w(\hat{\theta}), e(\hat{\theta}))$ for each possible $\hat{\theta}$
3. In every state θ , truth-telling is optimal for the agent (incentive compatible)

Definition 5.16 (Incentive Compatibility in Hidden-Information). In Hidden Information Problems, the agent's incentive compatibility (truth-telling) constraints are

$$\begin{aligned} w_H - g(e_H, \theta_H) &\geq w_L - g(e_L, \theta_H) \\ w_L - g(e_L, \theta_L) &\geq w_H - g(e_H, \theta_L) \end{aligned}$$

For simplicity, assume that the agent is infinite risk averse. In a state-contingent diagram, this corresponds to Leontief indifference curves (i.e. $\min\{x, y\}$). The agent's ex-ante expected utility is her lowest ex-post utility.

The ex-ante participation constraint becomes:

$$\min\{v(w_H - g(e_H, \theta_H)), v(w_L - g(e_L, \theta_L))\} \geq \bar{u}$$

or

$$\begin{aligned} v(w_H - g(e_H, \theta_H)) &\geq \bar{u} \\ v(w_L - g(e_L, \theta_L)) &\geq \bar{u} \end{aligned}$$

So the principal's problem can be written as

$$\begin{aligned}
\max_{w_H, e_H, w_L, e_L} \quad & \lambda[\pi(e_H) - w_H] + (1 - \lambda)[\pi(e_L) - w_L] \quad \text{s.t.} \\
& w_H - g(e_H, \theta_H) \geq v^{-1}(\bar{u}) \quad (\text{PC}_H) \\
& w_L - g(e_L, \theta_L) \geq v^{-1}(\bar{u}) \quad (\text{PC}_L) \\
& w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H) \quad (\text{IC}_H) \\
& w_L - g(e_L, \theta_L) \geq w_H - g(e_H, \theta_L) \quad (\text{IC}_L)
\end{aligned}$$

How do we reduce the number of constraints?

Lemma 5.17. We can ignore the PC_H constraint: $w_H - g(e_H, \theta_H) \geq v^{-1}(\bar{u})$

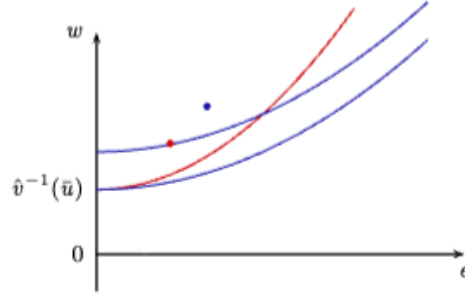
Proof. If the constraints (PC_L) and (IC_H) are satisfied, it must be that

$$w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H) \geq w_L - g(e_L, \theta_L) \geq v^{-1}(\bar{u})$$

The second inequality holds because we assume that a high type has lower effort cost for any given effort i.e.

$$g(e, \theta_H) \leq g(e, \theta_L) \quad \forall e$$

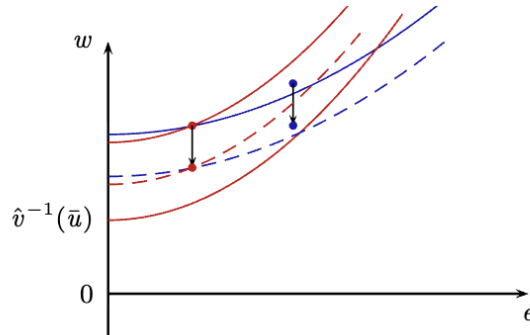
We can see this as follows



Graphically, in (e, w) -space the indifference curve of the θ_H -type through any point $(\hat{e}, \hat{w})$ is *flatter* than that of the θ_L -type through the same point (the single-crossing property). This means that the θ_H -type requires a smaller wage increase to compensate for a given increase in effort (since lower effort cost), so any (w_L, e_L) bundle that satisfies the θ_L -type's participation constraint will leave the θ_H -type strictly above her reservation utility. Hence, PC_H is automatically satisfied whenever PC_L and IC_H are, and can be ignored. ■

Lemma 5.18. The PC_L constraint binds: $w_L - g(e_L, \theta_L) = v^{-1}(\bar{u})$

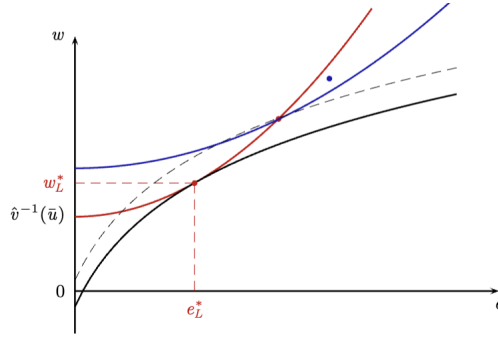
Proof. Suppose for contradiction that $w_L - g(e_L, \theta_L) > v^{-1}(\bar{u})$ at the optimal wage. Consider the modified wages $\hat{w}_L = w_L - \varepsilon$ and $\hat{w}_H = w_H - \varepsilon$ for some small $\varepsilon > 0$. The incentive constraints IC_H and IC_L are unaffected since the same ε is subtracted from both sides of each constraint. For ε sufficiently small, PC_L remains satisfied. This reduces the principal's expected wage bill by ε , contradicting optimality.



Graphically, the contract point (w_L, e_L) lies strictly above the indifference curve $u(w, e, \theta_L) = \bar{u}$ in (e, w) -space. We can shift both contract points (w_L, e_L) and (w_H, e_H) downward by ε — moving them along vertical lines in the diagram — preserving the relative positions of the two points and hence both IC constraints, while reducing the wage bill. We continue until (w_L, e_L) reaches the θ_L -type's reservation indifference curve, at which point PC_L holds with equality. ■

Lemma 5.19. In any optimal contract, $\hat{e}_L \leq e_L^*$

Proof. Recall that e_L^* is the first-best effort level for type θ_L , defined by $\pi'(e_L^*) = g_e(e_L^*, \theta_L)$, and that at the first-best contract point (w_L^*, e_L^*) , the isoprofit curve of the principal is tangent to the indifference curve of the θ_L -type. Now suppose the contract offers $(\hat{w}_L, \hat{e}_L)$ with $\hat{e}_L > e_L^*$. Since PC_L binds (Lemma 2), the point $(\hat{w}_L, \hat{e}_L)$ lies on the θ_L -type's reservation indifference curve. But for $e > e_L^*$, the isoprofit curve of the principal lies *below* this indifference curve, meaning the principal earns less than at (w_L^*, e_L^*) .



Graphically, in (e, w) -space the reservation indifference curve of the θ_L -type and the principal's isoprofit curves are tangent at e_L^* ; to the right of this tangency point, the indifference curve rises faster than any isoprofit curve, so moving $\hat{e}_L$ above e_L^* along the reservation indifference curve reduces the principal's payoff. Hence any contract with $\hat{e}_L > e_L^*$ is dominated, and in any optimal contract $\hat{e}_L \leq e_L^*$. ■

Lemma 5.20. In any optimal contract $\hat{e}_H = e_H^*$

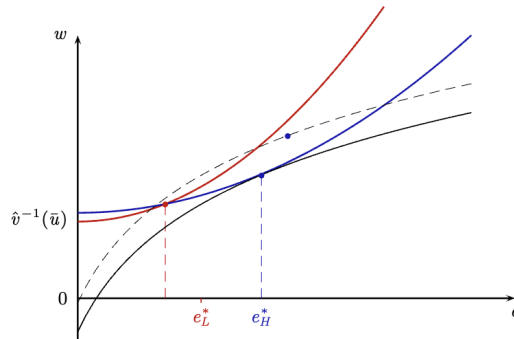
Proof. We show that the effort of the θ_H -type is undistorted. Consider any contract $(\hat{w}_H, \hat{e}_H)$ for the high type. Since IC_H binds (it must bind in any optimal contract, as slackness would allow the principal to raise $\hat{w}_H$ and extract more surplus), we have

$$\hat{w}_H - g(\hat{e}_H, \theta_H) = \hat{w}_L - g(\hat{e}_L, \theta_H)$$

so the wage $\hat{w}_H$ is pinned down by $\hat{e}_H$ and the low-type contract. The principal's problem with respect to $\hat{e}_H$ is therefore to maximise

$$\pi(\hat{e}_H) - \hat{w}_H = \pi(\hat{e}_H) - g(\hat{e}_H, \theta_H) - [\hat{w}_L - g(\hat{e}_L, \theta_H)]$$

Since the bracketed term does not depend on $\hat{e}_H$, this is maximised by choosing $\hat{e}_H$ to maximise $\pi(\hat{e}_H) - g(\hat{e}_H, \theta_H)$, which gives the first-order condition $\pi'(\hat{e}_H) = g_e(\hat{e}_H, \theta_H)$, exactly the condition defining e_H^* .



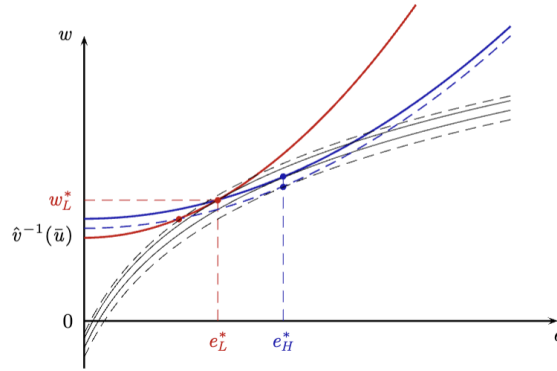
Graphically, varying $\hat{e}_H$ while keeping IC_H binding traces out the θ_H -type's reservation indifference curve shifted up by $\hat{w}_L - g(\hat{e}_L, \theta_H)$; the principal's isoprofit is tangent to this curve at e_H^* , and there is no binding IC or PC constraint on the high type that would force a distortion away from this tangency. Hence $\hat{e}_H = e_H^*$. ■

Lemma 5.21. In any optimal contract, $\hat{e}_L < e_L^*$

Proof. By Lemma 5.19 we know $\hat{e}_L \leq e_L^*$. It remains to rule out equality. Suppose for contradiction that $\hat{e}_L = e_L^*$. Then $(\hat{w}_L, \hat{e}_L) = (w_L^*, e_L^*)$, the first-best contract for the θ_L -type. Since IC_H binds,

$$\hat{w}_H = \hat{w}_L + g(\hat{e}_H, \theta_H) - g(\hat{e}_L, \theta_H) = w_L^* + g(e_H^*, \theta_H) - g(e_L^*, \theta_H)$$

Now consider reducing $\hat{e}_L$ slightly below e_L^* while keeping PC_L binding, so $\hat{w}_L = v^{-1}(\bar{u}) + g(\hat{e}_L, \theta_L)$ decreases (since $g_{e\theta} < 0$ implies the cost of effort is lower for the high type, so reducing effort reduces the required wage). This allows $\hat{w}_H$ to decrease as well via IC_H , reducing the principal's expected wage bill. The loss in profit from the θ_L -state is second-order (since e_L^* is an interior optimum of the first-best problem), while the gain from paying a lower $\hat{w}_H$ in the θ_H -state is first-order. Hence $\hat{e}_L = e_L^*$ cannot be optimal, and in any optimal contract $\hat{e}_L < e_L^*$: there is a *distortion at the bottom*.



Graphically, starting from the first-best allocation for both types, a small leftward movement of the θ_L -type's contract point along her reservation indifference curve relaxes the IC_H constraint, allowing the θ_H -type's contract point to shift downward and saving the principal money at a first-order rate. ■

In sum, the optimal contract $(\hat{w}_H, \hat{e}_H), (\hat{w}_L, \hat{e}_L)$ with unobservable θ and infinitely risk averse agent satisfies

- $\hat{e}_H = e_H^*$: No distortion at the top
- $\hat{e}_L < e_L^*$: Distortion at the bottom
- Imperfect risk sharing: $u_H > \bar{u} = u_L$
- Principal loses

Remark. We assume that the agent is infinitely risk averse and hence the ex-ante participation constraint becomes:

$$\begin{aligned} w_H - g(e_H, \theta_H) &\geq v^{-1}(\bar{u}) \\ w_L - g(e_L, \theta_L) &\geq v^{-1}(\bar{u}) \end{aligned}$$

These are the same as the participation constraints of a (finitely risk averse) agent who signs a contract only after she observes θ i.e. at the interim stage. Hence an infinitely risk averse agent's ex-ante participation constraints just are the finitely risk averse agent's interim participation constraint

Example 5.22 (Monopolistic Screening). Problem: There are two types of rail traveller: L and H ; $\lambda =$ fraction of H -types.

- H -types utility from travelling: $u_H(m, q) + \theta_H q$
- L -types utility from travelling: $u_L(m, q) = m + \theta_L q$

where m is the net money income, q is the quality of travel, and θ is a taste parameter with $\theta_H > \theta_L > 0$. Assume $\lambda < \theta_L/\theta_H$. Suppose that the cost per passenger of providing quality q is $q^2/2k$. The monopolist charges $p(q)$ when offering quality q .

First Best: If θ were commonly known, the monopolist would solve

$$\max_{q_i, p_i} p_i - q_i^2/2k \text{ s.t. } \theta_i q_i - p_i \geq 0$$

where the constraint is the participation constraint. Hence, the monopoly would choose qualities:

$$q_H^* = k\theta_H, q_L^* = k\theta_L$$

and charge

$$p_H^* = \theta_H q_H^*, p_L^* = \theta_L q_L^*$$

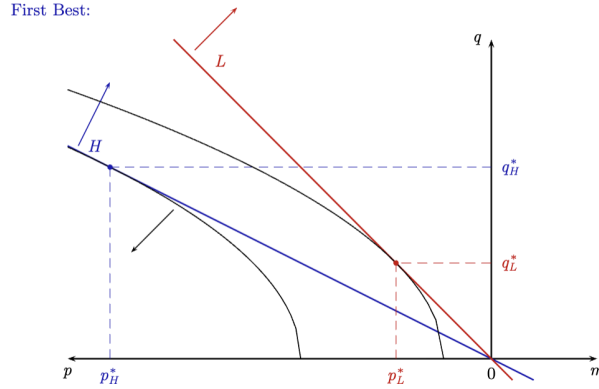
This can be derived via the lagrangian (both PC will hold with equality)

$$\mathcal{L}(p_i, q_i, \lambda) = p_i - q_i^2/2k - \lambda(\theta_i q_i - p_i)$$

with FOCs

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial p_i} &= 1 + \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial q_i} &= -q_i/k - \lambda\theta = 0 \end{aligned}$$

Solving gives us the solutions since $\lambda = -1$. Graphically, the first best is



Indifference curves. Agent i 's utility is $u_i = m + \theta_i q$, so indifference curves are straight lines with slope $-1/\theta_i$ in this diagram. Since $\theta_H > \theta_L$, the H -type's indifference curves (blue, labelled H) are *steeper* than the L -type's (red, labelled L): the high type is willing to sacrifice more money income per unit of quality, reflecting a higher willingness to pay. Higher indifference curves (further up and to the left) represent higher utility.

Isoprofit curves. The principal's profit from type i is $\pi_i = p_i - q_i^2/2k = (y - m_i) - q_i^2/2k$, so along an isoprofit curve, $m_i = y - q_i^2/2k - \pi_i$. These are concave, downward-curving arcs. The black arrows indicate the direction of increasing profit for the principal (down and to the right, since lower m means higher price extracted).

Optimal contracts. Since the PC constraint binds for each type at the optimum, each agent is held to their reservation utility $u = 0$, so they lie on their reservation indifference curve: the straight line through

the origin with slope $-1/\theta_i$. The optimal quality q_i^* is found where the principal's isoprofit curve is *tangent* to this reservation indifference curve.

Second Best: Suppose that θ is private information. Then the monopolist's problem is

$$\max_{(q_L, p_L), (q_H, p_H)} \lambda[p_H - q_H^2/2k] + (1 - \lambda)[p_L - q_L^2/2k]$$

such that the PCs hold:

$$\begin{aligned} \theta_H q_H - p_H &\geq 0 & (PC_H) \\ \theta_L q_L - p_L &\geq 0 & (PC_L) \end{aligned}$$

and the IC (truth-telling) constraints hold:

$$\begin{aligned} \theta_H q_H - p_H &\geq \theta_H q_L - p_L & (IC_H) \\ \theta_L q_L - p_L &\geq \theta_L q_H - p_H & (IC_L) \end{aligned}$$

Note that (IC_H) and (PC_L) imply (PC_H) hold as strict inequality when $q_L > 0$. Now, we proceed our analysis of the constraints to simplify

- Fix the difference between p_H and p_L . If (IC_H) and (IC_L) hold, then they still do with higher p_H and p_L . So increase p_L (and p_H) until (PC_L) binds.
Interpretation: L -types are indifferent between travelling and not, and given that, H -types definitely want to travel
- Fix p_L . If (IC_L) holds, then it will still do so with higher p_H . So increase p_H until (IC_H) binds.
Interpretation: H -types are indifferent between travelling Business and Tourist Class, and given that, L -types will definitely want to travel Tourist rather than Business

Therefore (PC_L) and (IC_H) bind with equality and we obtain $p_L = \theta_L q_L$ and $p_H = \theta_H q_H - (\theta_H - \theta_L)q_L$. Substitute these into the monopolist's maximisation problem gives

$$\begin{aligned} \max_{q_H, q_L} \lambda[\theta_H q_H - (\theta_H - \theta_L)q_L - q_H^2/2k] + (1 - \lambda)[\theta_L q_L - q_L^2/2k] \\ \text{s.t. } \theta_H q_H - p_H &\geq 0 & (PC_H) \\ \theta_L q_L - p_L &\geq \theta_L q_H - p_H & (IC_L) \end{aligned}$$

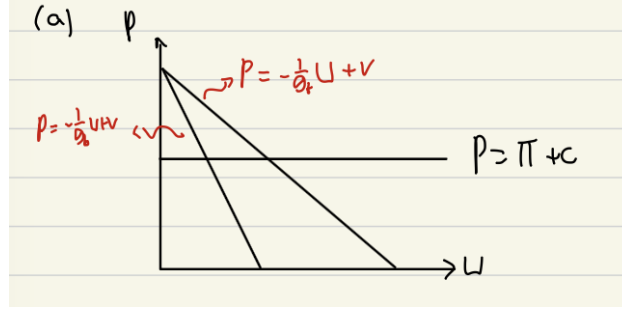
We know that (PC_H) does not bind if $q_L > 0$. Suppose none of the constraints bind, then the FOCs are

$$\begin{aligned} \lambda(\theta_H - q_H/k) &= 0 \\ -\lambda(\theta_H - \theta_L) + (1 - \lambda)(\theta_L - q_L/k) &= 0 \end{aligned}$$

Solving gives

$$\begin{aligned} \hat{q}_H &= k\theta_H = q_H^* \\ \hat{q}_L &= k\theta_L - k(\theta_H - \theta_L)\lambda/(1 - \lambda) < q_L^* \\ \hat{p}_H &= \theta_H \hat{q}_H - (\theta_H - \theta_L)\hat{q}_L < p_H^* \\ \hat{p}_L &= \theta_L \hat{q}_L < p_L^* \end{aligned}$$

Note that $\lambda < \theta_L/\theta_H$ ensures that $\hat{q}_L > 0$ and thus (PC_H) is slack. Also, $\hat{p}_H - \hat{p}_L = \theta_H(\hat{q}_H - \hat{q}_L)$ and so (IC_L) does indeed hold as a strict inequality. As $\lambda/(1 - \lambda)$ rises, the distortion on the L -type product gets larger since $\hat{q}_L$ becomes smaller i.e. even less than q_L^* . In turn, $\hat{p}_L$ also falls. Intuitively, it means more H -types and so the monopolist cares about attracting high types more and so worsens the quality of L -type product to make it less attractive to H -types. Graphically, we have that the H -types get the optimum (i.e. tangency) while L -types get a distorted contract (i.e. no tangency)



The airline faces

$$\max_{P_b, P_t} \lambda [P_t - c] + (1 - \lambda) [P_b - c]$$

with participation constraints

$$v - \theta_b P_b - W_b \geq 0$$

$$v - \theta_t P_t - W_t \geq 0$$

and Incentive Compatibility Constraints

$$v - \theta_b P_b - W_b \geq v - \theta_b P_t - W_t$$

$$v - \theta_t P_t - W_t \geq v - \theta_t P_b - W_b$$

and $P_b, P_t \geq 0$. The problem becomes

$$\max \quad \lambda (P_t - c) + (1 - \lambda) (P_b - c)$$

$$v - \theta_b P_b - W_b \geq 0 \quad (PC_b)$$

$$v - \theta_t P_t - W_t \geq 0 \quad (PC_t)$$

$$\theta_b (P_t - P_b) \geq W_b - W_t \quad (IC_b)$$

$$\theta_t (P_b - P_t) \geq W_t - W_b \quad (IC_t)$$

$$P_b, P_t \geq 0$$

- (b) Show that in the optimal solution, tourists are indifferent between travelling and not travelling. Fix the difference between P_t and P_b . If IC_b and IC_t hold, they hold at higher P_b and P_t . So increase both until PC_t binds s.t. $v - \theta_t P_t - W_t = 0$ since $\theta_t > \theta_b > 0$. So the tourist is indifferent between buying and not buying.
- (c) Show that in the optimal solution, businessmen never buy their ticket in advance, and are indifferent between doing this and buying when tourists buy. Fix P_t . If IC_t holds, then it will do so at higher P_b . So, increase P_b until IC_b binds such that

$$\theta_t (P_t - P_b) = W_b - W_t$$

Furthermore, if $W_b > 0$, the businessman can reduce W_b until 0 to increase his utility. Therefore, $W_b = 0$ i.e. $\theta_t (P_t - P_b) = -W_t$

- (d) Describe fully the optimal price-discrimination scheme. How does it depend on θ_b/θ_t and λ ? [Hint: There is a cut-off value for θ_b/θ_t in terms of λ on one side of which the airline prefers pooling to separating and *vice versa*.]

The new problem is

$$\begin{aligned} \max & \lambda(P_t - c) + (1 - \lambda)(P_b - c) \text{ s.t.} \\ & v - \theta_b P_b \geq 0 \\ & v - \theta_t P_t - W_t = 0 \\ & \theta_b(P_t - P_b) = -W_t \\ & \theta_t(P_b - P_t) \geq W_t - W_b \\ & P_b, P_t, W_t \geq 0 \end{aligned}$$

Manipulating give

$$P_t = \frac{\theta_t P_t - v}{\theta_b} + P_b$$

and

$$\left(1 - \frac{\theta_t}{\theta_b}\right) P_t = P_b$$

Hence, we have

$$\max \lambda(P_t - c) + (1 - \lambda) \left[\left(1 - \frac{\theta_t}{\theta_b}\right) P_t + \frac{v}{\theta_b} - c \right]$$

such that

$$v - \theta_b P_b \geq 0, \theta_t(P_b - P_t) \geq W_t, W_t, P_b, P_t \geq 0$$

Notice that

$$v - \theta_b \left(1 - \frac{\theta_t}{\theta_b} P_t + \frac{v}{\theta_b}\right) = (\theta_t - \theta_b) P_t \geq 0 \iff P_t \geq 0$$

and so the constraint is always satisfied. Furthermore, $v - \theta_t P_t - W_t \geq v - \theta_t P_b - W_b$ and since $v - \theta_t P_t - W_t = 0$ and $W_b = 0$. So $0 \geq v - \theta_t P_b$. Hence, $P_b \geq \frac{v}{\theta_t}$ which is always satisfied as

$$P_b = \frac{v}{\theta_t} + \left(1 - \frac{\theta_t}{\theta_b}\right) P_t$$

where the second term is less than 0 since $\theta_t > \theta_b$. Only constraint is

$$W_t \geq 0 \implies v - \theta_t P_t \geq 0$$

Hence

$$0 \leq P_t \leq \frac{v}{\theta_t}$$

Since $\pi(P_t) = \lambda(P_t - c) + (1 - \lambda) \left[\left(1 - \frac{\theta_t}{\theta_b}\right) P_t + \frac{v}{\theta_b} - c \right] = \left[\lambda + (1 - \lambda) \left(1 - \frac{\theta_t}{\theta_b}\right) \right] P_t + (1 - \lambda) \frac{v}{\theta_b} - c$, it is linear in P_t and maximising $\pi(P_t)$ depends on coefficient. Hence

- If $\left[\lambda + (1 - \lambda) \left(1 - \frac{\theta_t}{\theta_b}\right) \right] > 0$, then $P_t^* = \frac{v}{\theta_t}$ implies $W_t = 0$ and so $P_b^* = \frac{v}{\theta_t}$ i.e. pooling. Then $1 > (1 - \lambda) \frac{\theta_t}{\theta_b}$ which implies $\frac{\theta_b}{\theta_t} > 1 - \lambda$ and hence $(P_t^*, W_t^*) = (P_b^*, W_b^*) = (\frac{v}{\theta_t}, 0)$
- If $\left[\lambda + (1 - \lambda) \left(1 - \frac{\theta_t}{\theta_b}\right) \right] < 0$, then $P_t^* = 0 \implies W_t = v$ and $P_b^* = \frac{v}{\theta_b}$ i.e. separating. So the optimal contract is $(P_b^*, W_b^*) = (\frac{v}{\theta_b}, 0)$ and $(P_t^*, W_t^*) = (0, v)$. Then, $1 < (1 - \lambda) \frac{\theta_t}{\theta_b} \implies \frac{\theta_b}{\theta_t} < 1 - \lambda$.
- If $\left[\lambda + (1 - \lambda) \left(1 - \frac{\theta_t}{\theta_b}\right) \right] = 0$, then $P_t \in [0, \frac{v}{\theta_t}]$

(e) What should the airline do if c is higher than v/θ_b ? What should it do if c is between v/θ_t and v/θ_b ? For c below v/θ_t we might, in principle, see all passengers being served. However, even when costs are so low, find a condition in terms of c , v , θ_b , θ_t and λ under which the airline will not serve tourists.

- If $c > \frac{v}{\theta_b}$, even the best business ticket $(\frac{v}{\theta_b}, 0)$ results in negative profit and so Tourist ticket is worse at $\frac{v}{\theta_t} - c < 0$ since $\theta_t > \theta_b$ and so they will serve no one

- If $\frac{v}{\theta_t} < c < \frac{v}{\theta_b}$, then tourists can never be charged enough since $\frac{v}{\theta_t} - c < 0$ and so serve businessman only
- If $c < \frac{v}{\theta_t}$, under pooling, profit is

$$\pi_{\text{pool}} = \frac{v}{\theta_t} - c$$

and if serve businessman only by offering $(\frac{v}{\theta_b}, 0)$ we yield

$$\pi_B = (1 - \lambda) \left(\frac{v}{\theta_b - c} \right)$$

Hence $\pi_B \geq \pi_{\text{pool}}$ iff

$$(1 - \lambda) \left(\frac{v}{\theta_b} - c \right) \geq \left(\frac{v}{\theta_t} - c \right) \implies c \geq \frac{\frac{v}{\theta_t} - (1 - \lambda) \frac{v}{\theta_b}}{\lambda}$$

Rothschild-Stiglitz Model

See separate notes made by Claude Opus :)

6 Decision Theory

6.1 Basic Decision Theory: Preferences and Utility

We typically take preferences to be ordinal i.e. only tells us which options are better and not by how much

Binary Relations

Definition 6.1 (Binary Relation). A binary relation from a set A to set B , $R \subseteq A \times B$, is a set of ordered pairs (a, b) such that $a \in A$ and $b \in B$.

$$R = \{(a, b) : a \in A \text{ and } b \in B\}$$

Remark. In our setting, our binary relation is usually over the Cartesian product of the set of alternatives X with itself i.e X^2 . We usually denote our preference relation as $\succsim$

Definition 6.2 (Common Properties of Binary Relations). We have the following common properties

- Reflexivity: R is reflexive iff for every $x \in X$, $(x, x) \in R$
- Connectedness: R is connected iff for every $x, y \in X$ such that $x \neq y$, either $(x, y) \in R$ or $(y, x) \in R$ or both
- Completeness: R is complete iff for every $x, y \in X$, either $(x, y) \in R$ or $(y, x) \in R$ or both
- Symmetry: R is symmetric iff for every $x, y \in R$: if $(x, y) \in R$, then $(y, x) \in R$
- Transitivity: R is transitive iff for every $x, y, z \in X$: if $(x, y) \in R$, $(y, z) \in R$, then $(x, z) \in R$
- Anti-symmetry: R is antisymmetric iff for every $x, y \in X$: if $(x, y) \in R$ and $(y, x) \in R$, then $x = y$
- Asymmetric: R is asymmetric iff for all $x, y \in R$: if $(x, y) \in R$, then $(y, x) \notin R$
- Equivalence Relation: R is an equivalence relation iff it is reflexive, transitive and symmetric

Proposition 6.3. Consider a binary relation R over X . Then,

- If R is complete, it is reflexive
- If R is connected and reflexive, then it is complete

Definition 6.4 (Rational Preferences). The binary relation $\succsim$ is rational iff it is complete and transitive

Proposition 6.5. The preference relation of $\succsim_M$ (i.e. the majority prefers...to...) is not rational

Proof. Consider

$$\begin{aligned}\text{Ann: } & x \succ_A y \succ_B z \\ \text{Bob: } & z \succ_B x \succ_C y \\ \text{Charlie: } & y \succ_C z \succ_C x\end{aligned}$$

Then, the majority voting preference generated is

$$x \succ_M y, y \succ_M z, z \succ_M x$$

which is clearly not transitive since $x \not\succ_M z$ ■

Definition 6.6 (Less useful properties). We have the following further properties of a binary relation

- Irreflexivity: R is irreflexive iff for all $x \in X$, $(x, x) \notin R$
- Negative Transitivity: R is negative transitive iff for all $x, y, z \in X$: if $(x, y) \notin R$ and $(y, z) \notin R$, then $(x, z) \notin R$

Definition 6.7 (Strict Preference and Indifference). Given a rational preference $\succsim$, the strict part of $\succsim$ is the binary relation $\succ$ defined by

$$x \succ y \iff x \succsim y \text{ and not } y \succsim x$$

The indifference part of $\succsim$ is the binary relation $\sim$ defined by

$$x \sim y \text{ whenever } x \succsim y \text{ and } y \succsim x$$

Proposition 6.8. Given a rational preference $\succsim$, its strict part $\succ$ is (i) always transitive, (ii) may or may not be complete, (iii) always irreflexive, (iv) always asymmetric. Its indifference part $\sim$ is (i) reflexive and (ii) symmetric.

The interpretation of rational preferences is that the agent must compare every pair, is independent of how the preference is presented and does not allow for cumulative imperceptible differences i.e. ($x \sim y$ and $y \sim z$ but $x \succ z$)

Utility Representation

Definition 6.9. Given the rational preference $\succsim$ over X , we say that the real-valued utility function $u : X \rightarrow \mathbb{R}$ is a utility representation (or function) whenever, for every $x, y \in X$:

$$x \succsim y \iff u(x) \geq u(y)$$

Theorem 6.10. Let R be a binary relation on X and $u : X \rightarrow \mathbb{R}$ such that for every $x, y \in X$, $xRy \iff u(x) \geq u(y)$. Then, R must be a rational preference.

Proof. R must be complete because for any pair x, y , it is either $u(x) \geq u(y)$ meaning xRy or yRx or both. R must be transitive because if xRy and yRz , then $u(x) \geq u(y)$ and $u(y) \geq u(z)$ which imply $u(x) \geq u(z)$ which implies $x \succsim z$ ■

Remark. We may simply use the fact that $\geq$ is a total (linear) order on $\mathbb{R}$

The following theorem says that if we may form a bijection between the elements of X and $\mathbb{N}$, then we can construct utility functions that have a bounded range i.e. between 0 and 1.

Theorem 6.11. If X is a countable set (i.e. finite or countably infinite), then any rational preference $\succsim$ on X can be represented by a utility function with a bounded range.

Proof. Suppose $X = \{x_1, x_2, \dots\}$ is countable. Define two fictitious elements x_{-1} and x_0 and extend the preferences by imposing that

$$\forall x_i \in X : x_i \succ x_{-1} \text{ and } x_0 \succ x_1$$

and define $u(x_0) = 1$ and $u(x_{-1}) = 0$. WLOG, take elements in $X = \{x_1, x_2, \dots\}$ in their order. Define $u(x_k)$ equal to the average of utilities of “next weakly better” and “next weakly worse” elements that have already assigned utility:

$$u(x_k) = \frac{\max\{u(x_j) | x_k \succsim x_j, j < k\} + \min\{u(x_j) | x_j \succsim x_k, j < k\}}{2}$$

Thus, $u(x_1) = \frac{1}{2}$ and

- if $x_2 \succ x_1$, then $u(x_2) = 3/4$
- if $x_1 \succ x_2$, then $u(x_2) = 1/4$
- if $x_2 \sim x_1$, then $u(x_2) = 1/2$

and we can continue assigning utilities this way ■

The reason why we can only do this for countable sets is because the max and min operators are defined only for countable sets.

Most choices we see are uncountable sets i.e. budget sets. Intuitively, we were working with indifference curves, or for all $\mathbf{x} \in X \subseteq \mathbb{R}_+^2$, the set of alternatives that are weakly preferred to $\mathbf{x}$ according to the preference order. Standard indifference curves are closed sets i.e. whenever $\{\mathbf{s}_n\} \rightarrow \mathbf{s}$, then $\mathbf{s}^* \in S$. So geometrically, the boundary must belong to the set. We want to generate indifference curves and to do this we need preference continuity.

Definition 6.12 (Preference Continuity). A binary relation $\succsim$ on $X \subseteq \mathbb{R}^n$ is continuous iff for any sequence $\mathbf{x}_1, \mathbf{x}_2, \dots \in X$ which converges to $\mathbf{x}^*$ and $\mathbf{x}_k \succsim \mathbf{y}$ for all $k = 1, 2, \dots$, we also have $\mathbf{x}^* \succsim \mathbf{y}$

Theorem 6.13 (Debreu’s Theorem). Any rational and continuous preference $\succsim$ on $\mathbb{R}_+^n$ can be represented by a continuous utility function $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$

Proof. Suppose that $\succsim$ is rational and continuous on $\mathbb{R}_+^n$. To simplify the proof, suppose that any consumption bundle $(x_1, \dots, x_n)$ has a uniform consumption equivalent $(\hat{x}, \dots, \hat{x})$ i.e. $(x_1, \dots, x_n) \sim (\hat{x}, \dots, \hat{x})$. For each choice define $u(x_1, \dots, x_n) = \hat{x}$, such that $(x_1, \dots, x_n) \sim (\hat{x}, \dots, \hat{x})$. (the utility is a coordinate line where the indifference curve passing through $(x_1, \dots, x_n)$ crosses the 45 degree line. Suppose that $(x_1, \dots, x_n) \sim (\hat{x}, \dots, \hat{x})$ and $(y_1, \dots, y_n) \sim (\hat{y}, \dots, \hat{y})$, then

$$(x_1, \dots, x_n) \succsim (y_1, \dots, y_n) \iff (\hat{x}, \dots, \hat{x}) \succsim (\hat{y}, \dots, \hat{y}) \iff \hat{x} \geq \hat{y} \iff u(x_1, \dots, x_n) \geq u(y_1, \dots, y_n)$$

and therefore u represents $\succsim$ ■

Definition 6.14 (Uniform consumption equivalent). Let $X \subseteq \mathbb{R}^L$ be the (one-period) consumption set and let $\succsim$ be a preference relation on X .

Fix a reference bundle $\bar{x} \in X$. For any $x \in X$, a scalar $\lambda \in \mathbb{R}_+$ is called a *uniform consumption equivalent* of x (relative to $\bar{x}$) if

$$x \sim \lambda \bar{x},$$

i.e. if x is indifferent to receiving the reference bundle $\bar{x}$ scaled *uniformly in every good* by the factor λ .

Definition 6.15 (Lexicographic Preferences). The binary relation $\succsim$ is a lexicographic preference just when for any $(x_1, x_2), (y_1, y_2) \in X^2$

$$(x_1, x_2) \succsim (y_1, y_2) \iff x_1 > y_1 \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2)$$

Theorem 6.16. Lexicographic preferences over $\mathbb{R}^2$ cannot be represented.

Proof. Suppose for contradiction that $\succsim_L$ is representable. Fix some $x_1 \in [0, 1]$, for each x_1 it must be the case that $u(x_1, 1) > u(x_1, 0)$. Then, by the density of the rationals in the reals, there exists a $q(x_1) \in \mathbb{Q} : u(x_1, 1) > q(x_1) > u(x_1, 0)$. Such $q(x_1)$ must exist for all $x_1 \in [0, 1]$. For each $x_1 \neq x_2$, we have $[u(x_1, 0), u(x_1, 1)] \cap [u(x_2, 0), u(x_2, 1)] = \emptyset$, thus $q(x_1) \neq q(x_2)$. This allows us to form a mapping between $[0, 1]$ and some subset of $\mathbb{Q}$ but $[0, 1]$ is uncountable while the set of all q is countable, which is a contradiction. ■

We also know that utility representations of a rational preference $\succsim$ are not unique. The preference is ordinal and if u represents this preference, any monotone transformation of u represents the same preference (since monotonic transformations are order preserving).

Theorem 6.17. Let $\succsim$ be a rational preference over X . Suppose that u represents $\succsim$ and that $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing. Then, the real-valued utility function $v = f \circ u$ (i.e. $v(x) = f(u(x))$) is also a utility function representing $\succsim$

Proof. By definition of utility representation,

$$x \succsim y \iff u(x) \geq u(y)$$

Furthermore, since f is strictly increasing

$$u(x) \geq u(y) \iff v(x) = f(u(x)) \geq f(u(y)) = v(y)$$

Hence, $x \succsim y \iff v(x) \geq v(y)$. So $v = f \circ u$ represents the same preferences as u ■

Examples of monotonic transformations are $\ln(\cdot)$, and raising to some exponent on $\mathbb{R}_+$

We also want to choose the best option or maximise our utility. We want to prove existence of a best element in both finite and (uncountably) infinite settings.

Proposition 6.18. Let $\succsim$ be a rational preference over a finite set X . Every subset $A \subseteq X$ has a maximal element

Proof. Trivial ■

Theorem 6.19. If $\succsim$ is a rational and continuous preference relation over a compact set $X \subseteq \mathbb{R}_+^n$, then X has a maximal element

Proof. Since $\succsim$ is rational and continuous, then u is continuous on X , since for any sequence $\{\mathbf{x}\}_k \rightarrow \{\mathbf{x}^*\}$, we have $\mathbf{x}^k \succsim \mathbf{y} \iff u(\mathbf{x}_k) \geq u(\mathbf{y})$ and since by preference continuity, establish $\mathbf{x}^* \succsim \mathbf{y} \iff u(\mathbf{x}^*) \geq u(\mathbf{y})$. By the Weierstrass Extreme Value Theorem, since u is continuous on X and X is compact by assumption, it attains a maximum and minimum ■

Example 6.20. Suppose we have the following preferences

$$\text{For every } x, y \in \{0, 1, \dots, n\} : x \succsim y \iff x \leq y + 1$$

(a) Determine whether these preferences satisfy reflexivity, connectedness, completeness, transitivity, symmetry and antisymmetry.

- **Reflexive:** For any $x \in \{0, \dots, n\}$, $x \succsim x$ since $x \leq x + 1$.
- **Completeness:** Suppose $x \neq y$:
 - (a) $x < y \Rightarrow x < y + 1 \Rightarrow x \leq y + 1 \Rightarrow x \succsim y$
 - (b) $y < x \Rightarrow y < x + 1 \Rightarrow y \leq x + 1 \Rightarrow y \succsim x$
 Connectedness + Reflexivity = Completeness.
- **Not transitive:** Let $a = 2, b = 1, c = 0$. Then $2 \succsim 1$ since $2 \leq 1 + 1$, and $1 \succsim 0$ since $1 \leq 0 + 1$, but $2 \not\succsim 0$ since $2 \not\leq 0 + 1$.
- **Not symmetric:** Let $a = 0, b = 2$. Then $a \succsim b$ since $0 \leq 2 + 1$, but $b \not\succsim a$ since $2 \not\leq 0 + 1$.

- **Not anti-symmetric:** Let $a = 1, b = 0$. Then $a \succsim b$ since $1 \leq 0 + 1$, and $b \succsim a$ since $0 \leq 1 + 1$, but $a \neq b$ by construction.

(b) Conclude whether these preferences are rational or not.
 $\succsim$ is rational iff it is complete and transitive. By (a), it is not transitive and hence not rational.

(c) Are these preferences representable by a utility function?

AFC $\exists u$ s.t. $u : X \rightarrow \mathbb{R}$ and $x \succsim y \iff u(x) \geq u(y)$. By (a), let $a = 2, b = 1, c = 0$. Then

$$\begin{aligned} 2 \succsim 1 &\iff u(2) \geq u(1) \\ 1 \succsim 0 &\iff u(1) \geq u(0) \end{aligned}$$

Since $\geq$ is transitive on $\mathbb{R}$, then $u(2) \geq u(0)$. But from (a), $2 \not\succsim 0$ and hence $u(2) \not\geq u(0)$, a contradiction.

Example 6.21. Consider the space of bundles $\mathbb{R}_{++}^2$ and the rational preference

$$(x_1, x_2) \succsim (y_1, y_2) \iff x_1^3 x_2 \geq y_1^3 y_2$$

(a) Are these preferences representable by a continuous utility function?

Consider some arbitrary (y_1, y_2) . Suppose $(x_1^{(i)}, x_2^{(i)}) \succsim (y_1, y_2), \forall i \in \mathbb{N}$. Then,

$$x_1^{(i)2} x_2^{(i)} \succsim y_1^3 y_2 \text{ since } \mathbf{x} \succsim \mathbf{y} \iff u(\mathbf{x}) \geq u(\mathbf{y})$$

Then, since $x_1^{(i)2} x_2^{(i)}$ is continuous, if $x_1^{(i)2} x_2^{(i)} \rightarrow x_1^{*3} x_2^*$, then $x_1^{*3} x_2^* \geq y_1^3 y_2$ since ordering is preserved under limits. Then, $(x_1^*, x_2^*) \succsim (y_1, y_2)$

(b) Discuss whether or not $v(x_1, x_2) = x_1^{\frac{3}{4}} x_2^{\frac{1}{4}}$, $w(x_1, x_2) = 3\log x_1 + \log x_2$ and $\tau(x_1, x_2) = \frac{1}{x_1^3 x_2}$ are utility representations of these preferences.

Note that $v(x_1, x_2), w(x_1, x_2)$ are monotonic transformations since they are strictly increasing on $[0, \infty)$ and $(0, \infty)$ respectively. But $\tau(x_1, x_2)$ is not a utility representation of these preferences because it is not a monotonic transformation

(c) Construct the uniform consumption equivalent utility representation in which the utility (x_1, x_2) is the same amount x such that the bundle (x, x) is indifferent to (x_1, x_2) .

$(x_1, x_2) \sim (x, x)$ Then, $u(x_1, x_2) = x_1^3 x_2 = x^4$. Hence

$$x = (x_1^3 x_2)^{\frac{1}{4}} \implies u(\mathbf{x}) \geq u(\mathbf{y}) \iff (x_1^3 x_2)^{\frac{1}{4}} \geq (y_1^3 y_2)^{\frac{1}{4}} \iff x_1^3 x_2 \geq y_1^3 y_2$$

Choice

Definition 6.22 (Choice problem/menu). A choice problem/menu is a subset of alternatives $A \subseteq X$.

Definition 6.23 (Choice Data). Let $\Delta = \{A, B, \dots\}$ be a collection of menus. Choice Data is defined as a map of the form

$$c : \Delta \rightarrow X$$

with $c(K) \in K$ for all $K \in \Delta$.

Choice data is given by Δ and $c(\cdot)$:

- Δ is determined by the analyst or occurs naturally in the field
- c is determined by the agent

We have the following concepts for choice data

- Universal Choice Data: when $\Delta = \mathcal{X}$ where $\mathcal{X}$ is the collection of all subsets of X
- Binary Choice Data: when $\Delta = \mathcal{B}$, where $\mathcal{B}$ is the collection of all choice problems formed by two alternatives

- **Bundle Consumption Choice Data:** Let $X = \mathbb{R}_+^n$. Bundle Consumption Choice Data is such that Δ is formed by some budget sets of the form $p_1x_1 + p_2x_2 + \dots + p_nx_n \leq m$. Typically, the variations of price and income may be given by the environment

The goal of this section is to deduce the complete preference structure/rankings of the agent given his choices for a menu A . This is not easy since even if we observe $x = c(A)$ and $y \in A$ we cannot deduce anything (not even that x is preferred to y) since he might be indifferent among all options. Consider the case where there are no indifferences other than reflexivity.

Definition 6.24. A rational choice preference relation of a finite choice set $c(A, \succ)$ is the unique maximal element of A according to $\succ$

The notions and techniques can be extended to the general case using a set-valued c .

Definition 6.25 (Direct Revelation). Given choice data c , we say that x is Directly Revealed Preferred to $y \neq x$, and write $x \succ_c y$ whenever $x = c(A)$ and $y \in A$

The upshot of this is that if there is another alternative choice problem A' with $x, y \in A'$ we may predict that $y \neq c(A')$ because x is a better available option.

Theorem 6.26. Suppose that X is finite. Then Universal Choice Data or Binary Choice Data are enough to learn all the preferences of the individual.

Proof. Trivial because we observe every binary menu $\{x, y\}$ and can learn directly from this whether $x \succ y$ or $y \succ x$ ■

We want to motivate indirect revealed preference by giving an example where we must impose some structure of the preference relation to *infer* the entire preference ranking

Example 6.27. A college offers repeatedly the same four meals in hall $\{w, x, y, z\}$. Depending on the time of arrival to hall some options may be unavailable

- Monday: student chooses x from $\{x, z\}$
- Tuesday: student chooses w from $\{w, x, y\}$
- Wednesday: student chooses x from $\{x, y, z\}$

From this, it is directly revealed that $x \succ_c z$, $w \succ_c x$ and $w \succ_c y$, $x \succ_c y$ and $x \succ_c z$. NOTE: (i) we cannot compare y and z , (ii) we cannot say $w \succ_c z$ directly unless we impose transitivity from which we infer

Definition 6.28. Given a choice data c , we say that x is Indirectly Revealed Preferred to $y \neq x$ whenever there is a sequence of menus $A_1, \dots, A_n$ and a sequence of choices $z_1 = x, z_2, \dots, z_n = y$ such that $z_i = c(A_i)$ and $z_{i+1} \in A_i, i = 1, \dots, n-1$ i.e. whenever $z_1 = x \succ_c z_2 \succ_c \dots \succ_c y = z_n$.

In our example,

- $A_1 = \{w, x, y\}$, $c(A_1) = w$ and $x \in A_1$
- $A_2 = \{x, z\}$, $c(A_2) = x$ and $z \in A_2$
- Hence $w \succ_c x \succ_c z$

What is the minimum number of choice problems that we need to observe to learn all the preferences of the individual? Which is the needed structure? If X contains n elements, $n-1$ choice problems might be all we need to learn all the preferences since

- If we conjecture that $x_1 \succ x_2 \succ \dots \succ x_n$ is the preference, we just need to observe $n-1$ binary problems $\{x_i, x_{i+1}\}$ to confirm this

- We can run $n - 1$ experiments sequentially. Give the menu X and the chosen element x_1 must be the best. Ask for $X \setminus \{x_1\}$. The chosen element x_2 must be the second best and so on.

We want to test if the individual is being rational or not

Definition 6.29 (Rational Choice Data I). Choice Data $c : \Delta \rightarrow X$ is said to be rationalizable if there exists a rational preference $\succsim$ such that $c(A)$ is always the maximal element in A according to $\succsim$

Definition 6.30 (Rational Choice Data II). Choice Data $c : \Delta \rightarrow X$ is said to be rationalizable if there exists a utility representation u such that $c(A)$ is always the maximal element in A according to u

Definition 6.31 (Property α). We say that c satisfies property α if for any pair of choice problems $B \subseteq A$, we have

$$c(A) \in B \implies c(B) = c(A)$$

The intuition is that if someone chooses x from $A = \{x, y, z\}$ and does not choose x from $B = \{x, z\}$, we would think there is somethin wrong. Analogous to *Independence of Irrelevant Alternatives*.

Theorem 6.32. If choice data is rationalizable, it must satisfy property α

Proof. If c is rationalizable, there exists a preference relation $\succsim$ explaining all the choices i.e. for any $x \in c(A)$ and $y \in A$, we have $x \succsim y$. Then, for each $B \subseteq A$, we have $x \succsim y$ ■

Note that property α implies rationalizable preferences only for the strict domain. In general, property β is necessary

Definition 6.33 (Property β). We say that c satisfies property β if for any pair of choice problems $B \subseteq A$, if $x, y \in c(B)$ and $x \in c(A)$ then $y \in c(A)$

Example 6.34. We observe two students at Hall over an entire year. Given the variation in arrival time, we got universal choice data from them. It is as follows:

Menu	Ann	Bob
$\{M, F, V\}$	V	F
$\{M, F\}$	F	M
$\{M, V\}$	V	V
$\{F, V\}$	V	V

- Ann's choice data are clearly rational. The preference $V \succ_A F \succ_A M$ explains all choices.
- Bob's choice data are clearly not rational. No preference can explain all choices:
 - Any preference in which $M \succ_B F$ fails to explain the choice from menu $\{M, F, V\}$.
 - Any preference in which $F \succ_B M$ fails to explain the choice from menu $\{M, F\}$.

In particular, Bob's choices violate **Property α** : Bob chooses F from $\{M, F, V\}$, and F is also available in the smaller menu $\{M, F\} \subseteq \{M, F, V\}$, yet Bob does not choose F from $\{M, F\}$ — instead choosing M . Property α requires that if x is chosen from a larger set S , and $x \in T \subseteq S$, then x must be chosen from T .

Theorem 6.35. Let X be finite. If the universal choice data c satisfies property α , c is rationalizable

Proof. We construct the preference as follows:

- take $A_1 = X$ and $x_1 = c(A_1)$
- by property α , x_1 is chosen whenever $x_1 \in A \subseteq X$
- let $A_2 = A_1 \setminus \{x_1\}$ and $x_2 = c(A_2)$
- by property α , x_2 is chosen whenever $x_2 \in A$ and $x_1 \notin A$

- define $x_1 \succ_c x_2$
- let $A_3 = A_2 \setminus \{x_2\}$ and $x_3 = c(A_3)$
- ...
- in finite number of steps, reach $x_1 \succ_c x_2 \succ_c \dots \succ_c x_K$ where K is the number of elements in X

■

Here is an alternative condition for rational choice (and very famous)

Definition 6.36 (Weak Axiom of Revealed Preference). Let $x, y \in A$ and $x, y \in B$, then we cannot have $c(A) = x$ and $c(B) = y$

Remark. This is not exactly the same as property α : under WARP, we have $\{x, y\} \in A \cap B$ but not necessarily $B \subseteq A$. So it may not be true that $c(B) = x$ but it must be true that $c(B) \neq y$

Theorem 6.37. Choice induced by rational preference $\succ$ satisfy WARP. Furthermore, if X is finite and c satisfies WARP on universal choice data, then there exists a rational preference relation that induces c .

Example 6.38. Consider the following choice data

Menu	Ann	Bob
$\{M, F, V\}$	V	F
$\{M, F\}$	F	M
$\{M, V\}$	V	V
$\{F, V\}$	V	V

- Ann satisfies both properties:
 - **Property α :** Ann chooses V from $\{M, F, V\}$. In every smaller menu containing V , namely $\{M, V\}$ and $\{F, V\}$, Ann also chooses V . ✓
 - **WARP:** Ann's choices are rationalised by $V \succ_A F \succ_A M$, so whenever x is chosen over y in some menu, y is never chosen over x in any menu where x is available. Explicitly: V is chosen over F and M whenever available, and F is chosen over M whenever V is not available. No violation occurs. ✓
- You can see that Bob fails the properties by checking $\{M, F, V\}$ and $\{M, F\}$:
 - The choice in the largest menu differs from the choice in the smallest menu, so α fails: Bob chooses F from $\{M, F, V\}$, and $F \in \{M, F\} \subseteq \{M, F, V\}$, yet F is not chosen from $\{M, F\}$.
 - Equivalently, F is revealed preferred to M in the largest menu (since $F = c(\{M, F, V\})$ and $M \in \{M, F, V\}$), but M is revealed preferred to F in the smallest menu (since $M = c(\{M, F\})$ and $F \in \{M, F\}$). This directly violates **WARP**: WARP requires that if x is chosen over y when both are available, then y is never chosen over x in any menu where x is also available. Here F is chosen over M in $\{M, F, V\}$, yet M is chosen over F in $\{M, F\}$, with both M and F available in each menu.

Example 6.39. A gluttonous kid has been told to be polite and avoid taking the largest piece. The kid always takes the second-largest piece of cake:

- $c(\{1, 2, 3\}) = 2$
- $c(\{1, 2\}) = 2$
- $c(\{1, 3\}) = 3$
- $c(\{2, 3\}) = 3$

Do these choice data violate property α and/or WARP?

Property α : Consider $\{1, 2, 3\}$ and its subset $\{2, 3\}$. The kid chooses 2 from $\{1, 2, 3\}$, and $2 \in \{2, 3\} \subseteq \{1, 2, 3\}$, yet $c(\{2, 3\}) = 3 \neq 2$. Hence **property α is violated**.

WARP: From $\{1, 2, 3\}$, item 2 is chosen, so 2 is revealed preferred to 3 (since $3 \in \{1, 2, 3\}$). But from $\{2, 3\}$, item 3 is chosen, so 3 is revealed preferred to 2 (since $2 \in \{2, 3\}$). Both 2 and 3 are available in each menu, yet the choices contradict each other. Hence **WARP is also violated**.

Note that since property α and WARP are equivalent for choice functions, it is unsurprising that both fail simultaneously.

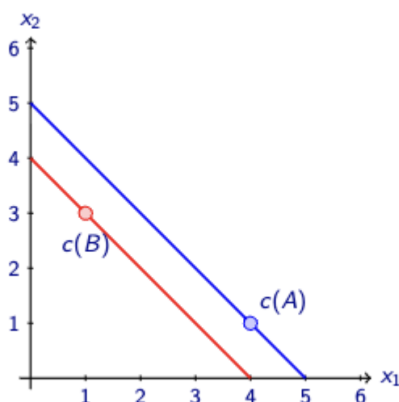
Bundles and Time

In the traditional framework of consumption of different goods, universal menus are not reasonable because variation of menus is usually due to changes in price or income or both.

The basic problem is that for general preferences, there is no natural order in the interior of the budget set. Continuity of preferences yields equivalent classes called indifference curves, where the utility does not change and does not put restrictions on the shape or direction of the utility change. We want to impose the extra condition of monotonicity, so that price-quantity data allows us to say more about the preferences.

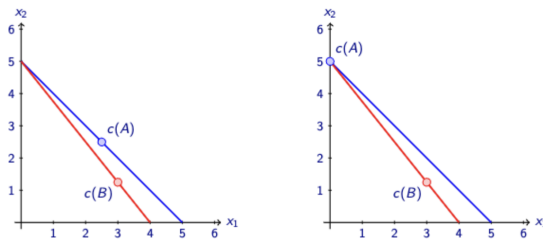
Definition 6.40 (Strict Monotonicity). A preference relation $\succ$ on $X \subseteq \mathbb{R}_+^n$ is strictly monotone if for all $\mathbf{x}, \mathbf{y} \in X$ such that $\mathbf{x} \succ \mathbf{y}$ we have $\mathbf{x} > \mathbf{y}$.

Remark. $\mathbf{x} > \mathbf{y}$ means that $x_i \geq y_i$ for all $i = 1, \dots, n$ with at least one j such that $x_j > y_j$

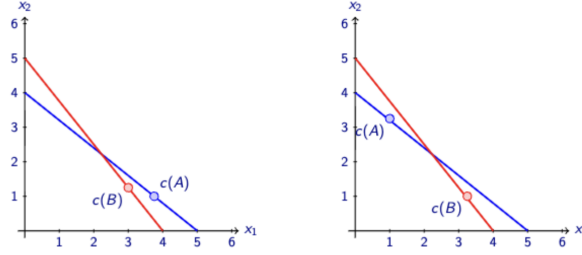


When we only change income, property α and WARP are unable to say anything because the choice in the big budget set A will never be available in the smaller budget set B .

When there is only a change in price, we can only reject property α in extreme cases such as the right-hand graph here



Dual variations in price and income are more useful. Here we use the weak axiom of revealed preference jointly with monotonicity, even accepting that indifferences may exist.



The right-hand graph violates WARP. By strict monotonicity, $x \succ c(A)$. By revealed preference, $c(B) \succsim x$. Again by revealed preference, $c(A) \succsim c(B)$ (since $c(B)$ lies in the interior of the budget set under which $c(A)$ was selected). Thus, we violate transitivity since

$$x \succ c(A) \succsim c(B) \succsim x$$

So, we have the following definition for WARP for bundles.

Definition 6.41 (WARP for Bundles). Suppose $\mathbf{x}_1$ was chosen given prices $\mathbf{p}_1$ and $\mathbf{x}_2$ was chosen given prices $\mathbf{p}_2$ and $\mathbf{p}_1 \cdot \mathbf{x}_2 \leq \mathbf{p}_1 \cdot \mathbf{x}_1$. Then it cannot be the case that $\mathbf{p}_2 \mathbf{x}_2 \geq \mathbf{p}_2 \mathbf{x}_1$

The following are some useful properties about preferences

Definition 6.42 ((strict) Convexity). Preferences are strictly convex if for every $x, y \in X \subseteq \mathbb{R}_+^n$ such that $x \sim y$ and $x \neq y$ and every $\alpha \in (0, 1)$:

$$\alpha x + (1 - \alpha)y \succ x \sim y$$

Remark. We require this property for uniqueness of demand

Example 6.43. Consider the space of bundles $\mathbb{R}_+^2$. A consumer has strictly monotone preferences and spends all income in the two goods considered. We observe the choices made in two situations, when prices for goods are $(p_1, p_2) = (3, 1)$ and when prices for goods are $(1, 3)$.

- (a) Show that the observed choices cannot be $(x_1, x_2) = (2, 1)$ in the first case, and $(1, 2)$ in the second. What if the choices are $(2, 1)$ and $(\frac{7}{4}, \frac{7}{4})$?

Case 1: $\mathbf{x}_A = (2, 1)$ at $\mathbf{p}_A = (3, 1)$ and $\mathbf{x}_B = (1, 2)$ at $\mathbf{p}_B = (1, 3)$. Incomes: $\mathbf{p}_A \cdot \mathbf{x}_A = 3(2) + 1(1) = 7$ and $\mathbf{p}_B \cdot \mathbf{x}_B = 1(1) + 3(2) = 7$. By Definition 6.41, check whether $\mathbf{p}_A \cdot \mathbf{x}_B \leq \mathbf{p}_A \cdot \mathbf{x}_A$:

$$\mathbf{p}_A \cdot \mathbf{x}_B = 3(1) + 1(2) = 5 \leq 7 = \mathbf{p}_A \cdot \mathbf{x}_A$$

so the antecedent of WARP is satisfied. But:

$$\mathbf{p}_B \cdot \mathbf{x}_A = 1(2) + 3(1) = 5 \leq 7 = \mathbf{p}_B \cdot \mathbf{x}_B$$

i.e. $\mathbf{x}_A$ was affordable when $\mathbf{x}_B$ was chosen. This violates WARP, so these choices are **inconsistent**.

Case 2: $\mathbf{x}_A = (2, 1)$ at $\mathbf{p}_A = (3, 1)$ and $\mathbf{x}_B = (\frac{7}{4}, \frac{7}{4})$ at $\mathbf{p}_B = (1, 3)$. Incomes: $\mathbf{p}_A \cdot \mathbf{x}_A = 7$ and $\mathbf{p}_B \cdot \mathbf{x}_B = 1(\frac{7}{4}) + 3(\frac{7}{4}) = 7$. Check whether $\mathbf{p}_A \cdot \mathbf{x}_B \leq \mathbf{p}_A \cdot \mathbf{x}_A$:

$$\mathbf{p}_A \cdot \mathbf{x}_B = 3(\frac{7}{4}) + 1(\frac{7}{4}) = 7 \leq 7 = \mathbf{p}_A \cdot \mathbf{x}_A$$

so the antecedent of WARP is satisfied. But:

$$\mathbf{p}_B \cdot \mathbf{x}_A = 1(2) + 3(1) = 5 \leq 7 = \mathbf{p}_B \cdot \mathbf{x}_B$$

i.e. $\mathbf{x}_A$ was affordable when $\mathbf{x}_B$ was chosen. This violates WARP, so these choices are also **inconsistent**.

- (b) Show that observed choices can be $(1, 3)$ in the first case and $(1, 2)$ in the second.

Let $\mathbf{x}_A = (1, 3)$ at $\mathbf{p}_A = (3, 1)$ and $\mathbf{x}_B = (1, 2)$ at $\mathbf{p}_B = (1, 3)$. Incomes: $\mathbf{p}_A \cdot \mathbf{x}_A = 3(1) + 1(3) = 6$ and $\mathbf{p}_B \cdot \mathbf{x}_B = 1(1) + 3(2) = 7$. Check whether $\mathbf{p}_A \cdot \mathbf{x}_B \leq \mathbf{p}_A \cdot \mathbf{x}_A$:

$$\mathbf{p}_A \cdot \mathbf{x}_B = 3(1) + 1(2) = 5 \leq 6 = \mathbf{p}_A \cdot \mathbf{x}_A$$

so the antecedent is satisfied and $\mathbf{x}_A \succsim \mathbf{x}_B$ by revealed preference. Now check the reverse:

$$\mathbf{p}_B \cdot \mathbf{x}_A = 1(1) + 3(3) = 10 > 7 = \mathbf{p}_B \cdot \mathbf{x}_B$$

i.e. $\mathbf{x}_A$ was **not** affordable when $\mathbf{x}_B$ was chosen, so the antecedent of WARP in the reverse direction fails. WARP is **not violated**. Furthermore, by strict monotonicity, $(1, 3)$ has weakly more of good 1 and strictly more of good 2 than $(1, 2)$, so $\mathbf{x}_A \succ \mathbf{x}_B$, consistent with the revealed preference above.

- (c) For bundles in part (b), discuss whether preferences can be strictly convex or not, and whether they can be Cobb-Douglas or not.

Strict convexity: The indifference curve through $\mathbf{x}_A = (1, 3)$ is tangent to the budget line $3x_1 + x_2 = 6$ at $(1, 3)$, giving $\text{MRS} = -3$. The indifference curve through $\mathbf{x}_B = (1, 2)$ is tangent to $x_1 + 3x_2 = 7$ at $(1, 2)$, giving $\text{MRS} = -\frac{1}{3}$. Since MRS decreases in magnitude as we move along the indifference curve (more x_1 , less x_2), this is consistent with **strictly convex preferences**.

Cobb-Douglas: A Cobb-Douglas utility $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ generates demands $x_1 = \frac{\alpha I}{p_1}$ and $x_2 = \frac{(1-\alpha)I}{p_2}$. At $\mathbf{p}_A = (3, 1)$, $I = 6$, $\mathbf{x}_A = (1, 3)$:

$$1 = \frac{6\alpha}{3} \implies \alpha = \frac{1}{2}, \quad 3 = \frac{6(1-\alpha)}{1} \implies \alpha = \frac{1}{2}$$

At $\mathbf{p}_B = (1, 3)$, $I = 7$, $\mathbf{x}_B = (1, 2)$:

$$1 = \frac{\frac{1}{2} \cdot 7}{1} = 3.5$$

The expenditure shares differ: $\frac{p_1 x_1}{I} = \frac{3}{6} = \frac{1}{2}$ at $\mathbf{p}_A$, but $\frac{p_1 x_1}{I} = \frac{1}{7}$ at $\mathbf{p}_B$. Since Cobb-Douglas demands always allocate fixed shares α and $1-\alpha$ of income regardless of prices, these choices are **not consistent with Cobb-Douglas preferences**.

Definition 6.44 (Quasi-linearity). The preference is quasi-linear on $(\mathbf{x}, m) \in \mathbb{R}_+^{n+1}$ if for all $\mathbf{x}, \mathbf{y}, m, m', n$ we have

$$(\mathbf{x}, m) \succsim (\mathbf{y}, m') \iff (\mathbf{x}, m+n) \succsim (\mathbf{y}, m'+n)$$

We also want to think about discounted utility which gets discounted in a compounding way as time goes by. The standard model involves an increasing utility over money $u : \mathbb{R}_+$ and a discount factor $\delta \in (0, 1]$.

Definition 6.45 (Discounted Utility). $(x, t) \succsim (y, s) \iff U(x, t) = \delta^t u(x) \geq \delta^s u(y) = U(y, s)$

Discounted Utility is a rational preference and can be represented by a utility function. We can use property α or WARP to test rationality when evaluating choices of dated prizes. It has the following additional property to be tested

Definition 6.46 (Stationarity). We say that the preference $\succsim$ over dated prizes satisfies stationarity if for every x, y, t, s, r :

$$(x, t) \succsim (y, s) \iff (x, t+r) \succsim (y, s+r)$$

This gives us the following theorem for exponential discounting utility representation.

Theorem 6.47 (Fishburn–Rubinstein (1982)). Let $X \subseteq \mathbb{R}$ be a set of outcomes, and let $\succsim$ be a preference relation on the set of dated outcomes

$$D := X \times \{0, 1, 2, \dots\}.$$

Interpret $(x, t) \in D$ as receiving outcome x at date t .

Suppose $\succsim$ satisfies:

1. **Weak order:** $\succsim$ is complete and transitive.

2. **Monotonicity:** for all $x, y \in X$ and all t ,

$$x > y \implies (x, t) \succ (y, t).$$

3. **Continuity:** the preference relation is continuous.

4. **Impatience:** for all $x > 0$ and all $t < s$,

$$(x, t) \succ (x, s).$$

5. **Stationarity:** for all $x, y \in X$ and all $t, s, \tau \in \{0, 1, 2, \dots\}$,

$$(x, t) \succsim (y, s) \iff (x, t + \tau) \succsim (y, s + \tau).$$

Then there exist a strictly increasing function $u : X \rightarrow \mathbb{R}$ and a discount factor $\delta \in (0, 1)$ such that, for all $(x, t), (y, s) \in D$,

$$(x, t) \succsim (y, s) \iff \delta^t u(x) \geq \delta^s u(y).$$

Hence $\succsim$ admits an exponential discounted utility representation.

Other useful properties for preferences include

Definition 6.48 (Single-Peakedness). Let $X = [0, 1]$. We say that $\succsim$ is single-peaked if there exists x^* such that:

- $y_2 < y_1 < x^* \implies y_1 \succ y_2$
- $x^* < y_1 < y_2 \implies y_1 \succ y_2$

Remark. We can generalise this to $\mathbb{R}^n$

Definition 6.49 (Single-peakedness on $\mathbb{R}^n$). Let $X \subseteq \mathbb{R}^n$ be convex. A preference relation $\succsim$ on X is *single-peaked* if there exists $x^* \in X$ such that for every $x \in X$ with $x \neq x^*$ and every $\lambda \in (0, 1)$,

$$\lambda x^* + (1 - \lambda)x \succ x.$$

Definition 6.50 (Local Non-satiation). Let $x \subseteq \mathbb{R}_+^n$. Preferences $\succsim$ are called locally non-satiated, if for any $x \in X$ there exists $\varepsilon > 0$ and $y \in X$ such that

$$\|y - x\| < \varepsilon \text{ and } y \succ x$$

Definition 6.51 (Local non-satiation in $\mathbb{R}^n$). Let $X \subseteq \mathbb{R}_+^n$. A preference relation $\succsim$ on X is *locally non-satiated* if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in X$ such that

$$y \in B_\varepsilon(x) \text{ and } y \succ x,$$

where

$$B_\varepsilon(x) := \{z \in \mathbb{R}^n : \|z - x\| < \varepsilon\}$$

is the open ball of radius ε centred at x .

Proposition 6.52. Local non-satiation does not imply monotone preferences

Example 6.53 (Locally non-satiated but not monotone). Let $X = \mathbb{R}_+^2$ and define preferences $\succsim$ on X by the utility representation

$$u(x_1, x_2) = x_1 - x_2.$$

That is,

$$(x_1, x_2) \succsim (y_1, y_2) \iff x_1 - x_2 \geq y_1 - y_2.$$

These preferences are *locally non-satiated*. Fix any $x = (x_1, x_2) \in X$ and any $\varepsilon > 0$. Let

$$y = (x_1 + \varepsilon/2, x_2).$$

Then $y \in X$ and

$$\|y - x\| = \varepsilon/2 < \varepsilon.$$

Moreover,

$$u(y) = x_1 + \varepsilon/2 - x_2 > x_1 - x_2 = u(x),$$

so $y \succ x$. Hence, for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in X$ with $\|y - x\| < \varepsilon$ and $y \succ x$. However, these preferences are *not monotone*. Let

$$z = (x_1 + \varepsilon/2, x_2 + \varepsilon).$$

Then $z \geq x$ componentwise and $z \neq x$, so monotonicity would require $z \succ x$. But

$$u(z) = x_1 + \varepsilon/2 - (x_2 + \varepsilon) = u(x) - \varepsilon/2 < u(x),$$

so in fact $z \prec x$.

Therefore, $\succsim$ is locally non-satiated but not monotone.

6.2 Expected Utility Theory

Basic Probability Theory

This section is a rehash of basic probability theory (pay special attention to the notions of stochastic dominance)

Definition 6.54 (Probability Space). A probability space is a triple $\langle \Omega, \mathcal{F}, \mathbb{P} \rangle$ where

- Ω is the sample space i.e. the set of all possible outcomes
- $\mathcal{F}$ is the event space i.e. a collection of subsets of Ω satisfying
 - $\Omega \in \mathcal{F}$
 - if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$
 - if $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$
- A function (measure) $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ satisfying the Komogolrov axioms
 - $\mathbb{P}(\Omega) = 1$
 - $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$
 - For any countable sequence of mutually exclusive (disjoint) events $A_1, A_2, \dots$ (i.e. $A_i \cap A_j = \emptyset$ for $i \neq j$),

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

Definition 6.55 (Random Variable). Let $\langle \Omega, \mathcal{F}, \mathbb{P} \rangle$ be a probability space. A random variable is a function

$$X : \Omega \rightarrow \mathbb{R}$$

such that for every $x \in \mathbb{R}$

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$$

Definition 6.56 (Cumulative Distribution Function). For a random variable X , the cumulative distribution function is defined as

$$F_X(x) = \mathbb{P}(X \leq x)$$

Remark. The CDF fully describes the random variable, and everything else (PDF, moments) are derived from the CDF

Definition 6.57 (Support). Let X be a random variable. The support of X is

$$\text{supp}(X) = \{x \in \mathbb{R} : \mathbb{P}(X \in U) > 0 \text{ for every open set } U \text{ containing } x\}$$

Equivalently, it is the smallest closed set $S \subseteq \mathbb{R}$ such that $\mathbb{P}(X \in S) = 1$

Intuitively, the support is a set of “possible realisations” of the random variable.

Example 6.58 (Discrete Support). If X takes values in a countable set $\{x_1, x_2, \dots\}$ then

$$\text{supp}(X) = \{x_i : \mathbb{P}(X = x_i) > 0\}$$

The distribution is described by probabilities

$$p_i = \mathbb{P}(X = x_i), \quad \sum_i p_i = 1$$

Example 6.59 (Continuous Support). If X has density f , then

$$F_X(x) = \int_{-\infty}^x f(t)dt, \quad F_X(x) = \int_{\text{supp}(X)} f(t)dt = 1$$

The support is essentially the set

$$\{x \in \mathbb{R} : f(x) > 0\}$$

Definition 6.60 (Expectation). The first moment (expectation) of a random variable X is

$$\mathbb{E}[X] = \begin{cases} \sum_{i=1}^n x_i \mathbb{P}(X = x_i) & \text{if discrete} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{if continuous} \end{cases}$$

where in the continuous case, it is provided that the integral $\int_{-\infty}^{\infty} |x| f(x) dx$ is finite.

Definition 6.61 (Variance). If $\mathbb{E}[X]$ exists, then the variance of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Not all random variables have expectations (and so other (higher-order) moments also do not exist).

Example 6.62 (Cauchy Distribution). The Cauchy Distribution is defined as

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \arctan(x), \quad x \in \mathbb{R}$$

Then F is increasing so

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow \infty} F_X(x) = 1, \quad f(x) = F'(x) = \frac{1}{\pi} \frac{1}{1+x^2}$$

Consider the expectation:

$$\int_{-\infty}^{\infty} |x| f(x) dx = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{|x|}{1+x^2} dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx = \left[\frac{1}{\pi} \ln(1+x^2) \right]_0^{\infty} = \infty$$

Theorem 6.63 (Law of the Unconscious Statistician). If X is a r.v. and g is a function from $\mathbb{R}$ to $\mathbb{R}$, then

$$\mathbb{E}[g(X)] = \begin{cases} \sum_x g(x) \mathbb{P}(X = x) & \text{if discrete} \\ \int_{-\infty}^{\infty} g(x) f(x) dx & \text{if continuous} \end{cases}$$

Definition 6.64 (Affine Transformations). For $a, b \in \mathbb{R}$, define the function

$$h : \mathbb{R} \rightarrow \mathbb{R}, \quad h(x) = a + bx$$

and the transformed random variable $Y = h(X) = a + bX$.

Proposition 6.65 (Properties of the Expectation Operator). For any r.v.s X, Y and any constant $c \in \mathbb{R}$, we have

- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
- $\mathbb{E}[cX] = c\mathbb{E}[X]$

Proposition 6.66 (Properties of Variance). For any r.v.s X, Y , assuming the expectation exists, we have

1. $\text{var}(aX + b) = a^2 \text{var}(X)$
2. $\text{var}(aX + bY) = a^2 \text{var}(X) + b^2 \text{var}(Y) - 2ab \text{cov}(X, Y)$
3. If X and Y are independent, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$

Theorem 6.67 (Jensen's Inequality). Suppose X is a r.v. and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is a convex function. Then,

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$$

If φ is a concave function, then

$$\mathbb{E}[\varphi(X)] \leq \varphi(\mathbb{E}[X])$$

Proof. Set

$$m := \mathbb{E}[X].$$

Since h is concave on an interval, there exists a supporting affine function at m : that is, there exist constants $a, b \in \mathbb{R}$ such that

$$h(x) \leq a + bx \quad \text{for all } x \in I,$$

with equality at $x = m$. Hence

$$h(m) = a + bm,$$

and for every realization of X ,

$$h(X) \leq a + bX.$$

Taking expectations on both sides gives

$$\mathbb{E}[h(X)] \leq \mathbb{E}[a + bX] = a + b\mathbb{E}[X] = a + bm = h(m).$$

Therefore,

$$\mathbb{E}[h(X)] \leq h(\mathbb{E}[X]).$$

Now suppose h is strictly concave and X is not almost surely constant. Then $\mathbb{P}(X \neq m) > 0$. For a strictly concave function, the supporting line at m touches the graph only at m , so

$$h(x) < a + bx \quad \text{for all } x \neq m.$$

Thus

$$h(X) < a + bX \quad \text{on the event } \{X \neq m\},$$

which has positive probability. Taking expectations yields

$$\mathbb{E}[h(X)] < \mathbb{E}[a + bX] = a + b\mathbb{E}[X] = h(m).$$

Hence

$$\mathbb{E}[h(X)] < h(\mathbb{E}[X]).$$

■

Example 6.68. Lottery L pays $x_k \in \mathbb{R}$ with probability p_k , where $k \in \{1, \dots, K\}$. There are N agents with identical, strictly risk averse EU preferences on final wealth. Every agent has the same initial wealth, w .

Consider two possible arrangements. In the first arrangement, an agent is picked at random (each with probability $1/N$); then the outcome of the lottery is given to the selected agent while all other agents receive nothing. In the second arrangement agents share the outcome of the lottery equally. Which arrangement do the agents prefer?

Provide a mathematical proof that relies on the strict concavity of the agents' Bernoulli utility function.

Arrangement 1 (random allocation): Each agent is selected with probability $1/N$; the selected agent receives final wealth $w + x_k$, all others retain wealth w . The ex-ante expected utility of any given agent is:

$$V_1 = \frac{1}{N} \sum_{k=1}^K p_k u(w + x_k) + \frac{N-1}{N} \sum_{k=1}^K p_k u(w) = \frac{1}{N} \sum_{k=1}^K p_k u(w + x_k) + \frac{N-1}{N} u(w) \quad (1)$$

Arrangement 2 (equal sharing): Every agent receives final wealth $w + x_k/N$ when outcome k realizes. The ex-ante expected utility of any agent is:

$$V_2 = \sum_{k=1}^K p_k u\left(w + \frac{x_k}{N}\right) \quad (2)$$

Claim: $V_2 \geq V_1$, with strict inequality unless the lottery is degenerate (i.e. $x_k = 0$ for all k with $p_k > 0$).

Proof: By strict concavity of u , for any outcome x_k we can write $w + x_k/N$ as a convex combination of $w + x_k$ and w :

$$w + \frac{x_k}{N} = \frac{1}{N}(w + x_k) + \frac{N-1}{N} w \quad (3)$$

Applying strict concavity of u to this convex combination:

$$u\left(w + \frac{x_k}{N}\right) = u\left(\frac{1}{N}(w + x_k) + \frac{N-1}{N} w\right) \geq \frac{1}{N} u(w + x_k) + \frac{N-1}{N} u(w) \quad (4)$$

with strict inequality whenever $x_k \neq 0$. Multiplying both sides by $p_k \geq 0$ and summing over $k = 1, \dots, K$:

$$\sum_{k=1}^K p_k u\left(w + \frac{x_k}{N}\right) \geq \frac{1}{N} \sum_{k=1}^K p_k u(w + x_k) + \frac{N-1}{N} u(w) \quad (5)$$

That is, $V_2 \geq V_1$, with strict inequality if $x_k \neq 0$ for at least one k with $p_k > 0$. Hence all agents prefer the equal-sharing arrangement. ■

Definition 6.69 (Independence). Let $\Omega = \{\omega_1, \dots, \omega_n\}$ be a finite set. Assign numbers $p_i = \mathbb{P}(\{\omega_i\})$ such that

$$p_i \geq 0, \quad \sum_{i=1}^n p_i = 1$$

For any event $E \subseteq \Omega$, the probability of E is defined as

$$\mathbb{P}(E) = \sum_{\omega_i \in E} p_i$$

Two events $E, F \subseteq \Omega$ are independent if

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$$

Definition 6.70 (Conditional Probability). Let $F \subseteq \Omega$ with $\mathbb{P}(F) > 0$. The conditional probability of E given F is

$$\mathbb{P}(E|F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}$$

If $\{E_1, \dots, E_k\}$ is a partition of Ω , then

$$\mathbb{P}(E_j|F) = \frac{\mathbb{P}(F|E_j)\mathbb{P}(E_j)}{\sum_{i=1}^k \mathbb{P}(F|E_i)\mathbb{P}(E_i)}$$

Definition 6.71 (First Order Stochastic Dominance). A distribution F first order stochastically dominates G if

$$F(x) \leq G(x), \quad \text{for all } x$$

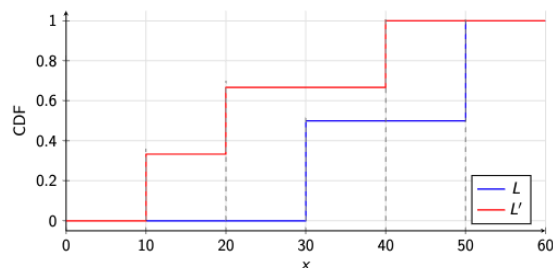
with strict inequality for some x

Definition 6.72 (Second Order Stochastic Dominance). A distribution F second order stochastically dominates G if

$$\int_{-\infty}^x F(t)dt \leq \int_{-\infty}^x G(t)dt, \quad \text{for all } x$$

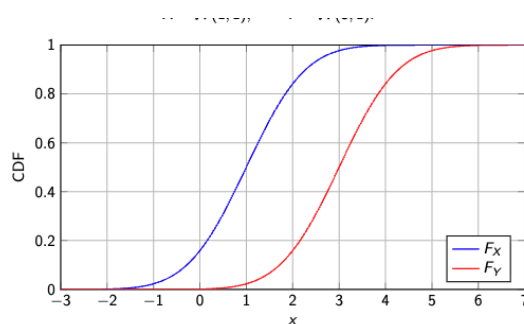
with strict inequality for some x

Example 6.73 (First Order Stochastic Dominance for Discrete Distributions). Consider two lotteries: $L = [1/2, 1/2; 30, 50]$ and $L' = [1/3, 1/3, 1/3; 10, 20, 40]$. Then,



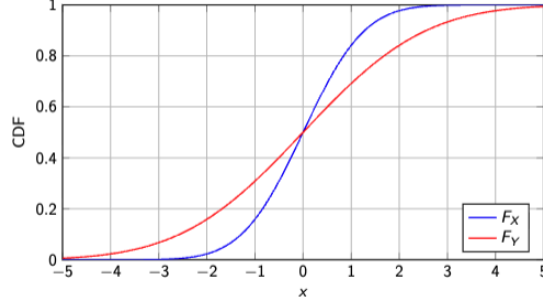
Thus $L \succ_{\text{FOSD}} L'$

Example 6.74 (First Order Stochastic Dominance for Continuous Distributions). Let $X \sim \mathcal{N}(1, 1)$, $Y \sim \mathcal{N}(3, 1)$. Then,



Since $F_Y(x) \leq F_X(x)$ for all x with some strict inequality, $Y \succ_{\text{FOSD}} X$

Example 6.75 (Second Order Stochastic Dominance for Continuous Variables). Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 2)$. Then,



Then, since for all x ,

$$\int_{-\infty}^x F_X(t)dt \leq \int_{-\infty}^x F_Y(t)dt$$

So, $X \succ_{\text{SOSD}} Y$.

Theorem 6.76. If $F \succ_{\text{FOSD}} G$, then $F \succ_{\text{SOSD}} G$

Proof. Suppose $F \succ_{\text{FOSD}} G$. Then,

$$F(x) \leq G(x), \quad \text{for all } x \in \mathbb{R}$$

Integrate both sides from $-\infty$ to x yields (since the integral is an increasing function over a non-negative function which holds by definition of the CDF)

$$\int_{-\infty}^x F(t)dt \leq \int_{-\infty}^x G(t)dt$$

Rearranging,

$$\int_{-\infty}^x (G(t) - F(t))dt \geq 0, \quad \text{for all } x$$

Since both F and G are CDFs,

$$\int_{-\infty}^{\infty} (G(t) - F(t))dt = 0$$

Therefore,

$$F \succ_{\text{SOSD}} G$$

■

Proposition 6.77. If $F \succ_{\text{FOSD}} G$, then for every increasing function $u : \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E}_F[u(X)] \geq \mathbb{E}_G[u(X)]$$

Proof. Write expectations

$$\mathbb{E}_F[u(X)] = \int_{-\infty}^{\infty} u(x)dF(x), \quad \mathbb{E}_G[u(X)] = \int_{-\infty}^{\infty} u(x)dG(x)$$

Apply integrating by parts:

$$\int_{-\infty}^{\infty} u(x)dH(x) = [u(x)H(x)]_{x=-\infty}^{x=+\infty} - \int_{-\infty}^{\infty} H(x)du(x)$$

Because $H(-\infty) = 0$ and $H(+\infty) = 1$, applying to F and G :

$$\mathbb{E}_F[u] - \mathbb{E}_G[u] = (u(+\infty) - \int_{-\infty}^{\infty} F(x)du(x)) - (u(+\infty) - \int_{-\infty}^{\infty} G(x)du(x)) = \int_{-\infty}^{\infty} (G(x) - F(x))du(x) > 0$$

■

Neither FOSD the other. Want to check SOSD. The CDFs are:

$$F_X(t) = \begin{cases} 0 & t < 0.8 \\ 0.3 & 0.8 \leq t < 1 \\ 0.94 & 1 \leq t < 4 \\ 1 & t \geq 4 \end{cases} \quad F_Y(t) = \begin{cases} 0 & t < 0.8 \\ 0.9 & 0.8 \leq t < 1 \\ 0.9 & 1 \leq t < 4 \\ 1 & t \geq 4 \end{cases}$$

where X is PhD and Y is No PhD. We compute $\int_0^x F_X(t) dt - \int_0^x F_Y(t) dt$ directly on each interval:

- $x \in [0, 0.8)$: Both CDFs are zero, so the difference is 0.
- $x \in [0.8, 1)$: The difference is $\int_{0.8}^x (0.3 - 0.9) dt = -0.6(x - 0.8) \leq 0$, with strict inequality for $x > 0.8$.
- $x \in [1, 4)$: The difference is $-0.6(0.2) + \int_1^x (0.94 - 0.9) dt = -0.12 + 0.04(x - 1)$. At $x = 1$ this is $-0.12 < 0$, and it reaches zero only when $x - 1 = 3$, i.e. $x = 4$. So the difference is strictly negative on $[1, 4)$.
- $x = 4$: The difference is $-0.12 + 0.04 \times 3 = 0$.

Therefore $\int_0^x F_X(t) dt \leq \int_0^x F_Y(t) dt$ for all x , with strict inequality for $x \in (0.8, 4)$. Hence No PhD $\succ_{\text{SOSD}}$ PhD.

- (d) Assume the career advisor is right. Furthermore, assume that earning a PhD eliminates the possibility that Ann ends up with £0.8m wealth (only £4m or £1m are possible). How likely should be the best outcome to convince Ann that a PhD is worth doing?
PhD becomes the lottery $L = [p, 1 - p; 4, 1]$. Then,

$$\begin{aligned} EU[\text{PhD}] &= pu(4) + (1 - p)u(1) = p + (1 - p)0.1 = 0.1 + 0.9p \\ EU[\text{No PhD}] &= 0.06u(4) + 0.3u(0.8) + 0.64u(1) = 0.06 + 0.3 \cdot 0 + 0.64 \cdot 0.1 = 0.124 \end{aligned}$$

$$\text{We need } 0.1 + 0.9p \geq 0.124 \implies p \geq \frac{2}{75}$$

We also have likelihood ratio dominance. This is motivated by the following. Suppose we have a uniform distribution over three outcomes $l < m < h$ with $[p_l, p_m, p_h] = [1/3, 1/3, 1/3]$.

Then, $[0, 0.6, 0.4]$ is an FOSD improvement since the probability of the worst outcome is lower and the probabilities of the other two are higher. Similarly, $[q_l, q_m, q_h] = [0, 0.4, 0.6]$ is also an FOSD improvement and the probabilities of better outcomes are increased in greater proportions i.e.

$$\frac{q_l}{p_l} < \frac{q_m}{p_m} < \frac{q_h}{p_h}$$

These are called likelihood ratios and when they are increasing in outcome x , the second lottery LR dominates the first one.

Definition 6.80 (Likelihood Ratio Dominance). Let $X = \{x_1 < x_2 < \dots < x_n\}$ be a finite ordered set. Let $p_i = \mathbb{P}_F(x_i)$ and $q_i = \mathbb{P}_G(x_i)$ with $p_i, q_i > 0$ for all i . We say that F likelihood-ratio dominates G ($F \succ_{\text{LR}} G$) if the sequence

$$\frac{p_i}{q_i}$$

is weakly increasing in i i.e.

$$\frac{p_{i+1}}{q_{i+1}} \geq \frac{p_i}{q_i}, \quad \text{for all } i = 1, \dots, n - 1$$

More generally, let F and G be two distributions with densities (or probability mass functions) f and g with common support S . We say that F likelihood-ratio dominates G (denoted $F \succ_{\text{LR}} G$) if the likelihood ratio

$$\frac{f(x)}{g(x)}$$

is weakly increasing in x on S

Theorem 6.81. Let F, G have densities $f, g > 0$ on a common support. If $\lambda(x) = \frac{f(x)}{g(x)}$ is increasing, then $F(x) \leq G(x)$ for all x

Proof. Since

$$\int \lambda(x)g(x)dx = \int f(x)dx = 1$$

the increasing function λ can cross the level 1 at most once. Hence, there exists c such that

$$\lambda(x) \leq 1 \text{ for } x \leq c, \quad \lambda(x) \geq 1 \text{ for } x \geq c$$

Equivalently,

$$f(x) \leq g(x) \text{ for } x \leq c, \quad f(x) \geq g(x) \text{ for } x \geq c$$

Thus,

$$F(x) - G(x) = \int_{-\infty}^x (f(t) - g(t))dt \leq 0, \text{ for all } x$$

because the integrand is nonpositive before c and any positive mass after c is needed to keep total mass equal. Therefore, $F(x) \leq G(x)$ for all x ■

Lotteries

We want to extend our rational preferences $\succsim$ over the distributions over state space:

Definition 6.82 (Simple lottery). A simple lottery is a probability distribution over a fixed set of outcomes $X = \{x_1, \dots, x_n\}$:

$$L = [p_1, \dots, p_n; x_1, \dots, x_n]$$

with $p_i \geq 0$ and $\sum_{i=1}^n p_i = 1$

Definition 6.83 (Compound lottery). Given a set of lotteries $L_1, L_2, \dots, L_K$ and probabilities $q_1, \dots, q_K$, $\sum_{k=1}^K q_k = 1$, a compound lottery

$$L = [q_1, \dots, q_K; L_1, \dots, L_K]$$

is the risky alternative that yields lottery L_k with probability q_k .

Thus, a compound lottery is a probability distribution over the lotteries, which can be reduced to a lottery over the set of outcomes.

We make the assumption that *decisions makers consider compound lotteries in their reduced form* i.e. where we readjust the probabilities such that it becomes a simple lottery.

Example 6.84 (Compound Lottery in Reduced Form). Bob owns \$4M at home, \$12M overseas, but the latter is confiscated with 50% chance before it is shipped, in which case it still perishes at sea with 25% chance; his final wealth is lottery

$$B = \left[\frac{1}{2}, \frac{1}{2}; 4, A \right]$$

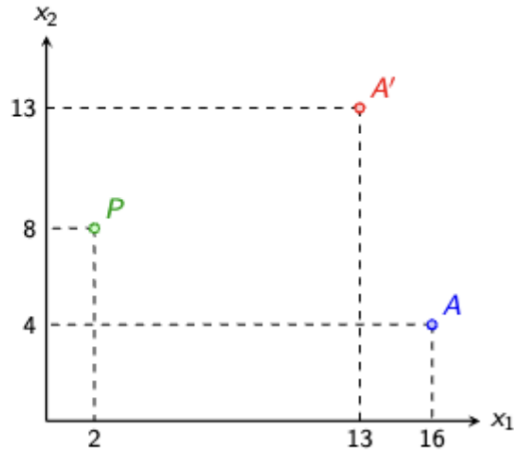
where

$$A = \left[\frac{1}{4}, \frac{3}{4}; 4, 16 \right]$$

so,

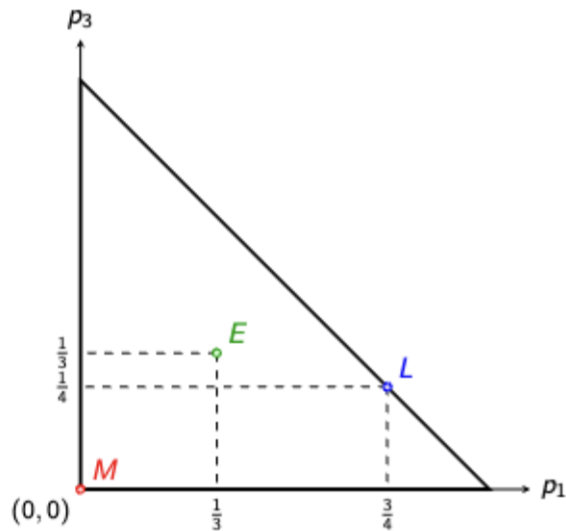
$$B = \left[\frac{1}{2}, \frac{1}{2}; 4, A \right] = \left[\frac{5}{8}, \frac{3}{8}; 4, 16 \right]$$

We want to plot lotteries with fixed probabilities and we can consider the space of lottery outcomes to represent the lottery. Referring to the previous example



- A indicates Ann's initial lottery
- A' is the lottery when Ann sells the goods for \$9M and wires money home
- P is the lottery of the pirate with initial wealth of \$2M who was hunting Ann's shipment, but recovers only half of it's value if succeeds
- We cannot plot Bob here because his probabilities are different

Alternatively, we can plot lotteries with fixed prizes (but varying probabilities). Furthermore, because probabilities add up to one, we can plot 3 probabilities on a 2 dimensional graph. This is called the Marschak-Machina triangle. Let $x_3 \succ x_2 \succ x_1$ and $L = [p_1, p_2, p_3; x_1, x_2, x_3]$



- Lottery L puts probability $3/4$ on the worst outcome and probability $1/4$ on the best outcome
- Lottery M delivers the median outcome for sure
- Lottery E puts equal probability on each outcome

Expected Utility

Now we want to construct preferences over lotteries.

We first have the very basic properties that are needed by Debreu's theorem, but they do not give enough to tell us about the structure of the utility function over lotteries

Definition 6.85 (Basic Properties of rational and continuous preferences over lotteries). Consider a space of simple lotteries over $X = \{x_1, \dots, x_n\}$:

$$\mathcal{L} = \{L = [p_1, \dots, p_n; x_1, \dots, x_n]\}$$

Then, a rational and continuous preference is

- complete: $\forall L, L' \in \mathcal{L}$ we have either $L \succsim L'$ or $L' \succsim L$ or both
- transitive: if $L \succsim L'$ and $L' \succsim L''$, then $L \succsim L''$
- continuous: if $L \succsim L' \succsim L''$ then there exists probability p such that

$$pL + (1-p)L'' \equiv [p, 1-p; L, L''] \sim L'$$

We need the additional axiom

Definition 6.86 (Independence Axiom). The preference relation on the space of simple lotteries satisfies the independence axiom if for all $L, L', L'' \in \mathcal{L}$ and $p \in (0, 1)$

$$L \succsim L' \text{ if and only if } pL + (1-p)L'' \succsim pL' + (1-p)L''$$

The basic intuition is that the decision maker does not worry about what could have happened but did not. The whole enterprise is to want to show that a preference relation can be represented by a von-Neumann-Morgenstern (expected) utility function (and you can provided it satisfies the four properties)

Definition 6.87 (Expected Utility). The utility function $U : \mathcal{L} \rightarrow \mathbb{R}$ has the (von-Neumann-Morgenstern) expected utility form if there exist numbers $u(x_i)$ such that for every $L = [p_1, \dots, p_n; x_1, \dots, x_n] \in \mathcal{L}$

$$U(L) = \sum p_i u(x_i)$$

Note that $U([1, x_i]) = u(x_i)$, that is, the value of the lottery which gives x_i for sure equals to the utility of x_i . An increasing utility function $u(x_i)$ defined on monetary payoffs is referred to as Bernoulli utility function.

Then, we have the following result. The key idea is that by the Independence Axiom, the utility of lottery L can be decomposed as the sum of utilities received in a mutually exclusive event when a particular outcome occurs.

Theorem 6.88 (Expected Utility Theorem). Suppose that agent's preferences over lotteries $\mathcal{L}$ are complete, transitive, continuous and satisfy the Independence Axiom. Then, there exists a function $u(x)$ such that the utility of any lottery $L = [p_1, \dots, p_n; x_1, \dots, x_n] \in \mathcal{L}$ can be written as

$$U(L) = p_1 u(x_1) + \dots + p_n u(x_n) = \sum_{i=1}^n p_i u(x_i)$$

i.e. for all $L \succsim L'$

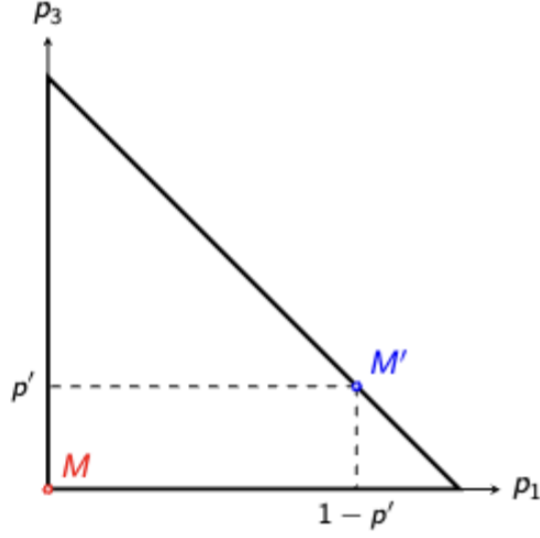
$$L \succsim L' \iff U(L) = \sum_{i=1}^n p_i u(x_i) \geq U(L') = \sum_{i=1}^n p'_i u(x'_i)$$

Proof. Consider a three outcome case $x_3 \succ x_2 \succ x_1$. Assign

$$U([1, x_1]) = u(x_1) = 0$$

$$U([1, x_3]) = u(x_3) = 1$$

Using continuous, there must be $M' = [1 - p', 0, p'; x_1, x_2, x_3]$ such that $M \sim M'$ so we can assign $U(M) = u(x_2) = p'$ since $x_2 \sim (1 - p')x_1 + p'x_3$. We can represent this on the Marschak-Machina triangle



Because $M \sim M'$, by the Independence Axiom, we must have that for any $q \in [0, 1]$,

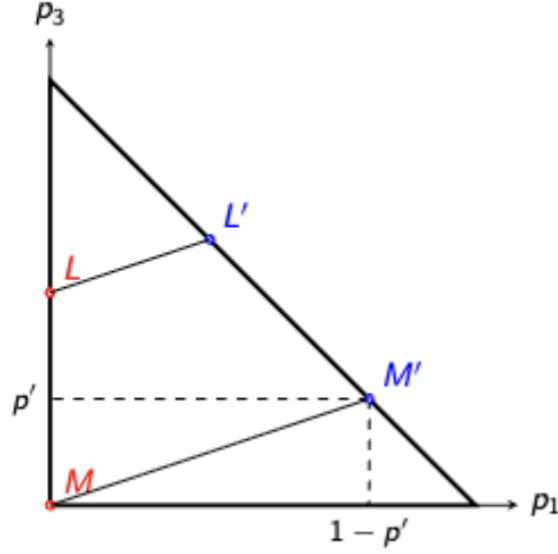
$$M \sim qM' + (1 - q)M$$

Hence, we have our first indifference curve. Now, fix some α and consider

$$L = \alpha M + (1 - \alpha)[1; x_3]$$

$$L' = \alpha M' + (1 - \alpha)[1; x_3]$$

Again, applying the Independence Axiom to L and L' , we get a parallel indifference curve with slope $p'/(1 - p')$.



Hence, $\forall L \in \mathcal{L}$, we have $U(L) = a + bp_1 + cp_3$. Checking corners: $a = p'$, $b = -p'$, $c = 1 - p'$. So,

$$U(L) = 0 \cdot p_1 + (1 - p_1 - p_3)p' + p_3 = p_1u(x_1) + p_2u(x_2) + p_3u(x_3)$$

■

How might we check if the utility function has an expected utility form? We have the following two properties.

Proposition 6.89 (Linearity of Expected Utility). $U : \mathcal{L} \rightarrow \mathbb{R}$ has the expected utility form if and only if it is linear:

$$U\left(\sum_{k=1}^K q_k L_k\right) = \sum_{k=1}^K q_k U(L_k)$$

for any $L_k \in \mathcal{L}$ and $\sum_{k=1}^K q_k = 1$

Proof. Suppose that U has the expected utility form, then

$$U\left(\sum_{k=1}^K q_k L_k\right) = \sum_{i=1}^n u(x_i) \left(\sum_{k=1}^K q_k p_i^k\right) = \sum_{k=1}^K q_k \left(\sum_{i=1}^n p_i^k u(x_i)\right) = \sum_{k=1}^K q_k U(L_k)$$

Suppose that U is linear. Then, any lottery $L = [p_1, \dots, p_n; x_1, \dots, x_n]$ can be written as a sum of degenerate lotteries $L^i = [1, x_i]$. Hence

$$U(L) = U\left(\sum_{i=1}^n p_i L^i\right) = \sum_{i=1}^n p_i u(x_i)$$

hence U has the expected utility form. ■

Proposition 6.90 (Invariance to Affine Transformation). Suppose that $U : \mathcal{L} \rightarrow \mathbb{R}$ is the expected utility function for the preference relation $\succsim$ on $\mathcal{L}$. Then V is another expected utility function representing $\succsim$ if and only if there exists $a > 0$ and b such that

$$V(L) = aU + b$$

Remark. In other words, the expected utility function is unique up to affine transformations (linear transformations), not ANY monotone transformations. Because non-linear transformations change the risk attitude and hence the preference over $\mathcal{L}$

Example 6.91. Determine whether the following preferences (decision criteria) are consistent with the EU hypothesis (Ariel Rubinstein's examples). Assume $L = [p_1, \dots, p_N; x_1, \dots, x_N]$ with $x_1 < \dots < x_N$

- (a) Preference for “less dispersion” $U(L) = -\sum (p_i - \frac{1}{N})^2$.

Let outcomes be $\{x_1, x_2\}$.

- $L = (1, 0)$: $U(L) = -(1 - \frac{1}{2})^2 - (0 - \frac{1}{2})^2 = -\frac{1}{2}$.
- $L' = (\frac{1}{2}, \frac{1}{2})$: $U(L') = 0$.

So $L' \succ L$. Now mix with $L'' = (0, 1)$ at $\alpha = \frac{1}{2}$:

$$\begin{aligned}\frac{1}{2}L' + \frac{1}{2}L'' &= (\frac{1}{4}, \frac{3}{4}), U = -(\frac{1}{4} - \frac{1}{2})^2 - (\frac{3}{4} - \frac{1}{2})^2 = -\frac{1}{8}. \\ \frac{1}{2}L + \frac{1}{2}L'' &= (\frac{1}{2}, \frac{1}{2}), U = 0.\end{aligned}$$

After mixing, $\frac{1}{2}L + \frac{1}{2}L'' \succ \frac{1}{2}L' + \frac{1}{2}L''$, **reversing** the original ranking. The Independence Axiom is violated. $\square$

- (b) Preference for “more certainty” $U(L) = \max\{p_1, \dots, p_N\}$.

Let outcomes be $\{x_1, x_2, x_3\}$.

- $L = (1, 0, 0)$: $U(L) = 1$.
- $L' = (0, 1, 0)$: $U(L') = 1$.

So $L \sim L'$. Mix with $L'' = (\frac{1}{2}, 0, \frac{1}{2})$ at $\alpha = \frac{1}{2}$:

$$\begin{aligned}\frac{1}{2}L + \frac{1}{2}L'' &= (\frac{3}{4}, 0, \frac{1}{4}), U = \frac{3}{4}. \\ \frac{1}{2}L' + \frac{1}{2}L'' &= (\frac{1}{4}, \frac{1}{2}, \frac{1}{4}), U = \frac{1}{2}.\end{aligned}$$

Independence requires the indifference to be preserved under mixing, but $\frac{3}{4} > \frac{1}{2}$, so the mixture breaks the tie. The Independence Axiom is violated. $\square$

- (c) More likely “good outcomes” $U(L) = \sum_{i \geq G} p_i$ for a given G .

Define

$$u(x_i) = \begin{cases} 1 & \text{if } i \geq G, \\ 0 & \text{if } i < G. \end{cases}$$

Then

$$U(L) = \sum_{i \geq G} p_i = \sum_{i=1}^N p_i u(x_i),$$

which is linear in the probabilities p_i . Hence U is an expected utility and the Independence Axiom is satisfied. $\square$

- (d) Better “worst-case” $U(L) = \min\{u(x_i) \text{ such that } p_i > 0\}$

Let $u(x_1) = 0$, $u(x_2) = 1$, $u(x_3) = 2$.

- $L = (0, 1, 0)$: $U(L) = 1$.
- $L' = (0, 0, 1)$: $U(L') = 2$.

So $L' \succ L$. Mix with $L'' = (1, 0, 0)$ at $\alpha = \frac{1}{2}$:

$$\begin{aligned}\frac{1}{2}L' + \frac{1}{2}L'' &= (\frac{1}{2}, 0, \frac{1}{2}), U = \min\{0, 2\} = 0. \\ \frac{1}{2}L + \frac{1}{2}L'' &= (\frac{1}{2}, \frac{1}{2}, 0), U = \min\{0, 1\} = 0.\end{aligned}$$

After mixing, $\frac{1}{2}L' + \frac{1}{2}L'' \sim \frac{1}{2}L + \frac{1}{2}L''$, but Independence requires $\frac{1}{2}L' + \frac{1}{2}L'' \succ \frac{1}{2}L + \frac{1}{2}L''$. The strict preference is not preserved. The Independence Axiom is violated. $\square$

- (e) Better “most-likely prize” $U(L) = \{\text{highest } k \text{ with } p_k \geq p_i, \forall i\}$

Let outcomes be $\{x_1, x_2, x_3\}$ with x_3 best.

- $L = (0, 0.6, 0.4)$: most likely prize is x_2 , so $U(L) = 2$.
- $L' = (0.6, 0, 0.4)$: most likely prize is x_1 , so $U(L') = 1$.

So $L \succ L'$. Mix with $L'' = (0, 0, 1)$ at $\alpha = \frac{1}{2}$:

$$\frac{1}{2}L + \frac{1}{2}L'' = (0, 0.3, 0.7), \text{ most likely: } x_3, \quad U = 3.$$

$$\frac{1}{2}L' + \frac{1}{2}L'' = (0.3, 0, 0.7), \text{ most likely: } x_3, \quad U = 3.$$

After mixing, both lotteries are tied ($U = 3$), but Independence requires the strict preference $L \succ L'$ to be preserved. The Independence Axiom is violated. $\square$

Theorem 6.92 (EU and FOSD). Let L and L' be lotteries with CDFs F and G . If $F \succ_{\text{FOSD}} G$ (i.e. $F(x) \leq G(x)$ for all x), then any expected-utility maximiser with an increasing utility $u : \mathbb{R} \rightarrow \mathbb{R}$ (for which the expectations below exist) prefers L to L' :

$$\mathbb{E}_F[u(X)] \geq \mathbb{E}_G[u(X)]$$

Proof. As $u(x)$ is increasing, the result follows by definition of FOSD ■

Note that one structure of the state space almost does not matter: we just need to be able to re-arrange the outcomes so that u is increasing on the outcome space and then check the FOSD property.

Note that the EU Theorem explicitly relies on continuity and independence. Continuity implies the existence of indifference curves while the Independence Axiom implies that the indifference curves, when drawn in probability space are:

- linear
- parallel

However, the EU hypothesis/theorem may not always hold in practice (i.e. independence axiom is not satisfied by the preferences of actual people) Maurice Allais (Nobel Prize 1988) proposed experiments producing consistent violations of the EU hypothesis.

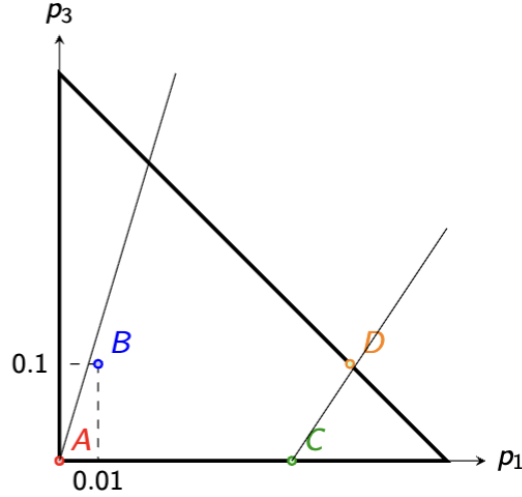
Allais paradox setup: choose either A or B and then either C or D:

- (A) win \$1 million for sure
- (B) win \$5 million with 10%, \$1 million with 89%, nothing with 1%
- (C) win \$1 million with 11% and nothing with 89%
- (D) win \$5 million with 10% and nothing with 90%

1. What would you choose between A and B?
2. What would you choose between C and D?

For most people $A \succ B$, $D \succ C$ (at least when \$1m was big money). We have three outcomes $\{0, \$1, \$5\}$, so we can represent this using the Marschak–Machina triangle.

Plotting A, B, C, D in the (p_1, p_3) plane (where p_1 is the probability of \$0 and p_3 is the probability of \$5m):



The indifference curve such that $A \succ B$ must be steeper than the indifference curve implying $D \succ C$. Algebraically, if $u(\$5m) = 1$, $u(\$1m) = z$, $u(0) = 0$, we get:

$$A \succ B \iff z > 0.1 + 0.89z \iff 0.11z > 0.1 \iff C \succ D$$

Hence $A \succ B$ and $D \succ C$ simultaneously is inconsistent with EU: under EU, $A \succ B$ if and only if $C \succ D$, so the common preference $A \succ B, D \succ C$ violates the EU hypothesis.

Risk Aversion

The motivating idea is from the St Petersburg Paradox

Example 6.93 (St. Petersburg Paradox). Toss a fair coin until the first head appears. If the first head occurs on toss n , the payoff is 2^n . That is, we have a lottery

$$L = \left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots; w+2, w+4, w+8, \dots \right]$$

where w is the initial wealth. How much would you pay to play? People are only willing to pay a modest amount even though the expected monetary value is infinite

$$E[X] = w + \sum_{i=1}^{\infty} \frac{1}{2^n} 2^n = w + \sum_{i=1}^{\infty} 1 = \infty$$

Bernoulli argued that we should evaluate the lottery using expected utility rather than expected value. For instance, if we set $u(x) = \log_2 x$ and $w = 0$,

$$U(L) = \mathbb{E}[u(X)] = \sum_{i=1}^{\infty} \frac{1}{2^n} \log_2(2^{n-1}) = \sum_{i=1}^{\infty} \frac{n-1}{2^n} = 1 \implies L \sim [1; 2]$$

This expresses risk aversion, which implies that expected utility *converges*

Definition 6.94 (Risk Aversion). Suppose that the agent has EU preferences:

$$U(L) = \mathbb{E}[u(x)]$$

where $u(x)$ is a Bernoulli utility function. An agent is risk averse if they prefer the expected value of a lottery over final wealth to the lottery itself i.e. for all $L \in \mathcal{L}$

$$[1; EV(L)] \succ L \iff u(\mathbb{E}(x)) > \mathbb{E}[u(X)]$$

Proposition 6.95. The expected utility maximiser is risk averse iff their u is concave

Proof. Follows immediately from Jensen's Inequality ■

Proposition 6.96. Consider two lotteries L_F and L_G over monetary outcomes and suppose that $F \succ_{\text{SOSD}} G$. Then, any risk averse expected utility maximiser has $L_F \succ L_G$

Proof. We have shown that a risk averse expected utility maximiser must have a concave Bernoulli utility function $u(x)$. We have shown that if u is increasing and strictly concave and $F \succ_{\text{SOSD}} G$, then

$$\mathbb{E}_F[u(x)] > \mathbb{E}_G[u(x)] \iff U(L_F) > U(L_G)$$

Thus $L_F \succ L_G$ ■

Definition 6.97 (Certainty Equivalent). The certainty equivalent of lottery L for the sure outcome that the agent finds exactly as good as the lottery $[1; CE(L)] \sim L$, or equivalently

$$u(CE(L)) = \mathbb{E}[u(x)]$$

Definition 6.98 (Risk Premium). The risk premium of lottery L is the difference between the monetary equivalent and the certainty equivalent of L :

$$RP(L) = \mathbb{E}[X] - CE(L)$$

Positive risk premium implies that the agent is risk averse since they need greater expected monetary value to hold a risky asset compared to a risk free asset (certainty equivalent).

How might we know that the certainty equivalent exists?

Theorem 6.99 (Existence of the Certainty Equivalent). Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and (weakly) increasing. Let X be a random final wealth such that $\mathbb{E}[u(X)]$ exists i.e. is finite. Then, there exists a unique $CE \in \mathbb{R}$ with

$$u(CE) = \mathbb{E}[u(X)]$$

Proof. Let $m \equiv \mathbb{E}[u(X)]$. Since u is real-valued on $\mathbb{R}$, set

$$u_{\min} := \inf_{x \in \mathbb{R}} u(x), \quad u_{\max} := \sup_{x \in \mathbb{R}} u(x)$$

Because $u(X)$ takes values in $[u_{\min}, u_{\max}]$, its expectation satisfies $u_{\min} \leq m \leq u_{\max}$. If $m = u_{\min}$ or $m = u_{\max}$ then set CE with $u(CE) = m$. Otherwise, $m \in (u_{\min}, u_{\max})$. Then, the continuous image of $u(\mathbb{R})$ is a (connected) interval containing $[u_{\min}, u_{\max}]$. By the Intermediate Value Theorem, there exists $CE \in \mathbb{R}$ with $u(CE) = m$. If u is strictly increasing then u is invertible, so the solution CE is unique. ■

Theorem 6.100. The following are equivalent definitions for an agent (with von-Neumann-Morgenstern preferences represented by a strictly increasing Bernoulli utility function u on final wealth) being risk averse:

- $u(\mathbb{E}[x]) > \mathbb{E}[u(x)]$ agent prefers “ $EV(L)$ for sure” to L
- $RP(L) > 0$
- agent's Bernoulli utility $u(x)$ is strictly concave

In most of our settings we refer to the agent as ‘risk averse’ because the risk attitudes do not change on the space of lotteries $\mathcal{L}$ which implies that u is globally concave.

Example 6.101. Consider the lottery $A = [\frac{1}{4}, \frac{3}{4}; 4, 16]$ and suppose the Bernoulli utility function is $u(x) = \sqrt{x}$. her expected utility is

$$U(A) = \frac{1}{4}\sqrt{4} + \frac{3}{4}\sqrt{16} = 3.5$$

Thus, her certainty equivalent is

$$CE(A) = u^{-1}(U(A)) = 3.5^2 = 12.25$$

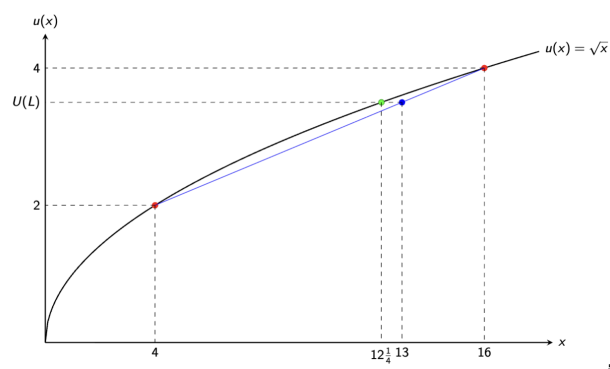
The monetary equivalent of the lottery is

$$\mathbb{E}[X] = \frac{1}{4}4 + \frac{3}{4}16 = 13$$

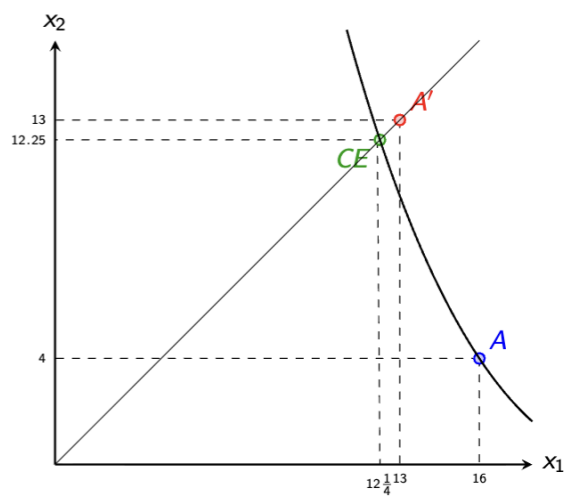
The risk premium is thus

$$RP(A) = \mathbb{E}[X] - CE(A) = 0.75 > 0$$

We can illustrate the utility function as



And we can illustrate in the state space



To find this, we write $U(A) = CE$ as

$$\frac{3}{4}\sqrt{x_1} + \frac{1}{4}\sqrt{x_2} = 3.5$$

which gives

$$x_2 = (14 - 3\sqrt{x_1})^2$$

and $CE(A)$ is the intersection of the indifference curve with the 45-degree line.

Example 6.102. Consider lotteries, all of them with payoffs in $[0, 1]$, that have density functions of the form $f(z) = (\beta + 1)z^\beta$, with $\beta \in [0, 1]$ (notice that $\beta = 0$ corresponds to the uniform distribution).

- (a) Show that the expected value of the lotteries is increasing in β . Can you easily conclude from there that the best lottery is that with $\beta = 1$? Can you maybe use an argument based on first order stochastic dominance?

The expected value is:

$$E[L] = \int_0^1 z(\beta + 1)z^\beta dz = (\beta + 1) \left[\frac{z^{\beta+2}}{\beta + 2} \right]_0^1 = \frac{\beta + 1}{\beta + 2}$$

Differentiating: $\frac{d}{d\beta} \frac{\beta+1}{\beta+2} = \frac{1}{(\beta+2)^2} > 0$, so $E[L]$ is increasing in β . ■

We cannot directly conclude $\beta = 1$ is best since optimality depends on the entire distribution, not just the mean. Instead, use **FOSD**: the CDF is $F_\beta(t) = t^{\beta+1}$. For $\beta' > \beta$ and $t \in (0, 1)$:

$$F_{\beta'}(t) = t^{\beta'+1} < t^{\beta+1} = F_\beta(t)$$

since $t \in (0, 1)$ and $\beta' + 1 > \beta + 1$. Hence lottery β' FOSD lottery β , so $\beta = 1$ FOSD all others and is preferred by any EU maximiser with increasing u . ■

- (b) An individual has risk averse CRRA preferences $u(z) = z^{1-r}$, with $r \in [0, 1]$. Compute the Expected Utility and the Certainty Equivalent of each lottery.

The expected utility is:

$$EU(\beta, r) = \int_0^1 z^{1-r}(\beta + 1)z^\beta dz = (\beta + 1) \int_0^1 z^{\beta+1-r} dz = \frac{\beta + 1}{\beta + 2 - r}$$

Since $u(CE) = EU$, the certainty equivalent satisfies $CE^{1-r} = \frac{\beta+1}{\beta+2-r}$, so:

$$CE = \left(\frac{\beta + 1}{\beta + 2 - r} \right)^{\frac{1}{1-r}}$$

Check with CRRA(1), i.e. $u(z) = \ln z$:

$$EU = \int_0^1 \ln(z)(\beta + 1)z^\beta dz = (\beta + 1) \int_0^1 z^\beta \ln z dz = (\beta + 1) \cdot \frac{-1}{(\beta + 1)^2} = \frac{-1}{\beta + 1}$$

so $CE = e^{-1/(\beta+1)}$.

We also want to say how much more risk averse an agent is compared to another. The fundamental intuition is that an agent is more risk averse if their Bernoulli utility function is more concave. But just comparing u'' is a bad idea since preferences are unique up to affine transformations. So, if $v(x) = bu(x)$ then $v''(x) = bu''(x)$ and $v'' > u''$ but they have the same degree of risk aversion.

Definition 6.103 (Arrow-Pratt measure of absolute risk aversion). The Arrow-Pratt measure of absolute risk aversion is defined as

$$A(x) = -\frac{u''(x)}{u'(x)}$$

Remark. Note the following:

- the second derivative of u is normalised by the first derivative
- the negative sign upfront makes $A(x)$ positive for concave $u(x)$
- invariant to affine transformations: $v(x) = a + bu(x)$ with $b > 0$ since

$$-\frac{v''(x)}{v'(x)} = -\frac{bu''(x)}{bu'(x)} = -\frac{u''(x)}{u'(x)}$$

There are many different ways to say that u_1 is more risk averse than u_2 (and hence many methods to show)

Theorem 6.104 (Arrow-Pratt Theorem). The following statements are equivalent when comparing Bernoulli utility function $u_1(x)$ and $u_2(x)$

1. an agent with utility function u_1 rejects any lottery rejected by an agent with Bernoulli utility function u_2 (and at the same initial wealth level)
2. for every lottery, the risk premium of the lottery under Bernoulli utility function u_1 is larger than that under u_2
3. u_1 is more concave than u_2 , specifically, there is an increasing and strictly concave function g such that $u_1(x) = g(u_2(x))$ for all x
4. the Arrow-Pratt Absolute Measure of risk aversion of u_1 is, as a function of x , greater than that of u_2 , for all x

Proof. We prove the equivalence between (3) and (4). Difference $u_1(x) = g(u_2(x))$ twice:

$$u_1'(x) = g'(u_2(x))u_2'(x)$$

and

$$u_1''(x) = g''(u_2(x))(u_2'(x))^2 + g'(u_2(x))u_2''(x)$$

Compute the Arrow-Pratt measure for u_1 :

$$\begin{aligned} A_1(x) &= -\frac{u_1''(x)}{u_1'(x)} = -\frac{g''(u_2(x))(u_2'(x))^2 + g'(u_2(x))u_2''(x)}{g'(u_2(x))u_2'(x)} \\ &= -\frac{g''(u_2(x))}{g'(u_2(x))}u_2'(x) - \frac{u_2''(x)}{u_2'(x)} > A_2(x) \end{aligned}$$

because $g' > 0, g'' < 0$ and $u_2' > 0$ ■

We can also approximate the Risk Premium for any twice-differentiable utility function

Example 6.105 (Risk Premium under Second-Order Approximation). Consider wealth x subject to additive risk:

$$X = x + \varepsilon, \quad \mathbb{E}[\varepsilon] = 0, \quad \text{Var}(\varepsilon) = \sigma^2.$$

Under expected utility, $CE = x - RP$ satisfies:

$$u(x - RP) = \mathbb{E}[u(X)]$$

Taking second-order approximations around x :

$$\begin{aligned} u(x - RP) &\approx u(x) - RP u'(x), \\ \mathbb{E}[u(X)] &= \mathbb{E}[u(x + \varepsilon)] \approx u(x) + u'(x)\mathbb{E}[\varepsilon] + \frac{1}{2}u''(x)\mathbb{E}[\varepsilon^2]. \end{aligned}$$

Since $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{E}[\varepsilon^2] = \sigma^2$:

$$\mathbb{E}[u(X)] \approx u(x) + \frac{1}{2}u''(x)\sigma^2.$$

Equating both sides:

$$u(x) - RP u'(x) \approx u(x) + \frac{1}{2}u''(x)\sigma^2 \implies RP \approx -\frac{1}{2}\sigma^2 \frac{u''(x)}{u'(x)} = \frac{1}{2}\sigma^2 A(x),$$

where $A(x) = -\frac{u''(x)}{u'(x)}$ is the Arrow-Pratt coefficient of absolute risk aversion.

Definition 6.106 (Constant Absolute Risk Aversion). We say that preferences exhibit constant absolute risk aversion to degree a when $A(x) = a$. We denote this $CARA(a)$

Proposition 6.107. An agent exhibits constant absolute risk aversion to degree a just when

$$u(x) = -e^{-ax}$$

i.e. has a constant $A(x) = a$.

Example 6.108 (CARA with Normal Risk). Write out the expected utility:

$$U(L) = \mathbb{E}[-e^{-aX}] = - \int_{-\infty}^{\infty} e^{-ax} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

Combine exponents:

$$-ax - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{1}{2\sigma^2} [(x - (\mu - a\sigma^2))^2 - a^2\sigma^4].$$

Hence:

$$\mathbb{E}[e^{-aX}] = \exp\left(-a\mu + \frac{1}{2}a^2\sigma^2\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - (\mu - a\sigma^2))^2}{2\sigma^2}\right) dx.$$

The integral equals 1, so:

$$U(L) = -\mathbb{E}[e^{-aX}] = -\exp\left(-a\mu + \frac{1}{2}a^2\sigma^2\right) = -e^{-aCE} \implies CE = \mu - \frac{1}{2}a\sigma^2.$$

Definition 6.109 (Decreasing Absolute Risk Aversion). We say that preferences exhibit decreasing risk aversion if $A(x)$ is decreasing in x

One such example of this is $u(x) = \sqrt{x}$ since $A(x) = 1/2x$

Definition 6.110 (Arrow-Pratt measure of relative risk aversion). The Arrow-Pratt measure of relative risk aversion is

$$R(x) = A(x)x = -\frac{u''(x)}{u'(x)}x$$

Definition 6.111 (Constant Relative Risk Aversion). We say that preferences exhibit constant relative risk aversion to degree r when

$$A(x) = \frac{r}{x}$$

i.e. $R(x) = r$. We denote this $CRRA(r)$

Proposition 6.112. $CRRA(r)$ is $u = x^{1-r}$, for $r \in (0, 1)$ and $u(x) = \ln x$ for $r = 1$

Example 6.113. An individual has expected utility preferences. The individual has wealth w and has to choose an amount $x \leq w$, after which a lottery is conducted in which with probability α she gains $2x$ and with probability $(1 - \alpha)$ she loses x .

- (a) Show that, if the individual is risk averse, the higher the value of α , the higher the amount that she chooses.

$EU(L) = \alpha u(w + 2x) + (1 - \alpha)u(w - x)$. Under risk aversion $u(\cdot)$ is increasing and concave in x . Therefore

$$\frac{dEU(x, \alpha)}{dx} = 2\alpha u'(w + 2x) - (1 - \alpha)u'(w - x) = 0$$

and

$$\frac{d^2EU(x, \alpha)}{dx^2} = 4\alpha u''(w + 2x) + (1 - \alpha)u''(w - x) < 0$$

Hence, FOC maximises and (x^*, α) solves $F(x, \alpha) = 2\alpha u'(w + 2x) - (1 - \alpha)u'(w - x) = 0$. Since $\frac{\partial F}{\partial \alpha} \neq 0$ and F is C^1 in an open ball around (x^*, α) define $x^* = x^*(\alpha)$ by the IFT and

$$\frac{\partial x^*}{\partial \alpha} = -\frac{F_\alpha(x, \alpha)}{F_x(x, \alpha)} = -\frac{u'(w + 2x) + u'(w - x)}{4\alpha u''(w + 2x) + (1 - \alpha)u''(w - x)} \geq 0$$

since $u'(\cdot) > 0$, $u''(\cdot) < 0$ by strict concavity. So x^* increases as α increases under risk aversion.

- (b) Suppose the individual has CRRA(r) preferences with $r \in (0, 1)$. Compute the amount chosen and show that it decreases with the level of risk aversion of the individual. Check her optimal choice with CRRA(1), i.e., logarithmic preferences.

For CRRA(r), $u(w) = \frac{w^{1-r}}{1-r}$. The FOC becomes:

$$2\alpha(1-r)(w+2x)^{-r} = (1-r)(1-\alpha)(w-x)^{-r} \\ \implies \left(\frac{w-x}{w+2x} \right)^r = \frac{1-\alpha}{2\alpha} \implies \frac{w-x}{w+2x} = \left(\frac{1-\alpha}{2\alpha} \right)^{1/r}$$

Let $A = \left(\frac{1-\alpha}{2\alpha} \right)^{1/r}$. Then $w-x = A(w+2x)$, giving:

$$x^* = w \cdot \frac{1-A}{1+2A}$$

To show x^* decreases with r : $\frac{dA}{dr} = -\frac{1}{r^2} \ln\left(\frac{2\alpha}{1-\alpha}\right) \cdot A$. For $\alpha > \frac{1}{3}$, $\ln\left(\frac{2\alpha}{1-\alpha}\right) > 0$, so A increases with r , hence $\frac{1-A}{1+2A}$ decreases with r , so x^* decreases with r . ■

Check with CRRA(1), i.e. $u(w) = \ln w$: The FOC gives:

$$\frac{2\alpha}{w+2x} = \frac{1-\alpha}{w-x} \implies 2\alpha(w-x) = (1-\alpha)(w+2x) \implies x^* = w \cdot \frac{3\alpha-1}{2}$$

Consistent with the formula: as $r \rightarrow 1$, $A \rightarrow \frac{1-\alpha}{2\alpha}$ and:

$$x^* = w \cdot \frac{1 - \frac{1-\alpha}{2\alpha}}{1 + 2 \cdot \frac{1-\alpha}{2\alpha}} = w \cdot \frac{2\alpha - (1-\alpha)}{2\alpha + 2(1-\alpha)} = w \cdot \frac{3\alpha-1}{2}$$

7 Dynamic Optimisation

7.1 Theory of Dynamic Optimisation, Bellman Equation

The fundamental idea of dynamic optimisation (and the Bellman equation) is to transform a sequential problem to a recursive one. First we develop how to reach the sequential problem.

In general, we can characterise our optimisation problem as solving

$$\sup_{\{a_t\}_{t=0}^T} \sum_{t=0}^T \delta^t v(x_t, a_t)$$

whereby the initial condition $x_0 \in X$ and any state $x_t \in X \subset \mathbb{R}^n$ i.e. the set of possible states. Furthermore, we have actions which are *constrained* by the set of admissible actions in state x_t

$$a_t \in A(x_t)$$

and state dynamics/evolution following some rule

$$x_{t+1} = f(x_t, a_t)$$

Example 7.1. Suppose you have a cake which can be eaten at most in three days. Utility is defined as $u = \ln c_t$ and the discount factor equals δ . We can characterise the problem as

- Time: $t = 0, 1, 2$
- State: x_t – how much of the cake is left
- Control: consumption c_t and the admissible set $A(x_t) = \{c_t \leq x_t\}$

- State evolution: $x_{t+1} = x_t - c_t$
- Initial condition is $x_0 = 1$
- Problem:

$$\max_{c_0, c_1, c_2} [\ln c_0 + \delta \ln c_1 + \delta^2 \ln c_2]$$

To find the optimal strategy, we use backward induction

1. In Period 2, we have $c_2 = x_2$ and $V_2(x_2) = \ln(x_2)$
2. In Period 1,
We first take into account period two consumption which is

$$c_2 = x_2 = x_1 - c_1$$

and the utility from period 1 onwards is

$$V_1 = \ln c_1 + \delta V_2(x_2) = \ln c_1 + \delta \ln x_2 = \ln(x_1 - c_1) + \delta \ln x_2$$

Optimising with respect to state x_2 (or control c_1) yields

$$\frac{\partial V_1}{\partial x_2} = -\frac{1}{x_1 - x_2} + \frac{\delta}{x_2} = 0$$

Optimal consumption:

$$c_1 = \frac{x_1}{1 + \delta}, c_2 = x_2 = \frac{\delta x_1}{1 + \delta}, V_1(x_1) = \ln \frac{x_1}{1 + \delta} + \delta \ln \frac{\delta x_1}{1 + \delta}$$

3. In Period 0, we know that if we arrive to period 1 with x_1 of our cake, our utility given the optimal behaviour further on is $V_1(x_1)$. So we have

$$V_0 = \ln c_0 + \delta V_1(x_1) = \ln(x_0 - x_1) + \delta \ln \frac{x_1}{1 + \delta} + \delta^2 \ln \frac{\delta x_1}{1 + \delta}$$

Optimising with respect to x_1 yields

$$\frac{\partial V_0}{\partial x_1} = -\frac{1}{1 - x_1} + \delta \frac{1}{x_1} + \delta^2 \frac{1}{x_1} = 0$$

The Optimal solution is

$$x_1 = \frac{\delta(1 + \delta)}{1 + \delta + \delta^2}, x_2 = \frac{\delta x_1}{1 + \delta} = \frac{\delta^2}{1 + \delta + \delta^2}$$

with payoff function

$$V_0(1) = \ln \frac{1}{1 + \delta + \delta^2} + \delta \ln \frac{\delta}{1 + \delta + \delta^2} + \delta^2 \ln \frac{\delta^2}{1 + \delta + \delta^2}$$

The main insight is that on each step, we can review the continuation payoff as a function of state variable $V(x_t)$. What this means is that in period 0, we had

$$V_0 = \ln c_0 + \delta V_1(x_1)$$

and so is just current utility plus discounted continuation value if you arrive at period 1 with x_1 . Furthermore, instead of looking at optimal actions $\{a_t\}_{t=0}^T$, we can reformulate the problem as choice of optimal sequence of state variables $\{x_t\}_{t=1}^T$.

So, we can do the following to reach the sequence problem

- Reformulate the choice over actions to choice over sequences of states $\{x_t\}_{t=1}^T$
- Restrict dynamics to

$$x_{t+1} \in \Gamma(x_t) = \{f(x_t, a_t) : a_t \in A(x_t)\}$$

This incorporates both the feasible set restriction and the state dynamics rule by saying “ $x_{t+1} = f(x_t, a_t)$ where $a_t \in A(x_t)$ i.e. restricted by the feasible set”

- And we redefine instantaneous payoff as

$$u(x_t, x_{t+1}) = \max_{a_t \in A(x_t) : f(x_t, a_t) = x_{t+1}} v(x_t, a_t)$$

instead of $v(x_t, a_t)$. What this means is that a_t is no longer a choice variable. Instead, our instantaneous payoff $u(x_t, x_{t+1})$ is defined such that for any x_t , look at all the actions that produce x_{t+1} and pick the action that provides the highest payoff.

Definition 7.2 (Sequence Problem). Consider $T \in 1, \dots, \mathbb{N} \cup \infty$, the control set $a_t \in A(x_t)$, a set of possible states $x_t \in X \subseteq \mathbb{R}^n$, state dynamics $x_{t+1} = f(x_t, a_t)$ and a instantaneous payoff function

$$u(x_t, x_{t+1}) = \max_{a_t \in A(x_t) : f(x_t, a_t) = x_{t+1}} v(x_t, a_t)$$

Then, our sequence problem (SP) is

$$\sup_{\{x_t\}_{t=1}^T} \sum_{t=0}^T \delta^t u(x_t, x_{t+1})$$

subject to

$$x_{t+1} \in \Gamma(x_t), x_0 \in X$$

where $\Gamma(x_t) = \{f(x_t, a_t) : a_t \in A(x_t)\}$

Example 7.3 (Infinite Cake Problem). The consumption utility is $u = \ln c_t$, with production $y_t = k_t^\alpha$. Part is consumed and part is reinvested so that $c_t = y_t - k_{t+1}$. Hence, $\Gamma(k_t) = [0, k_t^\alpha]$. And so

$$V(k_0) = \sup_{\{k_t\}_{t=1}^\infty} \sum_{t=0}^\infty \delta^t \ln(k_t^\alpha - k_{t+1})$$

subject to

$$k_{t+1} \in \Gamma(k_t)$$

We can rewrite the problem recursively. This is known as the Bellman equation

Definition 7.4 (Bellman Equation). Consider $T \in 1, \dots, \mathbb{N} \cup \infty$, the control set $a_t \in A(x_t)$, a set of possible states $x_t \in X \subseteq \mathbb{R}^n$, state dynamics $x_{t+1} = f(x_t, a_t)$ and a instantaneous payoff function

$$u(x_t, x_{t+1}) = \max_{a_t \in A(x_t) : f(x_t, a_t) = x_{t+1}} v(x_t, a_t)$$

Then, our optimisation problem is

$$V(x_t) = \sup_{x_{t+1} \in \Gamma(x_t)} [u(x_t, x_{t+1}) + \delta V(x_{t+1})]$$

We need the following regularity assumptions to make SP well-defined.

Definition 7.5 (Feasible Plan). Call any sequence $\{x_t\}_{t=0}^\infty \in X$ a plan. Given $x_0 \in X$, let

$$\Pi(x_0) = \{\{x_t\}_{t=0}^\infty : x_{t+1} \in \Gamma(x_t), t = 0, 1, \dots\}$$

be the set of plans that are feasible from x_0 .

Assumption 7.6. $\Gamma(x)$ is non-empty, for all $x \in X$

Basically, we need it to be the case that there are feasible plans. Furthermore, we need an additional restriction that all feasible plans can be evaluated using the objective function u and the discount rate δ

Assumption 7.7. For all $x_0 \in X$ and $(x_0, x_1, \dots) \in \Pi(x_0)$, the limit $\lim_{n \rightarrow \infty} \sum_{t=0}^n \delta^t u(x_t, x_{t+1})$ exists (although it may be $\pm\infty$)

Remark. To have **Assumption 7.7** hold, it is sufficient for $u(\cdot)$ to be bounded.

Proposition 7.8. We say that V^* satisfies the Bellman equation if three conditions hold:

1. If $|V^*(x_0)| < \infty$, then

$$V^*(x_0) \geq u(x_0, y) + \delta V^*(y), \forall y \in \Gamma(x_0)$$

and for any $\varepsilon > 0$,

$$V^*(x_0) \leq u(x_0, y) + \delta V^*(y) + \varepsilon, \exists y \in \Gamma(x_0)$$

2. If $V^*(x_0) = +\infty$, then there exists a sequence $\{y^k\}$ in $\Gamma(x_0)$ such that

$$\lim_{k \rightarrow \infty} [u(x_0, y^k) + \delta V^*(y^k)] = +\infty$$

3. If $V^*(x_0) = -\infty$, then

$$u(x_0, y) + \delta V^*(y) = -\infty, \forall y \in \Gamma(x_0)$$

Theorem 7.9. Let X, Γ, u and δ satisfy **Assumption 7.6-7.7**. Then, if $V(x_t)$ is a solution to SP, it is a solution to the Bellman equation.

Furthermore, if $V(x_t)$ is a solution to the Bellman equation and satisfies

$$\lim_{n \rightarrow \infty} \delta^n V(x_n) = 0, \forall (x_0, x_1, \dots) \in \Pi(x_0), \forall x_0 \in X$$

then it is a solution to SP

Remark. Note that any solution to the SP is uniquely defined (assuming **Assumptions 7.6-7.7** hold) but that the solution to the Bellman equation might have zero, one or many solutions and thus requires the additional assumption to ensure that there is exactly one solution to the Bellman equation.

For our purposes, the regularity conditions can be ignored. All we need to know is that in our setting, the two problems yield identical (unique) solutions.

Solving Bellman Equation

In general, we solve a difference equation and we usually iterate a functional operator. One shortcut is to guess a solution i.e. typically guess the functional form, then solve for coefficients.

Example 7.10 (Infinite Cake Problem – Logarithmic Utility). You have a cake which can be eaten in your lifetime. Your utility from consuming c_t of the cake on day t is $u = \ln c_t$, and the discount factor is δ . What is the optimal strategy of eating the cake?

Our problem is

$$\max_{c_t} \sum_{t=0}^{\infty} \delta^t \ln c_t \quad \text{s.t.} \quad x_t - c_t = x_{t+1}$$

Rewriting it as a Sequence Problem:

$$V(x_0) = \max_{x_t} \sum_{t=0}^{\infty} \delta^t \ln(x_t - x_{t+1}) \quad \text{s.t.} \quad x_{t+1} \leq x_t$$

Our corresponding Bellman Equation:

$$V(x_t) = \max_{x_{t+1} \leq x_t} [\ln(x_t - x_{t+1}) + \delta V(x_{t+1})]$$

Note that the constraint is $x_{t+1} \in \Gamma(x_t) = \{x_t - c_t : c_t \leq x_t\}$.
Finite cases give us good guesses. For the $T = 2$ case

$$\begin{aligned} V_2(x_2) &= \ln x_2 \\ V_1(x_1) &= \ln \frac{x_1}{1+\delta} + \delta \ln \frac{\delta x_1}{1+\delta} = (1+\delta) \ln x_1 + [\delta \ln \delta - (1+\delta) \ln(1+\delta)] \\ V_0(x_0) &= \ln \frac{x_0}{1+\delta} + \delta \ln \frac{\delta x_0}{1+\delta+\delta^2} + \delta^2 \ln \frac{\delta^2 x_0}{1+\delta+\delta^2} \end{aligned}$$

The conjecture is then

$$V(x) = C_0 + C_1 \ln(x)$$

Substituting this into the Bellman Equation yields

$$C_0 + C_1 \ln(x) = \max_{x_{t+1} \leq x_t} [\ln(x_t - x_{t+1}) + \delta V(x_{t+1})]$$

Our FOC is

$$-\frac{1}{x_t - x_{t+1}} + \frac{\delta C_1}{x_{t+1}} = 0 \implies x_{t+1} = \frac{\delta C_1}{1 + \delta C_1} x_t$$

Plugging back into the Bellman Equation:

$$C_0 + C_1 \ln(x) = \ln \left(x_t - \frac{\delta C_1}{1 + \delta C_1} x_t \right) + \delta \left[C_0 + C_1 \ln \left(\frac{\delta C_1}{1 + \delta C_1} x_t \right) \right]$$

We then compute the free coefficient and coefficient in front of $\ln$, get two equations with two unknowns

$$C_0 + C_1 \ln(x) = (1 + \delta C_1) \ln x_t + \delta C_0 - (1 + C_1) \ln(1 + \delta C_1) + C_1 \ln(\delta C_1)$$

Thus,

$$\begin{aligned} C_1 &= 1 + \delta C_1 \implies C_1 = \frac{1}{1 - \delta} \\ C_0 &= \frac{1}{1 - \delta} \left(\frac{\delta}{1 - \delta} \ln \delta - \ln(1 - \delta) \right) \end{aligned}$$

We want the optimal path

$$x_{t+1} = \frac{\delta C_1}{1 + \delta C_1} x_t = \delta x_t$$

and then we can find the lifetime utility for $x_0 = 1$ which gives us $x_1 = \delta$

$$V(x_0) = \frac{1}{1 - \delta} \left(\frac{\delta}{1 - \delta} \ln \delta - \ln(1 - \delta) \right) + \frac{1}{1 - \delta} \ln \delta$$

Example 7.11 (Infinite Cake Problem – Squareroot utility).

Theorem 7.12 (Envelope Theorem). Consider the Bellman Equation defined as

$$V(x_t) = \sup_{x_{t+1} \in \Gamma(x_t)} [u(x_t, x_{t+1}) + \delta V(x_{t+1})]$$

Then, we have

$$\frac{\partial V(x_t)}{\partial x_t} = \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_t}$$

Proof. We have by definition of the Bellman Equation

$$\begin{aligned} \frac{V(x_t)}{\partial x_t} &= \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_t} + \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_{t+1}} \frac{dx_{t+1}}{dx_t} + \delta \frac{\partial V(x_{t+1}^*)}{\partial x_{t+1}^*} \frac{dx_{t+1}}{dx_t} \\ &= \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_t} + \left[\frac{\partial u(x_t, x_{t+1}^*)}{\partial x_{t+1}} + \delta \frac{\partial V(x_{t+1}^*)}{\partial x_{t+1}} \right] \frac{dx_{t+1}}{dx_t} \\ &= \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_t} \end{aligned}$$

since $\frac{\partial u(x_t, x_{t+1}^*)}{\partial x_{t+1}} = \frac{\partial V(x_{t+1}^*)}{\partial x_{t+1}} = 0$ by construction as they are evaluated at the optimal value x_{t+1}^* . ■

The Envelope Theorem can help us derive the Euler Equation by eliminating $V'(x_{t+1})$.

Example 7.13 (Euler Equation). Consider the standard Bellman Equation:

$$V(x_t) = \max_{x_{t+1} \in \Gamma(x_t)} \{u(x_t, x_{t+1}) + \delta V(x_{t+1})\}$$

Assuming V is differentiable and we have an interior solution, the FOC with respect to the choice variable is

$$\frac{\partial u(x_t, x_{t+1})}{\partial x_{t+1}} + \delta V'(x_{t+1}) = 0$$

Since we do not know what $V'(x_{t+1})$ is we need to eliminate the derivative of the value function. We can do this by shifting the Envelope Condition forward by one period to yield

$$V'(x_{t+1}) = \frac{\partial u(x_{t+1}, x_{t+2})}{\partial x_{t+1}}$$

and substituting this back into the original FOCs:

$$\frac{\partial u(x_t, x_{t+1})}{\partial x_{t+1}} + \delta \frac{\partial u(x_{t+1}, x_{t+2})}{\partial x_{t+1}} = 0$$

which is the discrete time Euler equation that is the second order difference equation in the state variable x . The Euler Equation represents an intertemporal “no-arbitrage” condition:

- $\frac{\partial u(x_t, x_{t+1})}{\partial x_{t+1}}$: the marginal cost (or utility loss) of increasing the state x_{t+1} today (e.g. saving more instead of consuming)
- $\delta \frac{\partial u(x_{t+1}, x_{t+2})}{\partial x_{t+1}}$: the discounted marginal benefit of having a higher state x_{t+1} starting tomorrow
- At the optimum, you cannot gain by shifting a small unit of resources between today and tomorrow

This is relevant in the Optimal Growth Problem (Solow model with micro-foundations)

Example 7.14 (Optimal Growth Problem). Consider an economy with $T = \infty$, consumption utility $u = \ln c_t$, production $y_t = k_t^\alpha$ with $\alpha \in (0, 1)$ and $c_t = y_t - k_{t+1}$ (part is consumed and part is reinvested), and initial condition k_0 (i.e. starting capital stock). Our Bellman equation is

$$V(k_t) = \max_{k_{t+1}} [\ln[k_t^\alpha - k_{t+1}] + \delta V(k_{t+1})]$$

In two periods, we have

$$V_1(k_1) = \ln k_1^\alpha$$

and

$$V_0 = \ln[k_0^\alpha - k_1] + \delta \ln k_1^\alpha$$

Our FOC is thus

$$-\frac{1}{k_0^\alpha - k_1} + \frac{\alpha \delta}{k_1} = 0 \implies k_1 = \frac{\alpha \delta}{1 + \alpha \delta} k_0^\alpha$$

Plugging back into the value function yields

$$V_0(k_0) = \ln \left(\frac{1}{1 + \alpha \delta} k_0^\alpha \right) + \delta \ln \left(\frac{\alpha \delta}{1 + \alpha \delta} k_0^\alpha \right)$$

So, in the infinite period case, we conjecture $V(k) = C_0 + C_1 \ln k$. Our Bellman equation is

$$C_0 + C_1 \ln k_t = \max_{k_{t+1}} [\ln(k_t^\alpha - k_{t+1}) + \delta C_0 + \delta C_1 \ln k_{t+1}]$$

The FOC is

$$-\frac{1}{k_t^\alpha - k_{t+1}} + \delta \frac{C_1}{k_{t+1}} = 0 \implies k_{t+1} = \frac{\delta C_1}{1 + \delta C_1} k_t^\alpha$$

Plug back into the Bellman

$$C_0 + C_1 \ln k_t = \ln \left(k_t^\alpha - \frac{\delta C_1}{1 + \delta C_1} k_t^\alpha \right) + \delta \left[C_0 + C_1 \ln \left(\frac{\delta C_1}{1 + \delta C_1} k_t^\alpha \right) \right]$$

Solving for the coefficients we have

$$C_0 + C_1 \ln k_t = \alpha(1 + \delta C_1) \ln k_t + \delta C_0 - (1 + C_1) \ln(1 + \delta C_1) + C_1 \ln(\delta C_1)$$

Thus,

$$C_1 = \alpha(1 + \delta C_1) \implies C_1 = \frac{\alpha}{1 - \alpha\delta}$$

The optimal path is

$$k_{t+1} = \frac{\delta C_1}{1 + \delta C_1} k_t^\alpha = \alpha \delta k_t^\alpha$$

The Euler Equation can also give us more insight into the growth rate of consumption. Let $u(k_t, k_{t+1}) = \ln(k_t^\alpha - k_{t+1})$ and

- $\frac{\partial u}{\partial k_{t+1}} = -\frac{1}{k_t^\alpha - k_{t+1}} = -\frac{1}{c_t}$
- $\frac{\partial u}{\partial k_t} = \frac{\alpha k_t^{\alpha-1}}{k_t^\alpha - k_{t+1}} = \frac{\alpha k_t^{\alpha-1}}{c_t}$

Substituting into the Euler Equation:

$$-\frac{1}{c_t} + \delta \left(\frac{\alpha k_{t+1}^{\alpha-1}}{c_{t+1}} \right) = 0 \implies \frac{c_{t+1}}{c_t} = \delta \alpha k_{t+1}^{\alpha-1}$$

This is the Keynes-Ramsey Rule in discrete time and governs the growth rate of consumption

In general, the Bellman Equation is used to solve discrete problems as well (especially in Computer Science).

Example 7.15 (Towers of Hanoi). In a temple in Hanoi monks are displacing 64 golden disks with a hole in the middle, all of different sizes on three ivory sticks. They observe the following rules

1. it is only allowed to move one disk at a time from one stick to another
2. it is forbidden to put a larger disk on top of a smaller disk

If monks move one disk per second and follow an optimal strategy, how long will it take to move all disks from one stick to another.

This is a minimisation rather than maximisation problem. Our goal is to come up with the Bellman equation. Let $V(n)$ be the minimum number of moves required to move n disks from one stick to another. In order to move $n + 1$ disks we need:

- move top n disks to another stick – $V(n)$ moves
- move the largest disk – one move
- put n disks back on top of the large one – $V(n)$ moves

So, our Bellman equation is

$$V(n + 1) = 2V(n) + 1$$

So, $V(n) = 2^n - 1$ and it will take $2^{64} - 1$ seconds

7.2 Stochastic Dynamic Optimisation, Search

Stochastic Optimisation tends to rely on Markov chains to switch between states. We restrict our attention to simple problems of search which do not rely on Markov Chains and the like.

Fixed Sample Search

Definition 7.16 (Fixed Sample Problem). Suppose there is a large number of boxes with unknown prizes where each box k contains a prize $u_k \sim \text{i.i.d. } F(u)$ where $F(u)$ is continuous, differentiable and has full support. Furthermore, $F(u)$ has finite support $[\underline{u}, \bar{u}]$ or finite moments of the distribution. Also usually assume $\underline{u} \geq 0$ and 0 is an outside option. Pandora has to decide how many boxes she wishes to open ex ante with a cost of s per box. After deciding the number of boxes opened, Pandora collects the largest prize.

Theorem 7.17 (Optimal Search Rule for Fixed Sample). Consider the Fixed Sample Search Problem. Then the optimal search rule is n^* that satisfies

$$s = g(n^*) = \int_{\underline{u}}^{\bar{u}} [1 - F(u)][F(u)]^{n-1} du$$

We give the intuition for this. The basic idea is that we stop searching a new box when the marginal gain equals the marginal cost of searching

1. First, we want to find the distribution of $\hat{u} = \max\{u_1, \dots, u_n\}$. The probability of $\hat{u} \leq u$ is

$$\mathbb{P}(\hat{u} \leq u) = \mathbb{P}(u_1 \leq u, \dots, u_n \leq u) = \mathbb{P}(u_1 \leq u) \times \dots \times \mathbb{P}(u_n \leq u) = [F(u)]^n$$

2. The Utility from Search is

$$\int_{\underline{u}}^{\bar{u}} u d[F(u)]^n = n \int_{\underline{u}}^{\bar{u}} u [F(u)]^{n-1} f(u) du$$

The reason why this equality holds is we are writing the Stieltjes differential $d[F(u)]^n$ as an ordinary differential equation using the chain rule. Letting $G(u) = [F(u)]^n$, we have

$$G'(u) = nu[F(u)]^{n-1}$$

and so

$$d[F(u)]^n = G'(u)du = n[F(u)]^{n-1} f(u)du$$

and substituting into the original integral gives us the equality. The search cost is ns .

3. The marginal effect: Define the difference between sampling n and $n - 1$ boxes is

$$\begin{aligned} g(n) &= \int_{\underline{u}}^{\bar{u}} u d[F(u)]^n - \int_{\underline{u}}^{\bar{u}} u d[F(u)]^{n-1} \\ &= [uF(u)^n]_{\underline{u}}^{\bar{u}} - \int_{\underline{u}}^{\bar{u}} [F(u)]^n du - [uF(u)^{n-1}]_{\underline{u}}^{\bar{u}} + \int_{\underline{u}}^{\bar{u}} [F(u)]^{n-1} du \\ &= \bar{u} - \underline{u} + \int_{\underline{u}}^{\bar{u}} [1 - F(u)][F(u)]^{n-1} du \\ &= \int_{\underline{u}}^{\bar{u}} [1 - F(u)][F(u)]^{n-1} du \end{aligned}$$

So, the optimal number of boxes is given by

$$s = g(n)$$

The second equality follows by using the Riemann-Stieltjes where

$$\int adb = ab - \int bda$$

Proposition 7.18 (Existence and Uniqueness of Optimal Search). If it is worth opening at least one box, i.e. if $\underline{u} = 0$,

$$s < g(1) = \int_{\underline{u}}^{\bar{u}} u dF(u)$$

Furthermore, if the solution for $g(n) = s$ exists, then it is unique

Briefly, the reason for why it is unique if the solution exists is because $g(n)$ is decreasing in n . Furthermore, the reason why the existence of a solution is guaranteed if it is worth opening at least one box is since

$$\lim_{n \rightarrow \infty} g(n) = 0$$

Remark. Let n^* be the solution to $g(n) = s, n^* \in \mathbb{R}$. If $n^* \in \mathbb{N}$, then Pandora is indifferent between opening n^* and $n^* - 1$ boxes. If $n^* \notin \mathbb{N}$, then Pandora prefers to open $\lfloor n^* \rfloor$ boxes.

Sequential Search

Definition 7.19 (Sequential Search Problem). Suppose there are T number of boxes with unknown prizes where each box k contains a prize $u_k \sim \text{i.i.d. } F(u)$ where $F(u)$ is continuous, differentiable and has full support. Furthermore, $F(u)$ has finite support $[\underline{u}, \bar{u}]$ or finite moments of the distribution. Also usually assume $\underline{u} \geq 0$ and 0 is an outside option. Opening each box costs s . Pandora discounts future by δ per search round, and opens boxes sequentially and can stop at any moment. There is free recall, where Pandora can return to rejected options for free and no recall, where Pandora cannot go back to discarded options.

Free Recall

The main goal is to find the reservation value where you accept immediately once you discover a prize that is higher than that value and continue searching if you don't. We denote the reservation value z .

In the simple two box case, the reservation value is when Pandora is indifferent between accepting a box with prize z and continuing to search i.e. z satisfies

$$\begin{aligned} z &= \mathbb{E} \max\{z, u\} - s \\ &= -s + \mathbb{P}(u < z)z + \mathbb{P}(u \geq z)\mathbb{E}(u|u \geq z) \\ &= -s + F(z)z + [1 - F(z)] \int_z^{\bar{u}} u \frac{f(u)}{1 - F(z)} du \\ &= -s + z - [1 - F(z)]z + \int_z^{\bar{u}} u dF(u) \\ &= -s + z + \int_z^{\bar{u}} (u - z) dF(u) \end{aligned}$$

Thus, the reservation value is defined by

$$s = H(z) = \int_z^{\bar{u}} (u - z) dF(u) = \mathbb{E} \max\{u - z, 0\}$$

since z cancels out on both sides. Note that $H(z)$ is essentially the gains from search. So, the optimal search rule for the two box case is as above. Note that there is only one reservation value, since Pandora only needs to decide to open the second box i.e. only needs to compare u_1 to z .

Some facts about $H(z)$

- $H(z)$ is non-negative
- $H(z)$ is decreasing since

$$\frac{\partial H}{\partial z} = -[1 - F(z)]$$

- $H(z)$ is convex since

$$\frac{\partial^2 H}{\partial z^2} = f(z)$$

- $\lim_{z \rightarrow \bar{u}} H(z) = 0$
- $\lim_{z \rightarrow \underline{u}} H(z) = \mathbb{E}u$

Therefore, the interior solution exists if $s < \mathbb{E}u$ and is unique

Theorem 7.20 (Optimal Search Rule for Sequential Search with $T = \infty$). Our reservation value z^* satisfies

$$s = H(z) - \frac{1 - \delta}{\delta} z$$

where $H(z) = \int_z^{\bar{u}} (u - z) dF(u)$ i.e. our reservation values are stationary $z_1 = z_2 = \dots = z$ and the same as in the two period model. Hence, the optimal stopping rule is

1. If $\hat{u} = \max\{u_1, u_2, \dots\} < z$, open the next box
2. If $\hat{u} = \max\{u_1, u_2, \dots\} < z$, stop searching and take $\hat{u} = \max\{u_1, u_2, \dots\} < z$

Consider the case with free recall and no discounting. We try to construct our Bellman Equation.

- Our state variable is

$$\hat{u}_t = \max\{u_1, \dots, u_t\}$$

- Let $V_t(\hat{u}_t)$ be the value function of our Bellman equation
- Reservation value is the value of the best option at which Pandora is indifferent so

$$z_t = V_t(z_t)$$

The expected value from searching $t + 1$ boxes is

$$\begin{aligned} \mathbb{E}[V_{t+1}|\hat{u}_t] &= -(t+1)s + \int_{\underline{u}}^{\bar{u}} \max\{\hat{u}_t, u\} dF(u) \\ &= -(t+1)s + \int_{\underline{u}}^{\hat{u}} \hat{u}_t dF(u) + \int_{\hat{u}}^{\bar{u}} u dF(u) \\ &= -(t+1)s + \hat{u}_t + \int_{\hat{u}}^{\bar{u}} (u - \hat{u}_t) dF(u) \\ &= -(t+1)s + \hat{u}_t + H(\hat{u}_t) \\ &= V_t(\hat{u}_t) + [H(\hat{u}_t) - s] \end{aligned}$$

Note, the second equality follows because we split the integral from integrating u over the region where it is smaller than $\hat{u}$ and over the region where it is greater than $\hat{u}$. Thus, Pandora only gains from searching if $H(\hat{u}_t) - s > 0$ and so there is no more net marginal gain if it holds with equality with 0 (since searching will just give her the same value $V_t(\hat{u}_t)$ as stopping). Thus, for all t we have

$$H(z_t) = s$$

where all reservation values are the same $z_1 = z_2 = \dots = z$ and the same as in the two period model. Note that as recall never happens on the optimal search path, the **reservation value without recall is exactly the same when $T = \infty$** . The probability of stopping is $1 - F(z)$ and thus the expected duration of search is

$$\tau^e = \frac{1}{1 - F(z)}$$

Remark. Interpret as a Geometric Distribution to see why

If we want to add in discounting, we can just consider the two-period case with free recall since it is exactly the same as the case with $T = \infty$. So,

$$\begin{aligned} z &= \delta[\mathbb{E}\max\{z, u\} - s] \\ &= \delta \left[-s + zF(z) + \int_z^{\bar{u}} u dF(u) \right] \\ &= \delta \left[-s + z + \int_z^{\bar{u}} (u - z) dF(u) \right] \\ &= \delta[H(z) - s + z] \end{aligned}$$

So, we yield

$$s = H(z) - \frac{1 - \delta}{\delta} z$$

No Recall

Theorem 7.21 (Optimal Search Rule for No Recall Sequential Search with Finite T). The reservation values satisfy

$$z_1 \geq z_2 \geq \dots \geq z_T = 0$$

Hence, our optimal search rule is

1. If $u_k < z_{k+1}$, continue opening the next box
2. If $u_k \geq z_{k+1}$, take u_k as the prize

Proof. Our Bellman Equation is

$$V_t(x) = \max \left\{ x, -s + \int_{\underline{u}}^{\bar{u}} V_{t+1}(u) dF(u) \right\}$$

with boundary condition

$$V_T(u) = u$$

Let $z_{t+1}, \dots, z_T$ be the optimal reservation values from period $t + 1$ onwards. Then, in period t , Pandora can commit to using the same reservation values shifted by one period and not opening box T . As this guarantees the same expected payoff as in period $t + 1$, we have $z_t \geq z_{t+1}$. Then,

$$\begin{aligned} z_t &= -s + \int_{\underline{u}}^{\bar{u}} V_{t+1}(u) dF(u) \\ &= -s + \int_{\underline{u}}^{\bar{u}} \max \left\{ u, -s + \int_{\underline{u}}^{\bar{u}} V_{t+2}(u) dF(u) \right\} dF(u) \\ &= -s + \int_{\underline{u}}^{\bar{u}} \max \{ u, z_{t+1} \} dF(u) \\ &= -s + z_{t+1} + \int_{z_{t+1}}^{\bar{u}} (u - z_{t+1}) dF(u) \\ &= z_{t+1} + [H(z_{t+1}) - s] \end{aligned}$$

Suppose that $z_{t+1} > z$, the infinite horizon value. Then $H(z_{t+1}) < s \implies z_t < z_{t+1}$ which is a contradiction. So the sequence $\{z_t\}_{t=1}^T$ is bounded above by z and as $T \rightarrow \infty$ all the values converge to z ■

Theorem 7.22 (Optimal Search Rule for Free Recall Sequential Search with Finite T). Our reservation value z^* satisfies

$$s = H(z) - \frac{1 - \delta}{\delta} z$$

where $H(z) = \int_z^{\bar{u}} (u - z) dF(u)$ i.e. our reservation values are stationary $z_1 = z_2 = \dots = z$ and the same as in the two period model. Hence, the optimal stopping rule is

1. If $\hat{u} = \max\{u_1, u_2, \dots\} < z$, open the next box
2. If $\hat{u} = \max\{u_1, u_2, \dots\} < z$, stop searching and take $\hat{u} = \max\{u_1, u_2, \dots\} < z$

Proof. We can use similar arguments in the no recall case to state that $z_1 \geq \dots \geq z_T$ and $V_1(u) \geq \dots \geq V_T(u)$. That means that if a Pandora is indifferent between accepting or rejecting $u = z_t$, then this u will be accepted in the next round as $u \geq z_{t+1}$ (since we can retrieve the best u). Thus,

$$z_t = -s + z_t F(z_t) + \int_{z_t}^{\bar{u}} u dF(u) = z_t + [H(z_t) - s]$$

Thus, the reservation is stationary and the same as in the infinite horizon setting

$$H(z) = s$$

■

Ordered Search

Definition 7.23 (Ordered Search Problem). Suppose there are T number of boxes with unknown prizes where each box k contains a prize $u_i \sim F_i(u)$ where $F_i(u)$ is continuous, differentiable and has full support. Furthermore, $F_i(u)$ has finite support $[\underline{u}, \bar{u}]$ or finite moments of the distribution. Also usually assume $\underline{u} \geq 0$ and 0 is an outside option. Opening each box i costs s_i (which is assumed to be different). Suppose no discounting and Pandora can direct her search towards specific boxes. There is free recall, where Pandora can return to rejected options for free.

Theorem 7.24 (Optimal Rule: Pandora's Rule). Suppose we have n options characterised by a pair (s_i, F_i) and assume free recall with 0 as an outside option. For each option i define reservation value z_i as satisfying:

$$\mathbb{E}_i \max\{u - z_i, 0\} = s_i$$

Then, optimal search is defined by

1. Selection Rule: If another box is to be opened, it should be the unopened box with the highest reservation value z_i
2. Stopping Rule: Terminate search whenever $\hat{u}_k = \max\{u_1, \dots, u_k\} > z = \max\{z_{k+1}, \dots, z_n\}$ (assuming that the index is descending) i.e. whenever the highest discovered payoff exceeds the maximum reservation value of unopened boxes

Theorem 7.25. Define $w_i \equiv \min\{z_i, u_i\}$. Pandora rule is equivalent to choosing the box with the highest w_i .

Notice that $w_i > w_j$ implies one of three options

1. $z_i > z_j$ and $u_i > \min\{z_j, u_j\}$; so i is inspected first (as $z_i > z_j$) and chosen (as $u_i > z_j$ or $u_i > u_j$)
2. $z_i < z_j$ and $u_j < \min\{z_i, u_i\}$; so j is opened first, but then i is opened (as $u_j < z_i$) and chosen (as $u_j < u_i$)
3. $z_i = z_j = z$ and $u_j < \min\{z, u_i\}$; so either just i or both boxes are opened, but i is chosen

So by trichotomy, choose highest w_i . We now can show the proof of the Pandora rule given that choosing the highest w_i is optimal

Proof. Let $\mathbb{A}_i \in \{0, 1\}$ be selection indicator (with $\sum_i \mathbb{A}_i \leq 1$) and $\mathbb{I}_i \in \{0, 1\}$ be the search indicator function. Then, Pandora's expected payoff is

$$\begin{aligned}\mathbb{E}\left(\sum \mathbb{A}_i u_i - \sum \mathbb{I}_i s_i\right) &= \mathbb{E}\left(\sum \mathbb{A}_i u_i - \sum \mathbb{I}_i \max\{u_i - z_i, 0\}\right) \\ &\leq \mathbb{E}\left(\sum \mathbb{A}_i [u_i - \max\{u_i - z_i, 0\}]\right) \\ &= \mathbb{E}\left(\sum \mathbb{A}_i \min\{u_i, z_i\}\right) \\ &= \mathbb{E}\left(\sum \mathbb{A}_i w_i\right) \\ &\leq \mathbb{E}(\max\{w_i\})\end{aligned}$$

■

Note that if Pandora opens a box i but does not select it, i.e. $\mathbb{A}_i < \mathbb{I}_i$, it must be the case that $u_i \leq z_i$. Note that Pandora rule selects the box with highest w_i and thus Pandora rule delivers highest possible payoff. The claim shows choice equivalence between the Pandora Problem and the following: searcher opens all boxes for free and observes realisations of u_i and then chooses the box with the largest w_i .

Example 7.26. Let $(z_1, u_1) = (4, 2), (z_2, u_2) = (5, 3), (z_3, u_3) = (2, 6)$. Notice that $w_1 = 2, w_2 = 3, w_3 = 2$ and so option 2 is chosen. Checking Pandora rule

- as $z_2 > \max\{z_1, z_3\}$, open box 2 first
- as $u_2 < z_1$, open box 1
- as $u_2 > z_3$ stop and take box 2

Worked Examples

The focus is going to be deriving the solution from first principles, rather than applying the theorems immediately.

Example 7.27. There are three boxes each containing a prize drawn from a uniform distribution with support $[0, 1]$. Let $0 < s < 1/2$ be the cost of opening a single box.

- (a) Suppose that you have to decide how many boxes you want to open and then you can choose the largest prize. Describe the optimal search rule for various values of s .

The central idea is that you want to open a box when the marginal gain for opening an additional box is greater than the marginal cost until they equate one another. To this end, consider the marginal gain function

$$g(n) = \mathbb{E}[\max\{u_1, \dots, u_n\}] - \mathbb{E}[\max\{u_1, \dots, u_{n-1}\}]$$

Because we get to choose the best prize, the marginal benefit is the expected gain of the best prize from n boxes over the best prize from $n - 1$ boxes. Then, since $u_i \sim U[0, 1]$, $\hat{u} = \max\{u_1, \dots, u_n\}$ is distributed according to $(U[0, 1])^n$. Hence,

$$\begin{aligned}g(n) &= \int_0^1 \hat{u}_n \cdot f(\hat{u}_n) d\hat{u}_n - \int_0^1 \hat{u}_{n-1} f(\hat{u}_{n-1}) d\hat{u}_{n-1} \\ &= \int_0^1 u \cdot n u^{n-1} du - \int_0^1 u(n-1) u^{n-2} du \\ &= \left[\frac{n}{n-1} u^{n+1} \right]_0^1 - \left[\frac{n-1}{n} u^n \right]_0^1 \\ &= \frac{n}{n+1} - \frac{n-1}{n} = \frac{1}{n(n+1)}\end{aligned}$$

Hence, since $s \in (0, \frac{1}{2})$

$$n^* = \begin{cases} 3, & 0 < s \leq \frac{1}{12} \\ 2, & \frac{1}{12} < s \leq \frac{1}{6} \\ 1, & \frac{1}{6} < s < \frac{1}{2} \end{cases}$$

- (b) Now suppose that you can open boxes sequentially, but cannot come back to previously opened boxes. Describe the optimal search rule for various values of s .

The basic idea here is to use the concept of reservation values where one is indifferent between accepting the next box and continuing (i.e. the outside value), and we use a bit of backward induction.

Consider the last stage. After opening the last box, you have no choice to accept for any nonzero value (since we assume that the outside option 0 is always available). Hence $z_3 = 0$. Now,

$$z_2 = -s + \mathbb{P}(u \leq z_3)z_3 + \mathbb{P}(u \geq z_3)\mathbb{E}[u|u \geq z_3]$$

Basically, z_2 is the minimum value you will accept box 2 for and this must equal the continuation value of proceeding to open box 3 (i.e. paying search cost and then either getting u or behaving optimally when it is less than the reservation value at stage k or in this case 3). But clearly, $u \leq z_3 = 0$ occurs with probability 0 and $u \geq z_3 = 0$ occurs with probability 1 and $\mathbb{E}[u|u \geq z_3] = \mathbb{E}[u] = \frac{1}{2}$. Hence,

$$z_2 = -s + \frac{1}{2}$$

Then, we can define z_1 similarly (and it is even more instructive than it is equal to the search cost plus acting optimally at stage 2 i.e. continuing to search if below reservation value at stage 2 or accepting is above). Plugging in the value of $z_2 = -s + \frac{1}{2}$ yields

$$\begin{aligned} z_1 &= -s + \mathbb{P}(u \leq z_2)z_2 + \mathbb{P}(u \geq z_2)\mathbb{E}[u|u \geq z_2] \\ &= -s + \left(\frac{1}{2} - s\right)^2 + \int_{1/2-s}^1 u du \\ &= \frac{1}{2} - s + \frac{1}{2}\left(\frac{1}{2} - s\right)^2 \\ &= \frac{5}{8} - \frac{3}{2}s + \frac{1}{2}s^2 \end{aligned}$$

Remark. It is instructive to see here how the reservation value is (recursively) defined for any stage k

$$z_k = -s + \mathbb{E}[\max\{u, z_{k+1}\}]$$

where the term $\mathbb{E}[\max\{u, z_{k+1}\}]$ is acting optimally at after opening the box i.e. if the prize is less than the reservation prize continue to open and if the prize is at least as much then stop. The natural interpretation of this is that you should stop if the realised prize at stage k is at least as good as the expected value of acting optimally if you continue.

Hence, the search rule is

- If $x_1 \geq z_1 = \frac{5}{8} - \frac{3}{2}s + \frac{1}{2}s^2$, stop
- If $x_1 < z_1$, open box 2 and if $x_2 \geq z_2 = \frac{1}{2} - s$, stop
- if $x_2 < z_2$, open box 3 and stop

- (c) Now suppose that you can open boxes sequentially, but now you can get back to previously opened boxes. Describe the optimal search rule for various values of s .

Using our understanding that z_k is the value of acting optimally at stage $k+1$, under free recall, we can write

$$z_k = -s + \max\{m_k, u\}$$

where $m_k = \max\{u_1, \dots, u_k\}$ i.e. the best prize until now. Since z_k is being indifferent between taking the best prize now (i.e. stopping) and optimal continuation, we can have $z_k = m_k$. Hence,

$$\begin{aligned} z_k &= -s + \max\{z_k, u\} \\ &= -s + \mathbb{P}(u < z_k)z_k + \mathbb{P}(u \geq z_k)\mathbb{E}[u|u \geq z_k] \\ &= -s + z_k^2 + \int_{z_k}^1 u du \\ &= -s + z_k^2 + \frac{1 - z_k^2}{2} \end{aligned}$$

Hence, since we argued this for an arbitrary k , we have

$$s = \frac{(1 - z)^2}{2} \iff 1 - \sqrt{2s} = z$$

In other words, the reservation value is stationary. We can compute this by backward induction first: we just have to start from $k = 2$ instead of $k = 3$ because $z_3 = \max\{u_1, u_2, u_3\}$. Alternative, this would require careful interpretation to simply state $z_2 = -s + \mathbb{E}[z_3] = -s + \mathbb{E}[\max\{m_2, u_3\}]$, where $z_2 = m_2$ by definition.

Hence, the Optimal stop rule is (written in extremely poor and wrong pseudocode)

- For $i = 1, 2$, if $x_i \geq z = 1 - \sqrt{2s}$, stop
- elif, Open
- At $i = 3$, open box 3 and take best prize

Notice that you will open the first box always since $E[u] = \frac{1}{2} > s$

- (d) For various values of s compare the expected number of sampled boxes in cases (1) and (3). For which s does case (1) result in more search than case (3)?

For (c), $\mathbb{P}(\text{open box 2}) = \mathbb{P}(u_1 < z) = z = 1 - \sqrt{2s}$, and $\mathbb{P}(\text{open box 3}) = \mathbb{P}(u_1 < z) \times \mathbb{P}(u_2 < z) = z^2 = (1 - \sqrt{2s})^2$. Hence,

$$\mathbb{E}[\text{no. of boxes open}] = 1 + z + z^2 = 3 - 3\sqrt{2s} + 2s$$

Define the above $f(s)$. Notice that for $s \in (0, \frac{1}{2})$, $f(s) \in (1, 3)$, Hence

- If $s \in (0, \frac{1}{12}]$, then in (a), open 3 boxes while (c) opens less than 3 in expectation
- If $s \in (\frac{1}{12}, \frac{1}{6}]$, then (a) opens 2 while $f(s) = 3 - 3\sqrt{2(1/12)} + 2(1/12) = 1.94 < 2$. Since $f(s)$ is decreasing for $s \in (\frac{1}{12}, \frac{1}{6}]$, opens less boxes
- If $s \in (\frac{1}{6}, \frac{1}{2})$, then (a) opens 1 box and $f(s) > 1$ as shown so (c) opens more in expectation

Intuition: high search cost means sequential search opens more but low/moderate implies fixed sample search opens more in expectation.

Example 7.28. Consider a model of a search engine which matches consumers with firms. Firms sell differentiated product, which with probability θ yields valuation v uniformly distributed on $[0, 1]$ and with probability $1 - \theta$ gives match value of 0. Suppose that $\Pr(\theta = 1) = \alpha$ and $\Pr(\theta = 1/2) = 1 - \alpha$. When a consumer runs a search query, the search engine informs firms of their relevance and charges a click-through fee r . After that firms make decisions whether to join the search pool or not and set price for their product, p , and consumers search for the product at search cost s per firm.

- (a) Suppose that the search engine induces expected relevance θ^e . Derive the consumer's optimal stopping rule.

This is a problem with free search recall. With probability θ^e , the firm is relevant i.e. yields positive match value $v \sim U[0, 1]$. The consumer gets a surplus v but only benefits from this if $v > z$ i.e. the reservation value. The indifference condition says that the reservation value satisfies the equation

such that the expected gain from search equals the expected cost of search. Then, we can write by stationarity

$$s = \theta^e \int_z^1 (u - z) du + (1 - \theta^e) \cdot 0 = \theta^e \cdot \frac{(1 - z)^2}{2}$$

Hence, $z = 1 - \sqrt{2s/\theta^e}$. So, keep searching until $v \geq 1 - \sqrt{2s/\theta^e}$

- (b) For fixed θ^e , what is the price charged by a firm of quality θ and its profit? How does the equilibrium price depend on search costs and average relevance? Explain the intuition behind your answer.

For a θ -type firm, profit is

$$\pi(\theta) = \theta p [1 - F(z - p^* + p)]$$

This is because the stopping rule is $v - p \geq z - p^*$ i.e. so the consumer stops if $v \geq z - p^* + p$. Our FOC is

$$1 - F(z - p^* + p) - pf(z - p^* + p) = 0$$

At equilibrium, $p = p^*$ and so

$$p^* = \frac{1 - F(z)}{f(z)} = 1 - F(z) = \sqrt{\frac{2s}{\theta^e}}$$

since $f(z) = 1$ since v is uniformly distributed on $[0, 1]$. Hence, profit is $\pi = \frac{\theta}{\theta^e} (2s)$ i.e. high search cost implies higher price (since higher search frictions for consumer) and higher average relevance means lower price since consumers expect better future offers.

- (c) Express the search engine's fee and profit in terms of marginal relevance θ^* , average relevance θ^e and s .

The search engine charges a per-click fee r that extracts the marginal firm's entire surplus. The marginal firm has relevance θ^* and earns profit per click of $\pi(\theta^*) = \theta^* p^* (1 - F(z))$. The entry condition binds at θ^* (i.e. indifferent between joining or not), so

$$r = \pi(\theta^*) = \theta^* p^* (1 - F(z)) = \theta^* (1 - z)^2 = \theta^* \cdot \frac{2s}{\theta^e}$$

where we used $p^* = 1 - z$ and $1 - z = \sqrt{2s/\theta^e}$.

To derive the search engine's profit, we need the expected number of clicks per consumer. At each visit, the consumer finds a relevant firm (probability θ^e) and then accepts the offer (probability $1 - F(z) = 1 - z$). So the probability of a transaction on any given click is

$$\theta^e (1 - F(z)) = \theta^e (1 - z) = \theta^e \sqrt{\frac{2s}{\theta^e}} = \sqrt{2s\theta^e}$$

Since each click is an independent draw, the number of clicks until a transaction follows a geometric distribution with success probability $\sqrt{2s\theta^e}$. The expected number of clicks is therefore

$$\mathbb{E}[\text{clicks}] = \frac{1}{\theta^e (1 - F(z))} = \frac{1}{\sqrt{2s\theta^e}}$$

The search engine collects r on every click, so its expected profit per consumer is

$$\Pi = r \cdot \mathbb{E}[\text{clicks}] = \frac{\theta^* \cdot \frac{2s}{\theta^e}}{\sqrt{2s\theta^e}} = \frac{\theta^*}{\theta^e} \cdot \frac{2s}{\sqrt{2s\theta^e}} = \frac{\theta^*}{\theta^e} \sqrt{\frac{2s}{\theta^e}}$$

Equivalently, noting that $p^* = \sqrt{2s/\theta^e}$, this can be written as

$$\Pi = \frac{\theta^*}{\theta^e} \cdot p^*$$

so that $\Pi/p^* = \theta^*/\theta^e$: the search engine's profit as a fraction of the product price equals the ratio of marginal to average relevance in the pool. When $\theta^* = \theta^e$ (no distortion), the search engine extracts exactly p^* per consumer. When $\theta^* < \theta^e$, the pool is "too inclusive" and the engine earns less than p^* .

- (d) For which values of α does the search engine prefer to distort the search pool? What r will it choose for various values of α ?

Recall that $\Pr(\theta = 1) = \alpha$ and $\Pr(\theta = 1/2) = 1 - \alpha$. The search engine's choice of r determines which firms enter the pool. There are only two possible pool compositions:

- (a) **Inclusive pool:** both types enter, so $\theta^* = 1/2$ and $\theta^e = \alpha \cdot 1 + (1 - \alpha) \cdot \frac{1}{2} = \frac{1+\alpha}{2}$.
- (b) **Exclusive pool:** only $\theta = 1$ firms enter, so $\theta^* = 1$ and $\theta^e = 1$.

Using $\Pi = \frac{\theta^*}{\theta^e} \sqrt{\frac{2s}{\theta^e}}$, we compute the profit in each case.

Inclusive pool.

$$\Pi_I = \frac{1/2}{(1+\alpha)/2} \cdot \sqrt{\frac{2s}{(1+\alpha)/2}} = \frac{1}{1+\alpha} \cdot \sqrt{\frac{4s}{1+\alpha}} = \frac{2\sqrt{s}}{(1+\alpha)^{3/2}}$$

Exclusive pool.

$$\Pi_E = \frac{1}{1} \cdot \sqrt{\frac{2s}{1}} = \sqrt{2s}$$

When does the search engine prefer to distort? The search engine distorts (excludes the low-quality firms) when $\Pi_E > \Pi_I$, i.e.

$$\sqrt{2s} > \frac{2\sqrt{s}}{(1+\alpha)^{3/2}}$$

Dividing both sides by $\sqrt{s}$:

$$\sqrt{2} > \frac{2}{(1+\alpha)^{3/2}}$$

Rearranging:

$$(1+\alpha)^{3/2} > \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$1+\alpha > 2^{1/3}$$

$$\alpha > 2^{1/3} - 1 \approx 0.26$$

Intuition. The search engine faces a trade-off when raising r to exclude low-quality firms:

- **Higher fee per click:** the marginal firm has higher relevance ($\theta^* = 1$ instead of $1/2$), so the engine can charge more per click.
- **Fewer clicks:** higher average relevance θ^e means consumers find matches faster, reducing the number of clicks (and hence fee payments).
- **Lower product price:** higher θ^e raises the reservation value z , making consumers pickier, which forces firms to charge a lower p^* .

When α is large (many high-quality firms), excluding the $\theta = 1/2$ types doesn't sacrifice much variety but allows a much higher fee. The gain from a higher r outweighs the loss in click volume. Conversely, when α is small, excluding the low types removes most of the market, and the inclusive pool generates more total revenue.

Optimal fee.

- If $\alpha > 2^{1/3} - 1$: the search engine sets r high enough to exclude $\theta = 1/2$ firms. Using $r = \theta^* \cdot \frac{2s}{\theta^e}$ with $\theta^* = \theta^e = 1$:

$$r = 1 \cdot \frac{2s}{1} = 2s$$

- If $\alpha < 2^{1/3} - 1$: the search engine sets r to make the $\theta = 1/2$ type just indifferent. Using $\theta^* = 1/2$ and $\theta^e = \frac{1+\alpha}{2}$:

$$r = \frac{1}{2} \cdot \frac{2s}{(1+\alpha)/2} = \frac{2s}{1+\alpha}$$

- If $\alpha = 2^{1/3} - 1$: the search engine is indifferent between the two pool configurations.

Example 7.29. Consider two boxes, each containing a sum of money drawn from the same continuous distribution $F(u)$ with support $[0, 1]$. You can open the first box for free. Then you can decide whether you want to open the second box or not. It costs s to open the second box. If you open it, you can either take the amount of money there or you can get back to box one, but it costs you b .

- (a) Assuming that s and b are sufficiently small, i.e. both search and return are possible, derive the optimal search rule.

Since there are only two boxes, we only need to consider one single reservation value i.e. after opening the first box (since opening first box is free). Then, the reservation value just is the indifference between taking it and continuing optimally

$$\begin{aligned} z &= -s + \mathbb{P}(u \leq z - b)(z - b) + \mathbb{P}(u > z - b)\mathbb{E}[u|u > z - b] \\ &= -s + F(z - b)(z - b) + \int_{z-b}^1 u dF(u) + (z - b) - (z - b) \\ &= -s + (z - b) - (1 - F(z - b))(z - b) + \int_{z-b}^1 u dF(u) \\ &= -s - b + z + \int_{z-b}^1 (u - z + b) dF(u) \end{aligned}$$

The last equality holds because

$$(1 - F(z - b))(z - b) = \mathbb{P}(u < z - b)(z - b) = \int_{z-b}^1 (z - b) dF(u)$$

Then, this yields

$$s + b = \int_{z-b}^1 (u - z + b) dF(u)$$

i.e. z must satisfy

$$s = \int_{z-b}^1 (u - z + b) dF(u) - b$$

And for this z , if $x_1 \geq z$, then stop, else continue and take $\max\{x_2, x_1 - b\}$

- (b) How does the reservation value depend on s and b ?

Notice that the optimal z must satisfy

$$s = \int_{z-b}^1 (u - z + b) dF(u) - b$$

Hence, we can let $G(z, s, b) = \int_{z-b}^1 (u - z + b) dF(u) - b - s = 0$. Since G is a C^1 function in an open ball around (z, b, s) with $\partial G / \partial z(z, s, b) \neq 0$, then by the Implicit Function Theorem we can define $z(s, b)$ and compute comparative statics via

$$\frac{\partial z}{\partial s} = -\frac{\partial G / \partial s}{\partial G / \partial z}, \quad \frac{\partial z}{\partial b} = -\frac{\partial G / \partial b}{\partial G / \partial z}.$$

Step 1: Compute $\partial G/\partial z$. Write $G = \int_{z-b}^1 (u - z + b) dF(u) - b - s$. Using the Leibniz rule for differentiation under the integral sign with lower limit $\ell(z) = z - b$ (so $\partial \ell/\partial z = 1$) and constant upper limit:

$$\begin{aligned}\frac{\partial G}{\partial z} &= \int_{z-b}^1 \frac{\partial}{\partial z} (u - z + b) dF(u) - [(z - b) - z + b] \cdot f(z - b) \cdot 1 \\ &= \int_{z-b}^1 (-1) dF(u) - 0 \\ &= -[1 - F(z - b)] < 0.\end{aligned}$$

Note the boundary term vanishes because the integrand evaluated at the lower limit gives $(z - b) - z + b = 0$.

Step 2: Compute $\partial G/\partial s$. Since s appears only in the $-s$ term:

$$\frac{\partial G}{\partial s} = -1.$$

Step 3: Compute $\partial G/\partial b$. Now b appears in the lower limit $\ell(b) = z - b$ (so $\partial \ell/\partial b = -1$), in the integrand through $(+b)$, and in the $-b$ term outside. By the Leibniz rule:

$$\begin{aligned}\frac{\partial G}{\partial b} &= \int_{z-b}^1 \frac{\partial}{\partial b} (u - z + b) dF(u) - [(z - b) - z + b] \cdot f(z - b) \cdot (-1) - 1 \\ &= \int_{z-b}^1 (+1) dF(u) + 0 - 1 \\ &= [1 - F(z - b)] - 1 \\ &= -F(z - b) < 0.\end{aligned}$$

Step 4: Apply the Implicit Function Theorem.

$$\frac{\partial z}{\partial s} = -\frac{\partial G/\partial s}{\partial G/\partial z} = -\frac{-1}{-[1 - F(z - b)]} = -\frac{1}{1 - F(z - b)} < 0.$$

A higher search cost lowers the reservation value: the worker becomes less selective.

$$\frac{\partial z}{\partial b} = -\frac{\partial G/\partial b}{\partial G/\partial z} = -\frac{-F(z - b)}{-[1 - F(z - b)]} = -\frac{F(z - b)}{1 - F(z - b)} < 0.$$

A higher benefit b lowers z . If z represents a net-of-benefit reservation value (e.g. $z = w^* - b$), this is consistent with the reservation wage $w^* = z + b$ increasing in b .

- (c) For this question only assume that $F(u) = u$ and $s < 1/2$. For which values of b does return never happen on the optimal search path?

With $F(u) = u$ (uniform on $[0, 1]$), the reservation value equation $G(z, s, b) = 0$ becomes

$$\int_{z-b}^1 (u - z + b) du - b - s = 0.$$

Evaluating the integral:

$$\begin{aligned}\int_{z-b}^1 (u - z + b) du &= \left[\frac{u^2}{2} - (z - b)u \right]_{z-b}^1 \\ &= \frac{1}{2} - (z - b) - \frac{(z - b)^2}{2} + (z - b)^2 \\ &= \frac{1}{2} - (z - b) + \frac{(z - b)^2}{2} \\ &= \frac{[1 - (z - b)]^2}{2}.\end{aligned}$$

So the reservation value equation is

$$\frac{[1 - (z - b)]^2}{2} = b + s.$$

Solving for z :

$$z = 1 + b - \sqrt{2(b + s)}.$$

Return never happens on the optimal search path when the consumer never prefers going back to box 1 after opening box 2. The consumer returns to box 1 if the draw u_2 from box 2 satisfies $u_2 < z - b$, i.e. the net value of returning exceeds taking box 2. Return never happens when $z - b \leq 0$, meaning the return option is never exercised (since $u_2 \geq 0$ always). This requires

$$z \leq b \iff 1 + b - \sqrt{2(b + s)} \leq b \iff 1 \leq \sqrt{2(b + s)} \iff b \geq \frac{1}{2} - s.$$

Since $s < 1/2$, this threshold is positive. So return never happens for $b \geq \frac{1}{2} - s$.

Intuition: When the cost of returning b is high enough, the option to go back to box 1 is so expensive that it is never worth exercising. The threshold decreases in s because a higher search cost s lowers the reservation value, making the return region smaller.

- (d) Let $\bar{s}$ be the smallest level of search costs for which consumers never prefer opening the second box. How does $\bar{s}$ depend on b ?

The consumer opens the second box if and only if the expected gain from search exceeds the cost, i.e. if the value of the first draw u_1 is below the reservation value z . The consumer *never* prefers opening the second box when $z \leq 0$, meaning even the worst possible draw from box 1 does not justify searching.

From the general reservation equation $G(z, s, b) = 0$, set $z = 0$:

$$\int_{-b}^1 (u + b) dF(u) - b - \bar{s} = 0.$$

Since F has support $[0, 1]$ and $b \geq 0$, the lower limit $-b \leq 0$, so $F(-b) = 0$ and the integral effectively runs from 0 to 1:

$$\int_0^1 (u + b) dF(u) - b - \bar{s} = 0.$$

This gives

$$\bar{s} = \int_0^1 (u + b) dF(u) - b = \int_0^1 u dF(u) + b - b = \mathbb{E}[U].$$

Therefore

$$\bar{s} = \mathbb{E}[U],$$

which is *independent of b* .

Intuition: At the margin where search is just barely not worthwhile ($z = 0$), the consumer would never return to box 1 regardless (since $z - b = -b < 0$). So the return cost b is irrelevant at this margin. The consumer simply compares the search cost s against the expected value of a new draw. Search is never undertaken when $s \geq \mathbb{E}[U]$.

- (e) Let v_1 be the reservation value if $(s, b) = (x, 0)$ and v_2 be the reservation value if $(s, b) = (0, x)$. How does v_1 relate to v_2 ? In which case is there more search and why?

Case 1: $(s, b) = (x, 0)$, **reservation value v_1** . Here search costs x and return is free. The equation is

$$\frac{[1 - v_1]^2}{2} = x.$$

So $v_1 = 1 - \sqrt{2x}$.

Case 2: $(s, b) = (0, x)$, **reservation value** v_2 . Here search is free and return costs x . The equation is

$$\frac{[1 - (v_2 - x)]^2}{2} = x.$$

So $v_2 = 1 + x - \sqrt{2x}$.

Comparison. We have

$$v_2 - v_1 = (1 + x - \sqrt{2x}) - (1 - \sqrt{2x}) = x > 0.$$

Hence $v_2 > v_1$: the reservation value is *higher* when the return cost is x and search is free than when the search cost is x and return is free.

Which case has more search? There is more search (i.e. box 2 is opened more often) when the reservation value is higher, since the consumer opens box 2 whenever $u_1 < z$. Since $v_2 > v_1$, there is *more search in Case 2* (free search, costly return).

Intuition: When search is free ($s = 0$), there is no cost to opening box 2, so the consumer always opens it. The cost $b = x$ only matters *after* seeing box 2, making the consumer less willing to return but not less willing to search. In contrast, when $s = x$ and $b = 0$, the cost is paid upfront before seeing box 2, which directly deters search. The upfront cost is a stronger deterrent to search than an equal-sized contingent cost that may never be incurred.

In fact, when $s = 0$ we have $v_2 = 1 + x - \sqrt{2x} > 0$ for $x < 1/2$, so box 2 is *always* opened (since $u_1 \in [0, 1]$ and $v_2 > 0$ means even the best draws from box 1 can still justify free search). The cost x only affects the return decision, not the search decision.

Example 7.30. Pandora faces 10,000 boxes she could open. Suppose that the distribution of prizes in box n is given by $F_n = 1 - e^{-u/n}$, where e is the base of the natural logarithm, $u \geq 0$ is a prize value and $n = 1, \dots, 10000$. Each box is clearly labelled as “box n ”, so Pandora knows the distribution associated with each box. If Pandora decides not to open any of the boxes her utility is $u_0 = 0$. Suppose that Pandora can open box n at a cost s_n . Pandora may open boxes in any order she wants, and can stop at any moment and take the prize from the best opened box, or may decide not to open any boxes and take the outside option.

- (a) Find the expected value of the prize in box n . How does it depend on n ?

$f_n(u) = \frac{e^{-u/n}}{n}$. Then,

$$\begin{aligned} \mathbb{E}[u_n] &= \int_0^\infty u f_n(u) du \\ &= \int_0^\infty \frac{u}{n} e^{-u/n} du \\ &= \frac{1}{n} \left[-nu \cdot e^{-u/n} \right]_0^\infty + n \int_0^\infty e^{-u/n} du \\ &= n \end{aligned}$$

where we use integration by parts. It is linearly dependent on n (increasing linearly).

- (b) Denote z_n the value for which Pandora is indifferent between accepting it immediately and opening box n only. Derive z_n .

By definition,

$$z_n = -s_n + \mathbb{E} \max\{u_n, z_n\}$$

Since if u_n is worse than z_n he can just take z_n again (since in the problem it is always available to Pandora). Rearranging gives

$$\begin{aligned}
s_n &= \mathbb{E} \max\{u_n, z_n\} - z_n \\
&= \mathbb{E} \max\{u_n - z_n, 0\} \\
&= \int_{z_n}^{\infty} (u - z_n) f(u) du \\
&= \int_{z_n}^{\infty} (u - z_n) \cdot \frac{1}{n} e^{-u/n} du \\
&= \int_0^{\infty} t \cdot \frac{1}{n} e^{-(t+z_n)/n} dt \\
&= e^{-z/n} \int_0^{\infty} \frac{1}{n} t e^{-t/n} dt \\
&= n e^{-z/n}
\end{aligned}$$

Some manipulation yields

$$z_n = n \ln \left(\frac{n}{s_n} \right)$$

- (c) Formulate the optimal search rule.

The Optimal Search rule is: Open boxes in descending order of z_n . Let $x = \max\{x_{(1)}, x_{(2)}, \dots\}$. If $x < \max\{z_n : \text{box } n \text{ unopened}\}$: continue opening; else stop.

- (d) Suppose that $s_n = an$, $a > 0$. Show that if $a < 1$ Pandora wants to open box 10,000 first. What is the optimal search rule in this case? What will Pandora do if (i) $a > 1$, (ii) $a = 1$?

Let $s_n = an$. Then $z_n = n \ln \left(\frac{1}{a} \right)$. If $a < 1$, then $z_n > 0$ so z_n is strictly increasing in n . Hence, Pandora opens from box 10,000 downwards.

Search rule: Open box from 10,000 downwards. Continue if $x < (n-1) \ln \left(\frac{1}{a} \right)$ and stop if $x \geq (n-1) \ln \left(\frac{1}{a} \right)$, where $x = \max\{x_1, x_2, \dots\}$.

- (a) If $a > 1$, then $\ln \left(\frac{1}{a} \right) < 0$ so z_n for all n and since outside option is 0, she would not open any box

- (b) If $a = 1$, then $\ln(1) = 0$ so $z_n = 0$ for all n and Pandora is indifferent between opening the first box and not opening at all.

- (e) Suppose that $s_n = \frac{1}{e}n$, and suppose that the actual prize in box n equals $u_n = \frac{10000}{n}$. Which box will Pandora end up choosing?

Let $s_n = \frac{1}{e}n$. Then, $z_n = n \ln e = n$. So z_n is increasing in n while $u_n = \frac{10000}{n}$ is decreasing in n . Hence, we just need to check when

$$z_n = u_n \iff n = \frac{10000}{n}$$

because of single crossing. Which implies $n = 100$. Pandora will open the 100th box and choose that box.

7.3 Applications of Search

Recall two categories of information consumers seek when searching:

- (i) **Product information:** availability, match quality, taste fit.
- (ii) **Price information:** what different sellers charge for comparable goods.

A related distinction is between *search goods* (full information is revealed upon visiting a seller) and *experience goods* (full information is revealed only upon consumption). The models below deal with search goods.

First Principles: Sequential Search and Reservation Values

Consider an agent who draws offers from a known distribution, one at a time, at cost $s > 0$ per draw. The agent's problem is: when should I stop searching and accept the current best offer?

The key insight is *stationarity*. Because the distribution of future offers does not depend on past draws, the agent's continuation value is the same at every decision node. Let z denote the **reservation value**—the threshold at which the agent is indifferent between accepting the current offer and paying s to draw again. By the indifference condition, the option value of one more search must exactly equal its cost:

$$s = \int_z^{\bar{u}} (u - z) dF(u). \quad (6)$$

The right-hand side is the expected gain from search conditional on the new draw exceeding z , weighted by the probability of that event. The optimal rule is: *accept any offer $u \geq z$; reject and continue searching otherwise.*

This reservation-value principle underpins every model that follows. Two properties are worth noting:

- (a) The reservation value z is *decreasing* in the search cost s : higher costs make the agent less selective.
- (b) The reservation value is *independent* of the number of searches already conducted (stationarity), which greatly simplifies equilibrium analysis.

Diamond Search Model

Setting. Diamond (1971) provides the starkest illustration of how search frictions affect competition. The environment is as follows:

- n firms sell a *homogeneous* good and compete in prices; production cost is normalised to zero.
- A unit mass of consumers each value the product at $v > 0$.
- The first firm visit is free (a “local store”), but each subsequent search costs $s > 0$.
- Search is *sequential* and *random*: consumers sample firms one at a time in random order.

The competitive benchmark. With $s = 0$ (no search frictions), standard Bertrand logic applies: firms undercut each other until $p_i = 0$ for all i , and consumers capture the entire surplus v .

Why Bertrand breaks down with $s > 0$. Suppose all firms charge $p^* = 0$. A single firm can deviate to $p_i = s$ (or $p_i = s - \varepsilon$). A consumer visiting this firm compares the surplus from buying now, $v - s + \varepsilon$, with the expected surplus from paying s to search again and finding $p = 0$ elsewhere, which yields $v - s$. Since $v - s + \varepsilon > v - s$, the consumer prefers to buy immediately. The deviating firm therefore captures all its visitors at a strictly positive price—a profitable deviation. Hence $p^* = 0$ is *not* an equilibrium when $s > 0$.

No symmetric pure-strategy equilibrium below v . The same logic generalises. Suppose all firms charge some $p^* < v$. A firm can deviate to $p = \min\{p^* + s, v\}$. A consumer at the deviating firm would need to pay s to find another firm charging p^* , but the saving is at most s , so there is no strict gain from searching. The deviation is profitable.

The Diamond equilibrium. Setting $p^* = v$ is a Nash equilibrium, supported by the following logic:

1. If all firms charge v , a consumer's surplus from buying is $v - v = 0$, and the expected gain from searching (paying s to find another firm also charging v) is $-s < 0$. So consumers *never search*.
2. Given that consumers do not search, each firm is effectively a monopolist over its visitors. The optimal monopoly price is $p^* = v$.

Ruling out mixed strategies. Could firms play a mixed strategy $F(p)$ in equilibrium? No. If firms randomise, the expected price is $E[p]$. A firm can deviate to charging $E[p] + s$: a consumer who has drawn this price knows that the expected price elsewhere is $E[p]$, so the expected gain from searching is $E[p] + s - E[p] - s = 0$ —not enough to justify the cost s . The deviation is weakly profitable, ruling out interior mixing.

The Diamond Paradox. The striking conclusion:

Even an arbitrarily small search cost $s > 0$ causes the market outcome to jump from the competitive price ($p = 0$) to the monopoly price ($p = v$).

This discontinuity—known as the **Diamond Paradox**—arises because search frictions destroy the competitive mechanism entirely. Competition requires that consumers can costlessly compare offers; any positive friction insulates each firm from competitive pressure.

Further remarks.

- If the first visit were also costly (no free local store), the situation would be worse: consumers would anticipate paying v and gaining zero surplus, so they would not enter the market at all. The market *collapses* due to a hold-up problem.
- With downward-sloping demand, the monopoly price leaves consumers with some surplus, so market collapse is avoided—but prices still jump to monopoly levels.
- The predictions are arguably too stark to be realistic, motivating richer models of search.

Burdett and Judd (1983): Fixed-Sample Search and Price Dispersion

Diamond’s result hinges on *sequential* search: consumers observe one price at a time and decide whether to continue. Burdett and Judd (1983) consider an alternative protocol—**fixed-sample search**—in which consumers choose *how many* firms to sample before observing any prices, and then buy at the lowest observed price.

Setting.

- n firms sell a homogeneous good at zero cost, competing in prices.
- A unit mass of consumers value the product at $v > 0$.
- Each consumer decides how many firms k to sample, at cost s per firm.
- After sampling, the consumer buys at the lowest price in her sample (provided it is below v).

Diamond equilibrium persists. There still exists a “Diamond-style” equilibrium with $p^* = v$ and no search (all consumers sample one firm and accept v).

No symmetric pure-strategy equilibrium with $p^* < v$. Two cases arise:

- If all consumers sample at most one firm, firms face captive demand and should charge v —a contradiction with $p^* < v$.
- If some consumers sample two or more firms, a firm benefits from undercutting to $p^* - \varepsilon$, stealing all comparison shoppers—also a contradiction.

This rules out pure strategies, so we look for *mixed-strategy* equilibria where firms randomise over prices according to a CDF $F(p)$.

Equilibrium search behaviour. Let q_k denote the probability a consumer samples k firms.

Lemma 7.31 (Claim 1). $q_1 > 0$. If $q_1 = 0$, every active consumer compares at least two prices, so Bertrand competition drives $p^* = 0$. But if $p^* = 0$, sampling one firm is optimal (no gain from comparison), contradicting $q_1 = 0$.

Lemma 7.32 (Claim 2). $q_1 < 1$. If $q_1 = 1$, all consumers draw one quote, each firm is a monopolist, and $p^* = v$. This either collapses the market (if the first search is costly) or yields the Diamond monopoly equilibrium—not an equilibrium with $p^* < v$.

A key result is that the total expected cost of sampling k firms and buying at the minimum price,

$$sk + \int_{\underline{p}}^{\bar{p}} k p [1 - F(p)]^{k-1} dF(p),$$

is *convex* in k . Convexity implies that at most two adjacent integers can receive positive probability in the consumer's optimal mixing. Combined with $q_1 > 0$, this gives $q_2 > 0$ and $q_1 + q_2 = 1$. Consumers randomise between sampling one firm and sampling two.

Equilibrium price distribution. Denote $q = q_1$ and $1 - q = q_2$. A firm posting price p sells to:

- its share of one-sample consumers (who buy regardless), contributing demand $\frac{q}{n}$;
- its share of two-sample consumers for whom p is the lower draw, contributing $\frac{2(1-q)}{n} [1 - F(p)]$.

So each firm's expected profit is

$$\pi(p) = \frac{p}{n} [q + 2(1-q)(1 - F(p))].$$

In a mixed-strategy equilibrium, profit must be constant across the support. Using the boundary condition $\bar{p} = v$ (the highest price in the support must be the consumers' valuation), one can solve the equal-profit condition to obtain:

$$F(p) = 1 - \frac{q}{2(1-q)} \cdot \frac{v-p}{p}, \quad p \in \left[\frac{qv}{2-q}, v \right]. \quad (7)$$

Equilibrium profit per firm is $\pi = qv/n$.

Equilibrium search intensity. The mixing probability q is pinned down by the consumers' *indifference condition* between sampling one and two firms. The expected cost of sampling one firm is

$$s + \int_{qv/(2-q)}^v p dF(p),$$

while sampling two firms costs

$$2s + \int_{qv/(2-q)}^v p d[1 - (1 - F(p))^2].$$

Equating these and substituting (7) yields

$$g(q) \equiv \frac{q \left(\ln \frac{2-q}{q} - 2(1-q) \right)}{2(1-q)^2} = \frac{s}{v}. \quad (8)$$

The function $g(q)$ is hump-shaped on $(0,1)$ with $g(0^+) = 0$ and $g(1^-) = 0$. Hence, for s/v below the maximum of g , there exist two solutions for q ; for s/v above it, no interior equilibrium exists and only the Diamond equilibrium survives.

Key takeaways.

1. An equilibrium with active search and *price dispersion* exists when search frictions are sufficiently low relative to the product value.
2. In this equilibrium, firms play mixed strategies: identical firms charge different prices in equilibrium—a resolution of the Diamond Paradox.
3. Consumers benefit from search in two ways: (i) they sometimes observe two prices and pick the lower one, and (ii) the *entire price distribution* improves because firms compete for comparison shoppers.
4. Total industry profit is qv , which is *independent of n* . Adding firms does not reduce average prices paid—a surprising result driven by the fact that market structure adjusts through search intensity rather than price competition.
5. The Diamond collapsed-market equilibrium ($p^* = v$, $q_1 = 1$) coexists with the dispersed-price equilibrium—there is equilibrium multiplicity.

Wolinsky–Anderson–Renault (WAR): Sequential Search with Differentiated Products

The Diamond and Burdett–Judd models concern homogeneous goods, where consumers search over *prices* alone. In many markets, however, products are *differentiated*, and consumers search to learn both prices and how well a product matches their tastes. The Wolinsky (1986) and Anderson–Renault (1999) framework—hereafter WAR—embeds sequential search in a model of monopolistic competition with random match values.

Setting.

- n identical firms produce differentiated products at common marginal cost c .
- There is a unit mass of consumers per firm, each with unit demand.
- A consumer purchasing from firm i obtains surplus $u_i - p_i$, where u_i is a **match value** drawn i.i.d. from F on $[\underline{u}, \bar{u}]$ —independently across consumer–firm pairs.
- Firms simultaneously set prices. Consumers then search *sequentially* at cost $s > 0$ per firm visited, observing both u_i and p_i upon visiting firm i .
- The outside option yields payoff 0.

Reservation value with differentiated products. We derive the optimal stopping rule on the equilibrium path, where all firms charge p^* . By stationarity, the consumer’s reservation *match value* z satisfies

$$z - p^* = -s + F(z)(z - p^*) + \int_z^{\bar{u}} (u - p^*) dF(u).$$

The left-hand side is the net surplus from accepting an offer with match value exactly z . The right-hand side is the expected continuation value: with probability $F(z)$ the next draw is below z and the consumer falls back on the current offer (valued at $z - p^*$), and with probability $1 - F(z)$ the next draw exceeds z and the consumer takes it.

Crucially, the equilibrium price p^* *cancels out*, yielding

$$s = \int_z^{\bar{u}} (u - z) dF(u). \tag{9}$$

This is exactly the same form as the generic reservation-value equation (6). The reservation match value z depends only on the search cost s and the distribution of match values F —not on the price level. The intuition is that, on the equilibrium path, all firms charge the same price, so price differences cannot motivate search; only the prospect of a better match value drives continued search.

Off-path stopping rule. When a firm deviates to $p_i \neq p^*$, the consumer adjusts the stopping rule: buy from firm i if $u_i - p_i \geq z - p^*$, i.e. if the net surplus exceeds the reservation net surplus. If $z < p^*$ (search costs are so high that the reservation surplus is negative), consumers leave the market entirely.

Demand. A deviating firm charging p (while all others charge p^*) faces two sources of demand:

1. **Fresh demand** D_F : consumers visiting the firm for the first time whose match value clears the threshold, $u \geq z - p^* + p$, and who found nothing acceptable on their prior search path (all previous draws below z):

$$D_F(p, p^*) = [1 - F(z - p^* + p)] \cdot \frac{1 - F(z)^n}{1 - F(z)}.$$

2. **Return demand** D_R : consumers who visited the firm earlier, continued searching, but ultimately return because this firm offered the best net surplus among all visited firms:

$$D_R(p, p^*) = n \int_p^{z+p-p^*} [F(u - p + p^*)]^{n-1} dF(u).$$

Total demand is $D = D_F + D_R$.

Equilibrium price. At the symmetric equilibrium ($p = p^*$), demand simplifies to $D(p^*, p^*) = 1 - F(p^*)^n$. The first-order condition of profit maximisation $\pi = (p - c) D(p, p^*)$ evaluated at $p = p^*$ gives

$$p^* = c + \frac{1 - F(p^*)^n}{f(z) \frac{1 - F(z)^n}{1 - F(z)} - n \int_z^{p^*} F(u)^{n-1} f'(u) du}. \quad (10)$$

Together with the reservation-value equation (9), this characterises the equilibrium.

The limiting equilibrium ($n \rightarrow \infty$). As the number of firms grows, return demand vanishes (with infinitely many firms and a stationary reservation value, consumers never return to a previously visited firm). The equilibrium price converges to

$$p^* = c + \frac{1 - F(z)}{f(z)}. \quad (11)$$

This is a remarkably clean expression: the markup equals the inverse hazard rate of the match-value distribution evaluated at the reservation value. The price remains *bounded away from marginal cost* even with infinitely many firms—hence the title of Wolinsky’s paper, “True Monopolistic Competition.” Each firm retains market power because search frictions make consumers reluctant to keep searching for a better match.

Convergence to competition. However, if search frictions vanish simultaneously ($s \rightarrow 0$), the reservation value z approaches $\bar{u}$, so $F(z) \rightarrow 1$ and $p^* \rightarrow c$. Competition is restored only when both the number of firms is large *and* search is essentially costless.

Intuition from the infinite-firm case. With infinitely many firms, the demand faced by a single firm reduces to

$$D = \frac{1 - F(z - p^* + p)}{1 - F(z)},$$

which depends only on fresh visitors (no returns). Profit is $(p - c) \cdot D$, and the FOC at $p = p^*$ yields

$$-(p^* - c) \frac{f(z)}{1 - F(z)} + 1 = 0 \implies p^* = c + \frac{1 - F(z)}{f(z)},$$

confirming (11). This simpler derivation highlights the role of the hazard rate: a higher hazard rate $f(z)/[1 - F(z)]$ means more consumers are “on the margin” of accepting, so demand is more elastic and the markup is smaller.

Eliaz and Spiegler (2011): A Model of a Search Engine

The WAR framework assumes consumers search directly among firms. In practice, much consumer search is *intermediated* by platforms—most prominently, search engines. Eliaz and Spiegler (2011) model a monopoly search engine that controls which firms consumers can find, and study how the platform’s incentives interact with search frictions.

Setting: firms.

- A large (infinite) number of firms produce differentiated products.
- Firms are *vertically* differentiated by quality (“relevance”) $\theta \in [0, 1]$, distributed according to $G(\theta)$.
- A consumer visiting firm i with relevance θ_i obtains match value 0 with probability $1 - \theta_i$, and match value v with probability θ_i , where $v \sim F$ on $[0, 1]$.

Thus θ captures how likely it is that the firm is a good match at all (vertical quality), while v captures *how good* the match is conditional on being relevant (horizontal differentiation).

Setting: the search engine.

- A monopoly search engine controls access to the search pool by setting a **per-click fee** r .
- Firms that accept r enter the pool; those that do not are invisible to consumers.
- This is equivalent to the search engine choosing a quality threshold θ^* : only firms with $\theta \geq \theta^*$ enter.
- Consumers search sequentially at cost s through the pool.

Timing.

1. The search engine sets r .
2. Firms decide whether to accept r and enter the search pool.
3. Consumers search sequentially, make purchases, and the search engine collects per-click payments.

The model is solved by backward induction.

Consumer search and the reservation value. Define $\theta^e = \mathbb{E}[\theta \mid \theta \geq \theta^*]$ as the expected relevance of firms in the pool. The consumer’s reservation *match value* v^* satisfies

$$\theta^e \int_{v^*}^1 (z - v^*) dF(z) = s. \quad (12)$$

This has the same structure as (6), but scaled by θ^e : the expected relevance acts as a “discount” on the option value of search, because with probability $1 - \theta^e$ the next firm is entirely irrelevant (match value zero). A consumer stops and buys when $v - p \geq v^* - p^*$, and continues searching otherwise.

Firm pricing. A crucial feature of the per-click fee is that it is *sunk* at the moment of the pricing decision: the firm pays r whenever a consumer clicks, regardless of whether a sale occurs. Therefore r does not directly affect the pricing FOC. A firm with relevance θ solves

$$\max_p p \theta (1 - F(v^* - p^* + p)),$$

yielding the symmetric equilibrium price

$$p^* = \frac{1 - F(v^*)}{f(v^*)}. \quad (13)$$

Note this is exactly the WAR markup formula (11) with $c = 0$. The per-click fee affects prices only *indirectly*, through its effect on the composition of the search pool (which determines θ^e , which determines v^*).

Firm profits and search pool entry. A type- θ firm's expected profit per consumer visit (gross of the click fee) is

$$\pi(\theta) = p^* \cdot \theta \cdot (1 - F(v^*)) = \frac{\theta (1 - F(v^*))^2}{f(v^*)}.$$

A firm enters the search pool if and only if $\pi(\theta) \geq r$. The marginal (indifferent) firm has relevance θ^* satisfying

$$\frac{\theta^* (1 - F(v^*))^2}{f(v^*)} = r. \quad (14)$$

Search engine optimisation. The search engine's revenue is the per-click fee times the expected number of clicks per consumer. Expected search duration (number of clicks) is the inverse of the per-visit conversion rate:

$$\text{Expected clicks} = \frac{1}{\theta^e (1 - F(v^*))}.$$

So the search engine solves

$$\max_r \frac{r}{\theta^e (1 - F(v^*))}.$$

Substituting r from (14), this can be rewritten as a choice over the quality threshold θ^* :

$$\max_{\theta^*} \frac{\theta^*}{\theta^e} \cdot \frac{1 - F(v^*)}{f(v^*)}. \quad (15)$$

Trade-offs and potential distortion. Raising θ^* has two competing effects:

On search duration:

- Higher average relevance θ^e means consumers find matches faster, *reducing* clicks.
- But higher θ^e raises the option value of continued search, making consumers “pickier” (higher v^*), which *increases* clicks.

On the per-click fee:

- Higher θ^* raises v^* , which lowers the equilibrium price p^* (more selective consumers make demand more elastic).
- The conversion rate $\theta^*(1 - F(v^*))$ can move in either direction.

The search engine's equilibrium profit satisfies

$$\frac{\pi^*}{p^*} = \frac{\theta^*}{\theta^e}.$$

The left-hand side is a unit-free measure of the intermediation fee. The right-hand side is the ratio of marginal to average firm relevance in the pool. When the search engine does not distort quality, $\theta^*/\theta^e = 1$. Since the overall effect of raising θ^* is ambiguous, the search engine may find it optimal to set θ^* either above or below the efficient level, depending on the specific distributions F and G . This demonstrates that *a profit-maximising search platform can deliberately distort the relevance of its search results*—a central policy-relevant conclusion of the model.